



Lakshya JEE 2027

KPP

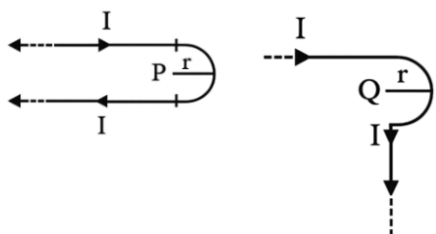
Moving Charges and Magnetism By: Saleem bhaiya

MAGNETIC FIELD DUE TO VARIOUS CURRENT CARRYING CONDUCTORS

1. A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm. The magnetic field at the mid point between the two wires is ____ μT .

$$\left(\frac{\mu_0}{4\pi} = 10^{-7} \text{ N/A}^2 \right). \quad [\text{JM April 6, 2026 (II)}]$$

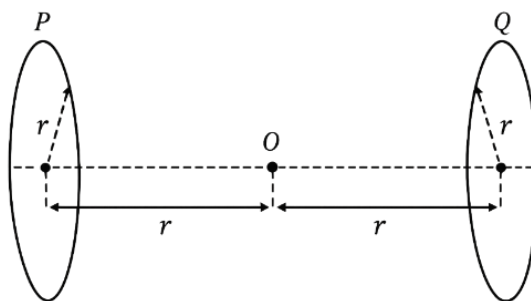
- (1) 30
(2) 300
(3) 150
(4) 0.0
2. Two identical long current carrying wires are bent into the shapes shown in the following figures. If the magnitude of magnetic fields at the centres P and Q of a semicircular arc are B_1 and B_2 respectively, then the ratio $\frac{B_1}{B_2}$ is _____. [JM April 2, 2026 (I)]



- (1) $\frac{2+\pi}{1+\pi}$ (2) $\frac{1+\pi}{1-\pi}$
(3) $\frac{2+\pi}{1-\pi}$ (4) $\frac{1+\pi}{2-\pi}$
3. The magnetic field at the centre of a current carrying circular loop of radius R is $16 \mu\text{T}$. The magnetic field at a distance $x = \sqrt{3}R$ on its axis from the centre is ____ μT . [JM Jan. 28, 2026 (I)]
- (1) 8 (2) 2
(3) 4 (4) $2\sqrt{2}$

4. Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O . The net magnetic field at O is:

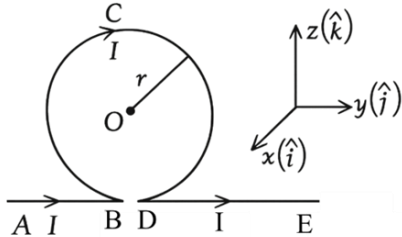
[JM Jan. 24, 2026 (II)]



- (1) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q
(2) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q
(3) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P
(4) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards P
5. The current passing through a conducting loop in the form of equilateral triangle of side $4\sqrt{3}$ cm is 2 A. The magnetic field at its centroid is $\alpha \times 10^{-5}$ T. The value of α is _____. (Given: $\mu_0 = 4\pi \times 10^{-7}$ SI units). [JM Jan. 23, 2026 (II)]
- (1) $3\sqrt{3}$
(2) $2\sqrt{3}$
(3) $\sqrt{3}$
(4) $\frac{\sqrt{3}}{2}$

6. An infinitely long straight wire carrying current I is bent in a planar shape as shown in the diagram. The radius of the circular part is r . The magnetic field at the centre O of the circular loop is:

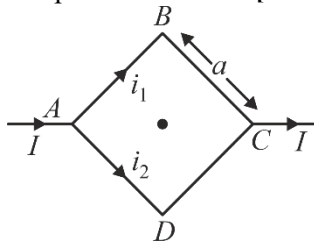
[JM Jan. 21, 2026 (II)]



- (1) $-\frac{\mu_0 I}{2\pi r}(\pi+1)\hat{i}$ (2) $-\frac{\mu_0 I}{2\pi r}(\pi-1)\hat{i}$
 (3) $\frac{\mu_0 I}{2\pi r}(\pi+1)\hat{i}$ (4) $\frac{\mu_0 I}{2\pi r}(\pi-1)\hat{i}$

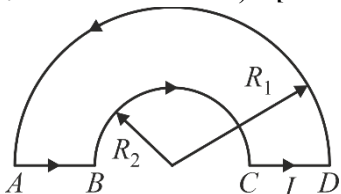
7. Figure shows a current carrying square loop $ABCD$ of edge length ' a ' lying in a plane. If the resistance of the ABC part is r and that of ADC part is $2r$, then the magnitude of the resultant magnetic field at centre of the square loop is

[JM April 8, 2025 (II)]



- (1) $\frac{3\pi\mu_0 I}{\sqrt{2}a}$ (2) $\frac{\mu_0 I}{2\pi a}$
 (3) $\frac{\sqrt{2}\mu_0 I}{3\pi a}$ (4) $\frac{2\mu_0 I}{3\pi a}$

8. A loop $ABCD$, carrying current $I = 12$ A, is placed in a plane, consists of two semi-circular segments of radius $R_1 = 6\pi$ m and $R_2 = 4\pi$ m. The magnitude of the resultant magnetic field at centre O is $k \times 10^{-7}$ T. The value of k is
 (Given $\mu_0 = 4\pi \times 10^{-7}$ TmA $^{-1}$) [JM April 3, 2025 (I)]



9. Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I . Let B_2 be the magnitude of magnetic field at an axial distance ' x ' from the center.

For $x : R = 3 : 4$ $\frac{B_2}{B_1}$ is: [JM April 2, 2025 (I)]

- (1) 4 : 5
 (2) 16 : 25
 (3) 64 : 125
 (4) 25 : 16

10. The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} . If the solenoid carries a current of 0.29 A, then the length of the solenoid is

[JM Jan. 29, 2025 (II)]

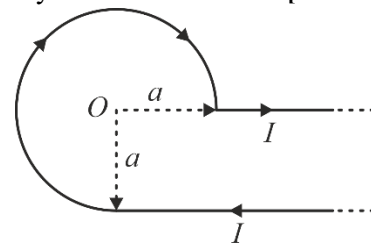
11. Consider a long straight wire of a circular cross-section (radius a) carrying a steady current I . The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be

[JM Jan. 29, 2025 (I)]

- (1) $\left[\frac{a}{4}, \frac{3a}{2}\right]$ (2) $\left[\frac{a}{2}, 2a\right]$
 (3) $\left[\frac{a}{2}, 3a\right]$ (4) $\left[\frac{a}{4}, 2a\right]$

12. An infinite wire has a circular bend of radius a , and carrying a current I as shown in figure. The magnitude of magnetic field at the origin O of the arc is given by:

[JM Jan. 28, 2025 (II)]



- (1) $\frac{\mu_0 I}{4\pi a} \left[\frac{\pi}{2} + 1 \right]$ (2) $\frac{\mu_0 I}{4\pi a} \left[\frac{3\pi}{2} + 1 \right]$
 (3) $\frac{\mu_0 I}{2\pi a} \left[\frac{\pi}{2} + 2 \right]$ (4) $\frac{\mu_0 I}{4\pi a} \left[\frac{3\pi}{2} + 2 \right]$

13. A tightly wound long solenoid carries a current of 1.5 A. An electron is executing uniform circular motion inside the solenoid with a time period of 75 ns. The number of turns per metre in the solenoid is _____. [Take mass of electron $m_e = 9 \times 10^{-31}$ kg, charge of electron $|q_e| = 1.6 \times 10^{-9}$ C, $\mu_0 = 4\pi \times 10^{-7} \frac{\text{N}}{\text{A}^2}$, $1 \text{ ns} = 10^{-9} \text{ s}$.]

[JM Jan. 24, 2025 (II)]

14. A current of 5 A exists in a square loop of side $\frac{1}{\sqrt{2}} \text{ m}$. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6} \text{ T}$. where, value of p is _____. [Take $\mu_0 = 4\pi \times 10^{-7} \text{ T mA}^{-1}$]. [JM Jan. 24, 2025 (I)]

15. A long straight wire of radius a carries a steady current I . The current is uniformly distributed across its cross section. The ratio of the magnetic field at $\frac{a}{2}$ and $2a$ from axis of the wire is:

[JM April 8, 2024 (II)]

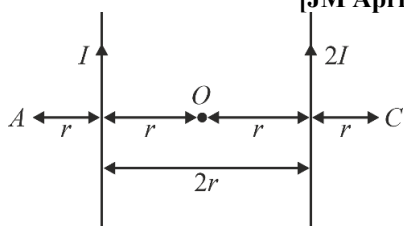
- (1) 1 : 4 (2) 4 : 1
(3) 1 : 1 (4) 3 : 4

16. A solenoid of length 0.5 m has a radius of 1 cm and is made up of ' m ' number of turns. It carries a current of 5 A. If the magnitude of the magnetic field inside the solenoid is $6.28 \times 10^{-3} \text{ T}$, then the value of m is:

[JM April 5, 2024 (II)]

17. Two parallel long current carrying wire separated by a distance $2r$ are shown in the figure. The ratio of magnetic field at A to the magnetic field produced at C is $\frac{x}{7}$. The value of x is ____.

[JM April 4, 2024 (II)]



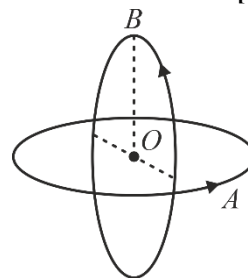
18. A regular polygon of 6 sides is formed by bending a wire of length 4π meter. If an electric current of $4\pi\sqrt{3} \text{ A}$ is flowing through the sides of the polygon, the magnetic field at the centre of the polygon would be $x \times 10^{-7} \text{ T}$. The value of x is ____.

19. Two circular coils P and Q of 100 turns each have same radius of $\pi \text{ cm}$. The currents in P and R are 1 A and 2 A respectively. P and Q are placed with their planes mutually perpendicular with their centers coincide. The resultant magnetic field induction at the center of the coils is $\sqrt{x} \text{ mT}$, where $x =$ ____.

[Use $\mu_0 = 4\pi \times 10^{-7} \text{ T mA}^{-1}$] [JM Jan. 31, 2024 (II)]

20. Two insulated circular loop A and B of radius ' a ' carrying a current of ' I ' in the anti clockwise direction as shown in the figure. The magnitude of the magnetic induction at the centre will be:

[JM Jan. 30, 2024 (I)]



- (1) $\frac{\sqrt{2}\mu_0 I}{a}$ (2) $\frac{\mu_0 I}{2a}$
(3) $\frac{\mu_0 I}{\sqrt{2}a}$ (4) $\frac{2\mu_0 I}{a}$

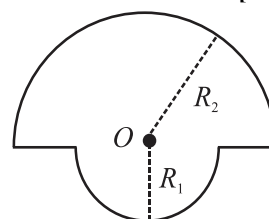
21. The current of 5 A flows in a square loop of sides 1 m is placed in air. The magnetic field at the centre of the loop is $X\sqrt{2} \times 10^{-7} \text{ T}$. The value of X is ____.

[JM Jan. 30, 2024 (II)]

22. The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2 \pi \text{ m}$, and $R_2 = 4 \pi \text{ m}$ carrying current $I = 4 \text{ A}$ as per figure given below is $\alpha \times 10^{-7} \text{ T}$. The value of α is ____.

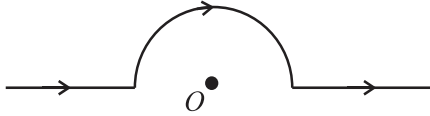
(Centre O is common for all segments)

[JM Jan. 27, 2024 (II)]



23. A straight wire carrying a current of 14 A is bent into a semicircular arc of radius 2.2 cm as shown in the figure. The magnetic field produced by the current at the centre (O) of the arc. Is $\text{_____} \times 10^{-4} \text{ T}$.

[JM April 10, 2023 (II)]

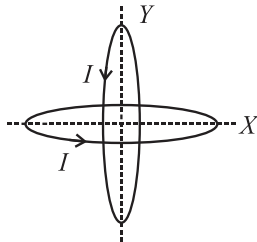


24. The ratio of magnetic field at the centre of a current carrying coil of radius r to the magnetic field at distance r from the centre of coil on its axis is $\sqrt{x}:1$. The value of x is _____ .

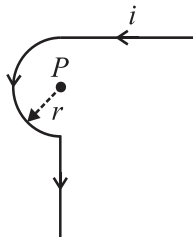
[JM April 8, 2023 (II)]

25. Two identical circular wires of radius 20 cm and carrying current $\sqrt{2}A$ are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wire is $\text{_____} \times 10^{-8} \text{ T}$. (Take $\pi = 3.14$)

[JM April 6, 2023 (I)]



26. Find the magnetic field at the point P in figure.



The curved portion is a semicircle connected to two long straight wires.

[JM Feb. 1, 2023 (I)]

- (1) $\frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{2\pi} \right)$ (2) $\frac{\mu_0 i}{2r} \left(1 + \frac{1}{\pi} \right)$
(3) $\frac{\mu_0 i}{2r} \left(1 + \frac{2}{\pi} \right)$ (4) $\frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{\pi} \right)$

27. As shown in the figure, a long straight conductor with semicircular arc of radius $\frac{\pi}{10} \text{ m}$ is carrying current $I = 3A$. The magnitude of the magnetic field, at the center O of the arc is:

(The permeability of the vacuum $= 4\pi \times 10^{-7} \text{ NA}^{-2}$)

[JM Feb. 1, 2023 (II)]



- (1) $6 \mu\text{T}$ (2) $1 \mu\text{T}$
(3) $3 \mu\text{T}$ (4) $4 \mu\text{T}$

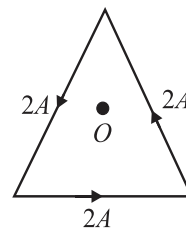
28. A long conducting wire having a current I flowing through it, is bent into a circular coil of N turns. Then it is bent into a circular coil of n turns. The magnetic field is calculated at the centre of coils in both the cases. The ratio of the magnetic field in first case to that of second case is:

[JM Jan. 31, 2023 (II)]

- (1) $N:n$
(2) $n^2:N^2$
(3) $N^2:n^2$
(4) $n:N$

29. As shown in the figure, a current of 2A flowing in an equilateral triangle of side $4\sqrt{3} \text{ cm}$. The magnetic field at the centroid O of the triangle is: (Neglect the effect of earth's magnetic field.)

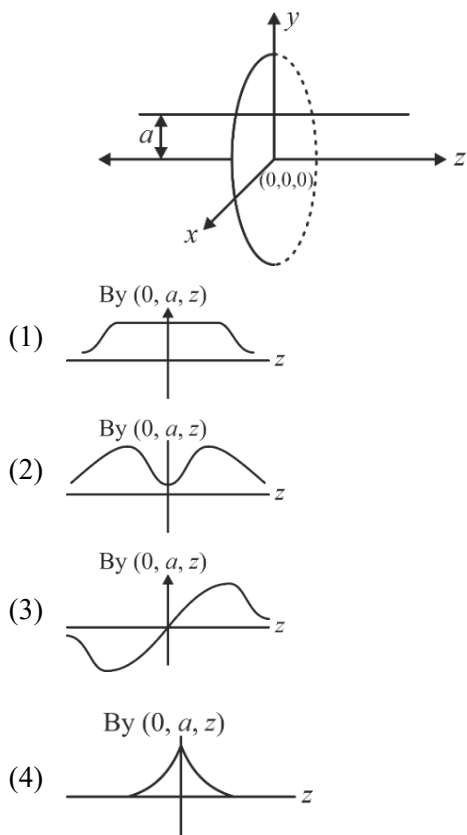
[JM Jan. 30, 2023 (II)]



- (1) $4\sqrt{3} \times 10^{-4} \text{ T}$
(2) $4\sqrt{3} \times 10^{-5} \text{ T}$
(3) $\sqrt{3} \times 10^{-4} \text{ T}$
(4) $3\sqrt{3} \times 10^{-5} \text{ T}$

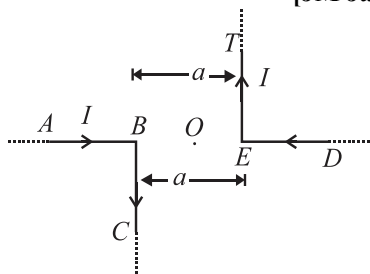
30. A single current carrying loop of wire carrying current I flowing in anticlockwise direction seen from +ve z direction and lying in xy plane in shown in figure. The plot of $\hat{j}$ component of magnetic field (B_y) at a distance ' a ' (less than radius of the coil) and on yz plane vs z coordinate look like

[JM Jan. 29, 2023 (I)]



31. The magnitude of magnetic induction at mid-point O due to current arrangement as shown in Fig will be:

[JM Jan. 29, 2023 (I)]



- (1) $\frac{\mu_0 I}{2\pi a}$ (2) 0
- (3) $\frac{\mu_0 I}{4\pi a}$ (4) $\frac{\mu_0 I}{\pi a}$

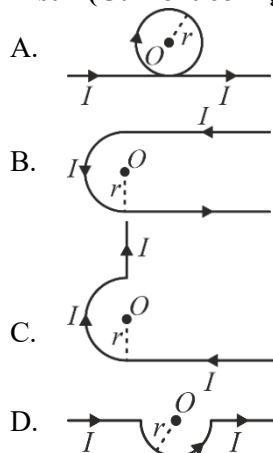
32. The electric current in a circular coil of four turns produces a magnetic induction 32 T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be:

[JM Jan. 29, 2023 (II)]

- (1) 8 T (2) 4 T
(3) 2 T (4) 16 T

33. Math List-I with List-II. [JM Jan. 25, 2023 (I)]

List-I (Current configuration)



List-II (Magnetic field at point O)

- I. $B_0 = \frac{\mu_0 I}{4\pi r} [\pi + 2]$
- II. $B_0 = \frac{\mu_0}{4} \frac{I}{r}$
- III. $B_0 = \frac{\mu_0 I}{2\pi r} [\pi - 1]$
- IV. $B_0 = \frac{\mu_0 I}{4\pi r} [\pi + 1]$

Choose the correct answer from the option given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-I, C-IV, D-III

34. A long solenoid is formed by winding 70 turns cm^{-1} . If 2.0 A current flows, then the magnetic field produced inside the solenoid is _____.

($\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$) [JM Jan. 24, 2023 (II)]

- (1) $1232 \times 10^{-4} \text{ T}$
(2) $176 \times 10^{-4} \text{ T}$
(3) $352 \times 10^{-4} \text{ T}$
(4) $88 \times 10^{-4} \text{ T}$

35. A closely wound circular coil of radius 5 cm produces a magnetic field of 37.68×10^{-4} T at its center. The current through the coil is _____ A.
[Given, number of turns in the coil is 100 and $\pi = 3.14$]
[JM July 29, 2022 (I)]

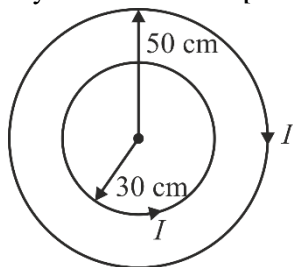
36. The magnetic field at the center of current carrying circular loop is B_1 . The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2 . The value of B_1/B_2 be
[JM July 28, 2022 (II)]

- (1) 9 : 4
(2) $12 : \sqrt{5}$
(3) 8 : 1
(4) $5 : \sqrt{3}$

37. B_X and B_Y are the magnetic fields at the centre of two coils X and Y respectively each carrying equal current. If coil X has 200 turns and 20 cm radius and coil Y has 400 turns and 20 cm radius, the ratio of B_X and B_Y is:
[JM July 26, 2022 (I)]

- (1) 1 : 1 (2) 1 : 2
(3) 2 : 1 (4) 4 : 1

38. Two concentric circular loops of radii $r_1 = 30$ cm and $r_2 = 50$ cm are placed in X-Y plane as shown in the figure. A current $I = 7$ A is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately:
[JM July 26, 2022 (II)]



- (1) $\frac{7}{2} \hat{k} \text{ Am}^2$ (2) $-\frac{7}{2} \hat{k} \text{ Am}^2$
(3) $7 \hat{k} \text{ Am}^2$ (4) $-7 \hat{k} \text{ Am}^2$

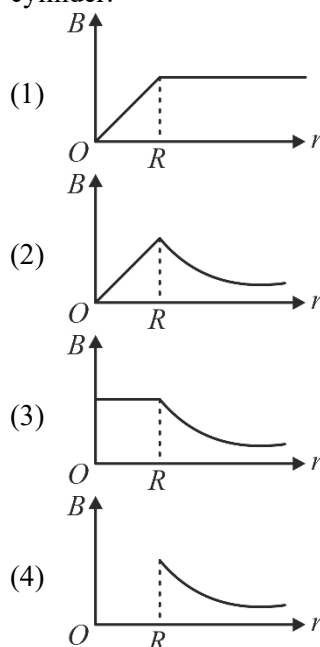
39. The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of B_2/B_1 is:
[JM July 25, 2022 (II)]

- (1) 5/2 (2) 25/4
(3) 5/4 (4) 25/2

40. A coil of n number of turns wound tightly in the form of a spiral with inner and outer radii r_1 and r_2 respectively. When a current of strength I is passed through the coil, the magnetic field at its centre will be:
[JM June 30, 2022 (I)]

- (1) $\frac{\mu_0 n I}{2(r_2 - r_1)}$
(2) $\frac{\mu_0 n I}{r_2}$
(3) $\frac{\mu_0 n I}{r_2 - r_1} \log_e \frac{r_1}{r_2}$
(4) $\frac{\mu_0 n I}{2(r_2 - r_1)} \log_e \frac{r_2}{r_1}$

41. An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface. Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.
[JM June 28, 2022 (I)]



42. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

[JM June 25, 2022 (II)]

- (1) B (2) $2B$
(3) $4B$ (4) $B/2$

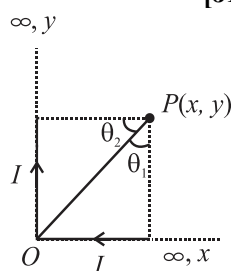
43. The magnetic field at the centre of a circular coil of radius r , due to current I flowing through it, is B . The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is:

[JM June 247, 2022 (I)]

- (1) $B/2$
(2) $2B$
(3) $\left(\frac{2}{\sqrt{5}}\right)^3 B$
(4) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

44. There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductors is 1 : 1. The magnetic field at point P is ____.

[JM Sep. 1, 2021 (II)]



- (1) $\frac{\mu_0 I}{4\pi xy} \left[\sqrt{x^2 + y^2} + (x + y) \right]$
(2) $\frac{\mu_0 I}{4\pi xy} \left[\sqrt{x^2 + y^2} - (x + y) \right]$
(3) $\frac{\mu_0 Ixy}{4\pi} \left[\sqrt{x^2 + y^2} - (x + y) \right]$
(4) $\frac{\mu_0 Ixy}{4\pi} \left[\sqrt{x^2 + y^2} + (x + y) \right]$

45. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii ' a ' and ' b ' respectively. Find the magnetic field at centre, when a current I passes through coil:

[JM Aug. 31, 2021 (I)]

- (1) $\frac{\mu_0 IN}{2(b-a)} \log_e \left(\frac{b}{a} \right)$
(2) $\frac{\mu_0 I}{8} \left[\frac{a+b}{a-b} \right]$
(3) $\frac{\mu_0 I}{4(a-b)} \left[\frac{1}{a} - \frac{1}{b} \right]$
(4) $\frac{\mu_0 I}{8} \left(\frac{a-b}{a+b} \right)$

46. A current of 1.5 A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is:

(Assume that the current is flowing in the clockwise direction.) [JM Aug. 31, 2021 (II)]

- (1) $3 \times 10^{-7} T$, outside the plane of triangle.
(2) $2\sqrt{3} \times 10^{-7} T$, outside the plane of triangle.
(3) $2\sqrt{3} \times 10^{-5} T$ inside the plane of triangle.
(4) $3 \times 10^{-5} T$, inside the plane of triangle.

47. A uniform conducting wire of length is $24a$, and resistance R is wound up as a current carrying coil in the shape of an equilateral triangle of side ' a ' and then in the form of a square of side ' a '. The coil is connected to a voltage source V_0 . The ratio of magnetic moment of the coils in case of equilateral triangle to that for square is $1 : \sqrt{y}$ where y is ____.

[JM Aug. 24, 2021 (I)]

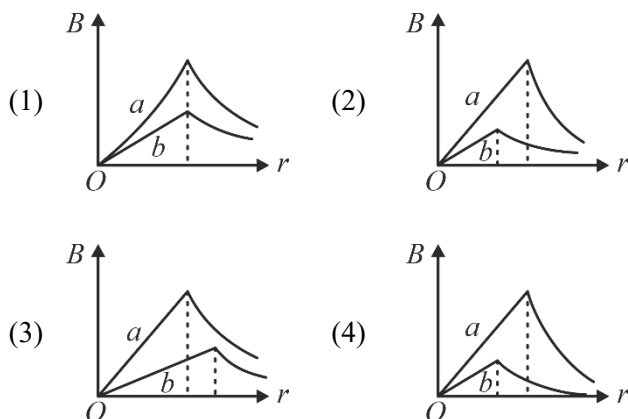
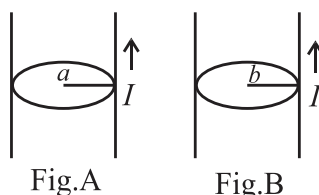
48. The fractional change in the magnetic field intensity at a distance ' r ' from centre on the axis of current carrying coil of radius ' a ' to the magnetic field intensity at the centre of the same coil is: (Take $r < a$)

[JM Aug. 26, 2021 (I)]

- (1) $\frac{3a^2}{2r^2}$ (2) $\frac{2a^2}{3r^2}$
(3) $\frac{2r^2}{3a^2}$ (4) $\frac{3r^2}{2a^2}$

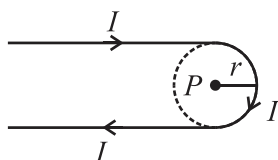
49. Figure A and B shown two long straight wires of circular cross-section (a and b with $a < b$), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as:

[JM July 27, 2021 (II)]



50. A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle?

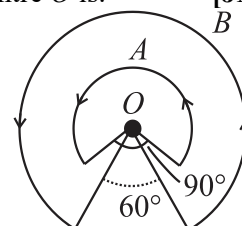
[JM March 17, 2021 (II)]



- (1) $\frac{\mu_0 I}{4\pi r} (2 + \pi)$
- (2) $\frac{\mu_0 I}{4\pi r} (2 - \pi)$
- (3) $\frac{\mu_0 I}{2\pi r} (2 - \pi)$
- (4) $\frac{\mu_0 I}{2\pi r} (2 + \pi)$

51. A wire A , bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire B , also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is:

[JM Sep. 04, 2020 (I)]



- (1) 4 : 6
- (2) 6 : 4
- (3) 2 : 5
- (4) 6 : 5

52. Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is:

[JM Sep. 03, 2020 (I)]

- (1) $250\sqrt{3}$
- (2) $50\sqrt{3}$
- (3) $500\sqrt{3}$
- (4) $5\sqrt{3}$

53. A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $a/3$ and $2a$, respectively from the axis of the wire is:

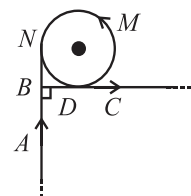
[JM 2007 S, 9 Jan 2020, (I)]

- (1) $2/3$
- (2) 2
- (3) $1/2$
- (4) $3/2$

54. A very long wire $ABDMNDC$ is shown in figure carrying current I . AB and BC parts are straight, long and at right angle. At D wire forms a circular turn $DMND$ of radius R .

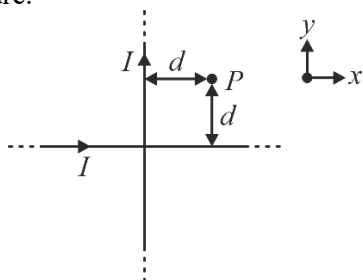
AB , BC parts are tangential to circular turn at N and D . Magnetic field at the centre of circle is:

[JM 8 Jan 2020, (II)]



- (1) $\frac{\mu_0 I}{2\pi R} \left(\pi + \frac{1}{\sqrt{2}} \right)$
- (2) $\frac{\mu_0 I}{2\pi R} \left(\pi - \frac{1}{\sqrt{2}} \right)$
- (3) $\frac{\mu_0 I}{2\pi R} (\pi + 1)$
- (4) $\frac{\mu_0 I}{2\pi R}$

55. Two very long, straight, and insulated wires are kept at 90° angle from each other in xy -plane as shown in the figure.

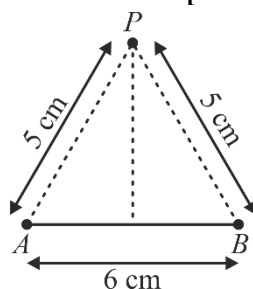


These wires carry currents of equal magnitude I , whose directions are shown in the figure. The net magnetic field at point P will be:

[JM 12 April 2019, (I)]

- (1) Zero (2) $-\frac{\mu_0 I}{2\pi d}(\hat{x} + \hat{y})$
 (3) $\frac{+\mu_0 I}{\pi d}(\hat{z})$ (4) $\frac{\mu_0 I}{2\pi d}(\hat{x} + \hat{y})$
56. A thin ring of 10 cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of $40\pi \text{ rad s}^{-1}$ about its axis, perpendicular to its plane. If the magnetic field at its centre is $3.8 \times 10^{-9} \text{ T}$, then the charge carried by the ring is close to ($\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$).
- [JM 12 April 2019, (I)]
- (1) $2 \times 10^{-6} \text{ C}$ (2) $3 \times 10^{-5} \text{ C}$
 (3) $4 \times 10^{-5} \text{ C}$ (4) $7 \times 10^{-6} \text{ C}$
57. Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5 A. (See figure) ($\mu_0 = 4\pi \times 10^{-7} \text{ N-A}^{-2}$)

[JM 12 April 2019, (II)]



- (1) $2.0 \times 10^{-5} \text{ T}$
 (2) $1.5 \times 10^{-5} \text{ T}$
 (3) $3.0 \times 10^{-5} \text{ T}$
 (4) $2.5 \times 10^{-5} \text{ T}$

58. The magnitude of the magnetic field at the center of an equilateral triangular loop of side 1 m which is carrying a current of 10 A is:

[Take $\mu_0 = 4\pi \times 10^{-7} \text{ N-A}^{-2}$]

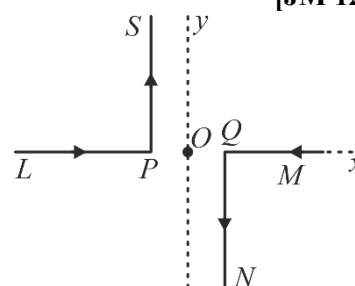
[JM April 15, 2018 S, 10 April 2019, II]

- (1) $18 \mu\text{T}$ (2) $9 \mu\text{T}$
 (3) $3 \mu\text{T}$ (4) $1 \mu\text{T}$

59. As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x -axis, while segments PS and QN are parallel to the y -axis.

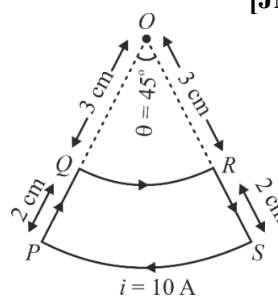
If $OP = OQ = 4 \text{ cm}$, and the magnitude of the magnetic field at O is 10^{-4} T and the two wires carry equal currents (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\mu_0 = 4\pi \times 10^{-7} \text{ N-A}^{-2}$):

[JM 12 Jan 2019, (I)]



- (1) 20 A, perpendicular out of the page
 (2) 40 A, perpendicular out of the page
 (3) 20 A, perpendicular into the page
 (4) 40 A, perpendicular into the page
60. A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to:

[JM 9 Jan. 2019 (I)]

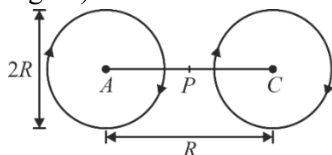


- (1) $1.0 \times 10^{-7} \text{ T}$
 (2) $1.5 \times 10^{-7} \text{ T}$
 (3) $1.5 \times 10^{-5} \text{ T}$
 (4) $1.0 \times 10^{-5} \text{ T}$

61. One of the two identical conducting wires length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the centre of the loop (B_L) to that at the centre of the coil (B_C), i.e., $\frac{B_L}{B_C}$ will be: [JM 2019]

- (1) N (2) $1/N$
(3) N^2 (4) $1/N^2$

62. A Helmholtz coil has pair of loops, each with N turns and radius R . They are placed coaxially at distance R and the same current I flows through the loops in the same direction. The magnitude of magnetic field at P , midway between the centres A and C , is given by (Refer to figure): [JM 2018]



- (1) $\frac{4N\mu_0 I}{5^{3/2} R}$ (2) $\frac{8N\mu_0 I}{5^{3/2} R}$
(3) $\frac{4N\mu_0 I}{5^{1/2} R}$ (4) $\frac{8N\mu_0 I}{5^{1/2} R}$

63. Two identical wires A and B , each length ' l ', carry the same current I . Wire A is bent into a circle of radius R and wire B is bent to form a square of side ' a '. If B_A and B_B are the values of magnetic field at the centres the circle and square respectively, then the ratio B_A/B_B is: [JM 2016]

- (1) $\frac{\pi^2}{16}$ (2) $\frac{\pi^2}{8\sqrt{2}}$
(3) $\frac{\pi^2}{8}$ (4) $\frac{\pi^2}{16\sqrt{2}}$

64. Consider two thin identical conducting wires covered with very thin insulating material. One of the wires is bent into a loop and produces magnetic field B_1 at its centre when a current I passes through it. The second wire is bent into a coil with three identical loops adjacent to each other produces ϕ magnetic field B_2 at the centre of loops when current $I/3$ passes through it. The ratio $B_1:B_2$ is: [JM April 12, 2014]

- (1) 1 : 1 (2) 1 : 3
(3) 1 : 9 (4) 9 : 1

65. A current i is flowing in a straight conductor of length L . The magnetic induction at a point on its axis at a distance $L/4$ from its centre will be:

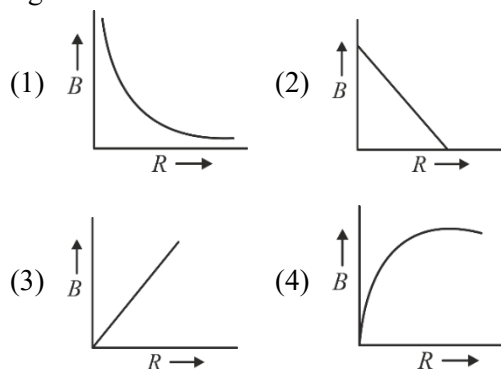
[JM April 22, 2013]

- (1) Zero (2) $\frac{\mu_0 i}{2\pi L}$
(3) $\frac{\mu_0 i}{\sqrt{2} L}$ (4) $\frac{4\mu_0 i}{\sqrt{5}\pi L}$

66. An electric current is flowing through a circular coil of radius R . The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil and on its axis is: [JM April 9, 2013]

- (1) $2\sqrt{2}$ (2) 27
(3) 36 (4) 8

67. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R . The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω . As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure: [JM 2012]

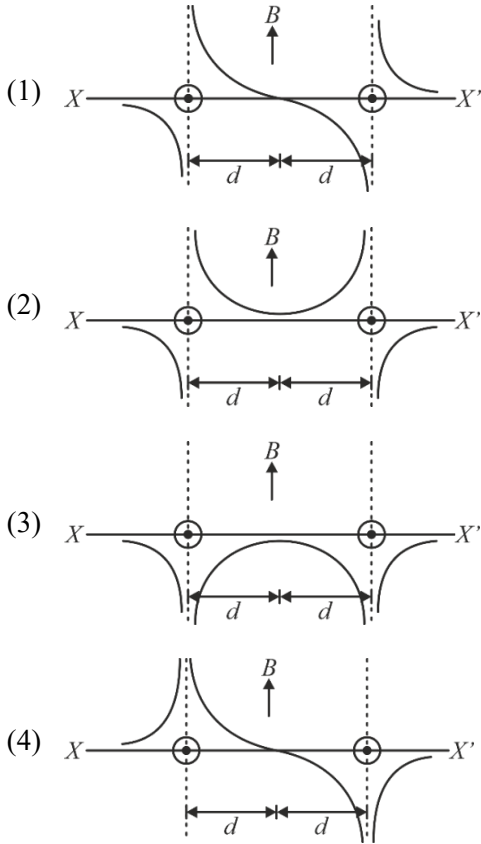


68. A current I flows in an infinitely long wire with cross section in the form of a semi-circular ring of radius R . The magnitude of the magnetic induction along its axis is: [JM 2011]

- (1) $\frac{\mu_0 I}{2\pi^2 R}$ (2) $\frac{\mu_0 I}{2\pi R}$
(3) $\frac{\mu_0 I}{4\pi R}$ (4) $\frac{\mu_0 I}{\pi^2 R}$

69. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by

[JM 2010]



70. A horizontal overhead power line is at height of 4m from the ground and carries a current of 100A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7} \text{ T A}^{-1}$)

[JM 2008]

- (1) $2.5 \times 10^{-7} \text{ T}$ south ward
 (2) $5 \times 10^{-6} \text{ T}$ north ward
 (3) $5 \times 10^{-6} \text{ T}$ south ward
 (4) $2.5 \times 10^{-7} \text{ T}$ north ward

71. Two identical conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries an electric current I_1 and COD carries a current I_2 . The magnetic field on point lying at a distance d from O , in a direction perpendicular to the plane of the wires AOB and COD , will be given by

[JM 2007]

- (1) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)$ (2) $\frac{\mu_0}{2\pi} \left(\frac{I_1 + I_2}{d} \right)^2$
 (3) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{\frac{1}{2}}$ (4) $\frac{\mu_0}{2\pi d} (I_1 + I_2)$

72. A long solenoid has 200 turns per cm and carries a current i . The magnetic field at its centre is $6.28 \times 10^{-2} \text{ Weber/m}^2$. Another long solenoid has 100 turns per cm and it carries a current $i/3$. The value of the magnetic field at its centre is:

[JM 2006]

- (1) $1.05 \times 10^{-2} \text{ Weber/m}^2$
 (2) $1.05 \times 10^{-5} \text{ Weber/m}^2$
 (3) $1.05 \times 10^{-3} \text{ Weber/m}^2$
 (4) $1.05 \times 10^{-4} \text{ Weber/m}^2$

73. A long wire carries a steady current I . It is bent into a circle of one turn and the magnetic field at the centre of the coil is B . It is then bent into a circular loop of n turns. The magnetic field at the centre of the coil will be:

[JM 2004]

- (1) $2nB$ (2) n^2B
 (3) nB (4) $2n^2B$

74. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is $54 \mu\text{T}$. What will be its value at the centre of loop?

[JM 2004]

- (1) $125 \mu\text{T}$ (2) $150 \mu\text{T}$
 (3) $250 \mu\text{T}$ (4) $75 \mu\text{T}$

75. If in a circular coil A of radius R , current I is flowing and in another coil B of radius $2R$ a current $2I$ is flowing, then the ratio of the magnetic fields B_A and B_B , produced by them at their centres will be:

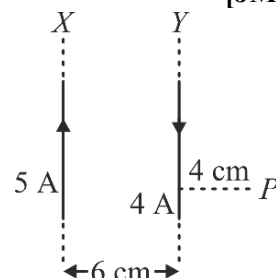
[JM 2002]

- (1) 1 (2) 2
 (3) $1/2$ (4) 4

76. Two long parallel wires X and Y , separated by a distance of 6 cm, carry currents of 5A and 4A, respectively, in opposite directions as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is:

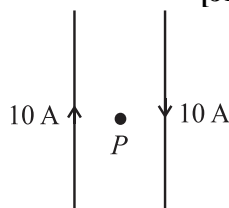
$x \times 10^{-5} \text{ T}$. The value of x is _____. (Take permeability of free space as $\mu_0 = 4\pi \times 10^{-7} \text{ SI units}$.)

[JM Jan. 22, 2025 (I)]



77. Two long, straight wires carry equal currents in opposite directions as shown in figure. The separation between the wires is 5.0 cm. The magnitude of the magnetic field at a point P midway between the wires is (Given: $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$).

[JM Jan. 27, 2024 (I)]



78. Two long parallel wires carrying currents 8A and 15A in opposite directions are placed at a distance of 7 cm from each other. A point P is at equidistant from both the wires such that the lines joining the point P to the wires are perpendicular to each other. The magnitude of magnetic field at P is $___ \times 10^{-6} \text{ T}$. (Given: $\sqrt{2} = 1.4$)

[JM Jan. 25, 2023 (II)]

79. Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is $300 \mu\text{T}$. The equal current flowing in the two conductors is:

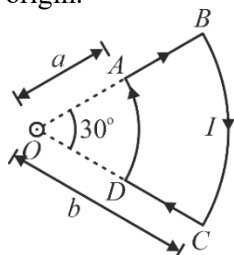
[JM June 29, 2022 (II)]

- (1) 30A in the same direction.
- (2) 30A in the opposite direction.
- (3) 60A in the opposite direction.
- (4) 300A in the opposite direction

Directions: Question numbers 80 and 81 are based on the following paragraph.

A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius = b) and DA (radius = a) of the loop are joined by two straight wires AB and CD . A steady current I is flowing in the loop. Angle made by AB and CD at the origin O is 30° . Another straight thin wire with steady current I_1 flowing out of the plane of the paper is kept at the origin.

[JM 2009]



80. The magnitude of the magnetic field (B) due to the loop $ABCD$ at the origin (O) is:

[JM 2009]

- (1) $\frac{\mu_0 I(b-a)}{24ab}$
- (2) $\frac{\mu_0 I}{4\pi} \left[\frac{b-a}{ab} \right]$
- (3) Zero
- (4) $\frac{\mu_0 I}{4\pi} \left[2(b-a) + \frac{\pi}{3}(a+b) \right]$

81. Due to the presence of the current I_1 at the origin:

[JM 2009]

- (1) The forces on AD and BC are zero.
- (2) The magnitude of the net force on the loop is given by $\frac{I_1 I}{4\pi} \mu_0 [2(b-a) + \pi/3(a+b)]$.
- (3) The magnitude of the net force on the loop is given by $\frac{\mu_0 I_1}{24ab} (b-a)$.
- (4) The forces on AB and DC are zero.

AMPERE'S CIRCUITAL LAW

82. A long cylindrical conductor with large cross section carries an electric current distributed uniformly over its cross-section. Magnetic field due to this current is:

- A. maximum at either ends of the conductor and minimum at the midpoint
- B. maximum at the axis of the conductor
- C. minimum at the surface of the conductor
- D. minimum at the axis of the conductor
- E. same at all points in the cross-section of the conductor

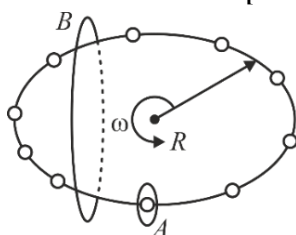
Choose the correct answer from the options given below:

[JM Jan. 28, 2026 (II)]

- (1) B, C Only
- (2) E Only
- (3) A, D Only
- (4) D Only

83. N equally spaced charges each of value q , are placed on a circle of radius R . The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The difference between enclosed currents $I_A - I_B$, for the given Amperian loops is

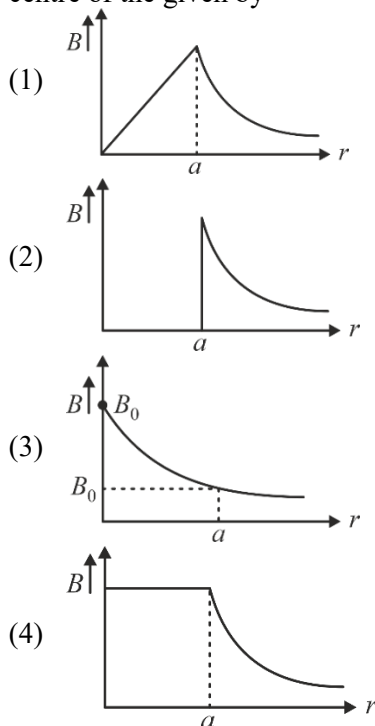
[JM Jan. 24, 2025 (II)]



- (1) $\frac{N^2}{2\pi} q\omega$ (2) $\frac{2\pi}{N} q\omega$
 (3) $\frac{N}{2\pi} q\omega$ (4) $\frac{N}{\pi} q\omega$

84. A long straight wire of a circular cross-section with radius ' a ' carries a steady current I . The current I is a uniformly distributed across this cross-section. The plot of magnitude field B with distance r from the centre of the given by

[JM Jan. 24, 2025 (II)]



85. A long straight wire of circular cross-section (radius a) is carrying steady current I . The current I is uniformly distributed across this cross-section. The magnetic field is [JM April 6, 2023 (I)]

- (1) Zero in the region $r < a$ and inversely proportional to r in the region $r > a$.
 (2) Inversely proportional to r in the region $r < a$ and uniform throughout in the region $r > a$.
 (3) Directly proportional to r in the region $r < a$ and inversely proportional to r in the region $r > a$.
 (4) Uniform in the region $r < a$ and inversely proportional to distance r from the axis, in the region $r > a$.

86. A long straight wire with a circular cross-section having radius R , is carrying a steady current I . The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r ($r < R$) from its centre will be: [JM June 25, 2022 (I)]

- (1) $B \propto r^2$ (2) $B \propto r$
 (3) $B \propto \frac{1}{r^2}$ (4) $B \propto \frac{1}{r}$

87. A coaxial cable consists of an inner wire of radius ' a ' surrounded by an outer shell of inner and outer radii ' b ' and ' c ' respectively. The inner wire carries an electric current i_0 which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly.

What will be the ratio of the magnetic field at a distance x from the axis when (i) $x < a$ and (ii) $a < x < b$? [JM Aug. 27, 2021 (II)]

- (1) $\frac{x^2}{a^2}$ (2) $\frac{a^2}{x^2}$
 (3) $\frac{x^2}{b^2 - a^2}$ (4) $\frac{b^2 - a^2}{x^2}$

88. A current I flows along the length of an infinitely long, straight, thin walled pipe. Then [JM 2008]

- (1) The magnetic field at all points inside the pipe is the same, but not zero
 (2) The magnetic field is zero only on the axis of the pipe
 (c) The magnetic field is different at different points inside the pipe
 (4) The magnetic field at any point inside the pipe is zero

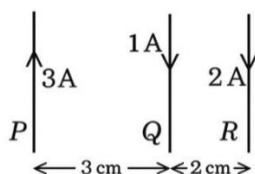
89. A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is: [JM 2004]

- (1) $\frac{\mu_0}{4\pi} \cdot \frac{2i}{r}$ tesla
- (2) Zero
- (3) Infinite
- (4) $\frac{2i}{r}$ tesla

MAGNETIC FORCE ON A CURRENT CARRYING CONDUCTOR

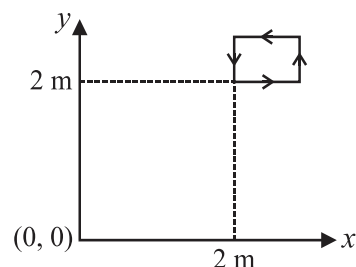
90. Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is _____. ($\mu_0 = 4\pi \times 10^{-7}$ T.m/A).

[JM Jan 28, 2026 (I)]



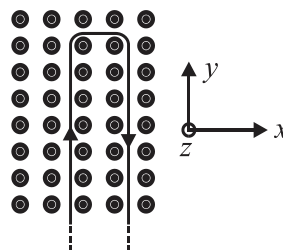
- (1) 6×10^{-6} N towards P
 - (2) 6×10^{-6} N towards R
 - (3) 6×10^{-7} N towards P
 - (4) 6×10^{-7} N towards R
91. A 4.0 cm long straight wire carrying a current of 8 A is placed perpendicular to an uniform magnetic field of strength 0.15 T. The magnetic force on the wire is ____ mN. [JM April 3, 2025 (II)]
92. A 2A current carrying straight metal wire of resistance 1Ω , resistivity $2 \times 10^{-6} \Omega\text{m}$, area of cross-section 10 mm^2 and mass 500 g is suspended horizontally in mid air by applying a uniform magnetic field $\vec{B}$. The magnitude of B is $___ \times 10^{-1}\text{T}$ (given, $g = 10 \text{ m/s}^2$) [JM April 5, 2024 (I)]

93. The magnetic field existing in a region is given by $\vec{B} = 0.2(1 + 2x)\hat{k}\text{T}$. A square loop of edge 50 cm carrying 0.5 A current is placed in x - y plane with its edges parallel to the x - y axes, as shown in figure. The magnitude of the net magnetic force experienced by the loop is ____ mN. [JM April 5, 2024 (I)]



94. A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immersed in a perpendicular magnetic field $B = B_0 \hat{k}$ as shown in figure. The magnetic force on the wire if it has a current i is:

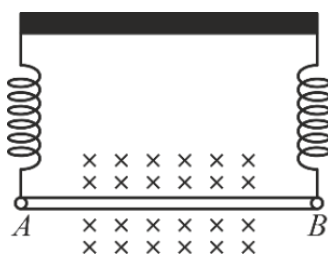
[JM Jan. 31, 2024 (I)]



- (1) $-2iBR \hat{j}$
 - (2) $2iBR \hat{j}$
 - (3) $iBR \hat{j}$
 - (4) $iBR \hat{j}$
95. The horizontal component of earth's magnetic field at a place is $3.5 \times 10^{-5} \text{ T}$. A very long straight conductor carrying current of $\sqrt{2}$ in the direction from South East to North-West is placed. The force per unit length experienced by the conductor is $___ \times 10^{-6} \text{ N/m}$. [JM Jan. 30, 2024 (I)]

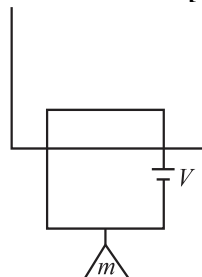
96. A straight wire AB of mass 40 g and length 50 cm is suspended by a pair of flexible leads in uniform magnetic field of magnitude 0.40 T as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is ____ A. (Take $g = 10 \text{ ms}^{-2}$).

[JM April 13, 2023 (II)]



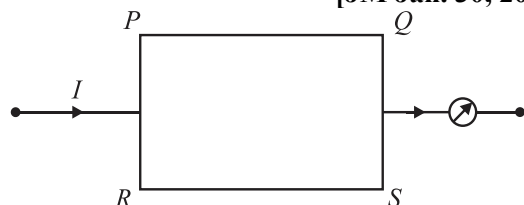
97. A massless square loop, of wire of resistance 10Ω , supporting a mass of 1 g, hangs vertically with one of its sides in a uniform magnetic field of 10^3 G , directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V , the magnetic force will exactly balance the weight of the supporting mass of 1g? (If sides of the loop = 10 cm, $g = 10 \text{ ms}^{-2}$)

[JM Jan. 30, 2023 (I)]



- (1) $1/10 \text{ V}$ (2) 100 V
(3) 1V (4) 10 V
98. A current carrying rectangular loop $PQRS$ is made of uniform wire. The length $PR = QS = S \text{ cm}$ and $PQ = RS = 100 \text{ cm}$. If ammeter current reading changes from I to $2I$, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $f_{PQ}^I : f_{PQ}^{2I}$ is:

[JM Jan. 30, 2023 (II)]



- (1) 1 : 2 (2) 1 : 4
(3) 1 : 5 (4) 1 : 3

99. Two long straight wires P and Q carrying equal current 10A each were kept parallel to each other at 5 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1 . If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be:

[JM Jan. 24, 2023 (I)]

- (1) $8 F_1$ (2) $10 F_1$
(3) $F_1/8$ (4) $F_1/10$

100. A single turn current loop in the shape of a right angle triangle with sides 5 cm, 12 cm, 13 cm is carrying a current of 2A. The loop is in a uniform magnetic field of magnitude 0.75 T whose direction is parallel to the current in the 13 cm side of the loop. The magnitude

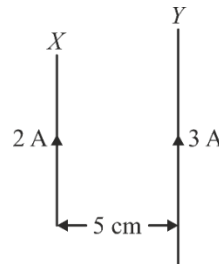
of the magnetic force on the 5 cm side will be $\frac{x}{130} \text{ N}$.

The value of x is ____.

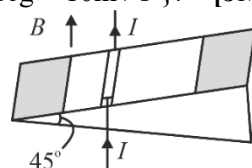
[JM Jan. 24, 2023 (II)]

101. A wire X of length 50 cm carrying a current of 2 A is placed parallel to a long wire Y of length 5 m. The wire Y carries a current of 3 A. The distance between two wires is 5 cm and currents flow in the same direction. The force acting on the wire Y is:

[JM July 29, 2022 (II)]



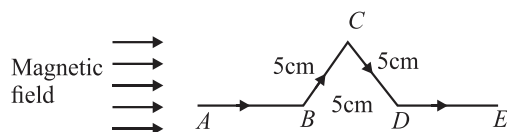
- (1) $1.2 \times 10^{-5} \text{ N}$ directed towards wire X .
(2) $1.2 \times 10^{-4} \text{ N}$ directed away from wire X .
(3) $1.2 \times 10^{-4} \text{ N}$ directed towards wire X .
(4) $2.4 \times 10^{-5} \text{ N}$ directed towards wire X .
102. As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth incline plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be: {Use $g = 10 \text{ m/s}^2$ }. [JM July 28, 2022 (I)]



- (1) 30 A (2) 15 A
(3) 10 A (4) 3 A

103. A triangular shaped wire carrying 10 A current is placed in a uniform magnetic field of 0.5 T, as shown in figure. The magnetic force on segment CD is (Given $BC = CD = BD = 5$ cm.)

[JM July 28, 2022 (II)]

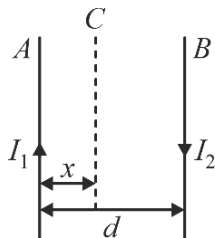


- (1) 0.126 N (2) 0.312 N
(3) 0.216 N (4) 0.245 N
104. Two parallel, long wires are kept 0.20 m apart in vacuum, each carrying current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately: [JM June 28, 2022 (II)]
- (1) 1 (2) 2.4
(3) 1.4 (4) 2

105. Two 10 cm long, straight wires, each carrying a current of 5 A are kept parallel to each other. If each wire experienced a force of 10^{-5} N, then separation between the wires is _____ cm.

[JM June 26, 2022 (II)]

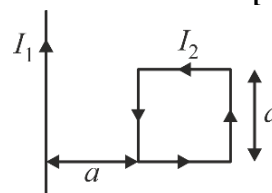
106. Two wires A and B are carrying currents I_1 and I_2 , as shown in the figure. The separation between them is d . A third wire C carrying a current I is to be kept parallel to them at a distance x from A such that the net force acting on it is zero. The possible values of x are: [10 April 2019, I]



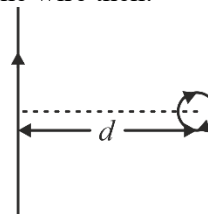
- (1) $x = \left(\frac{I_1}{I_1 - I_2} \right) d$ and $x = \left(\frac{I_2}{I_1 + I_2} \right) d$
(2) $x = \left(\frac{I_2}{I_1 + I_2} \right) d$ and $x = \left(\frac{I_2}{I_1 - I_2} \right) d$
(3) $x = \left(\frac{I_1}{I_1 + I_2} \right) d$ and $x = \left(\frac{I_2}{I_1 - I_2} \right) d$
(4) $x = \pm \frac{I_1 d}{(I_1 - I_2)}$

107. A rigid square of loop of side ' a ' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to the wire will be:

[JM 9 April 2019 I]

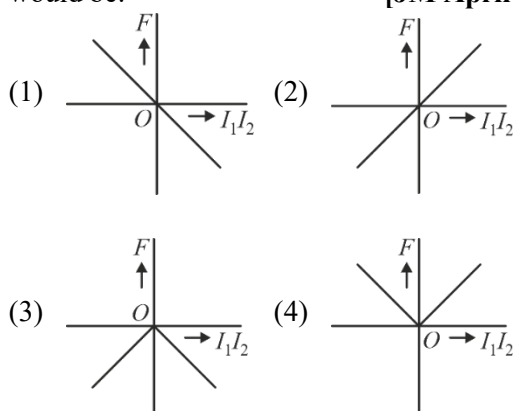


- (1) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{2\pi}$
(2) Attractive and equal to $\frac{\mu_0 I_1 I_2}{3\pi}$
(3) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{4\pi}$
(4) Zero
108. An infinitely long current carrying wire and a small current carrying loop are in the plane of the paper as shown. The radius of the loop is a and distance of its centre from the wire is d ($d \gg a$). If the loop applies a force F on the wire then: [JM 9 Jan. 2019 I]

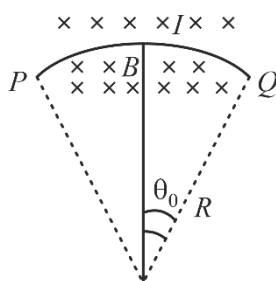


- (1) $F = 0$ (2) $F \propto \left(\frac{a}{d} \right)$
(3) $F \propto \left(\frac{a^2}{d^3} \right)$ (4) $F \propto \left(\frac{a}{d} \right)^2$
109. Two coaxial solenoids of different radius carry current I in the same direction. $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then: [JM 2015]
- (1) $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$
(2) $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$
(3) $\vec{F}_1 = \vec{F}_2 = 0$
(4) $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards

110. Two long straight parallel wires, carrying (adjustable) current I_1 and I_2 , are kept at a distance d apart. If the force ' F ' between the two wires is taken as 'positive' when the wires repel each other and 'negative' when the wires attract each other, the graph showing the dependence of ' F ', on the product $I_1 I_2$, would be: [JM April 11, 2015]

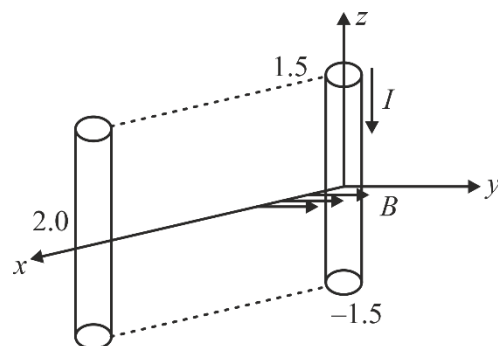


111. A wire carrying current I is tied between points P and Q and is in the shape of a circular arc of radius R due to a uniform magnetic field B (perpendicular to the plane of the paper, shown by $\times \times \times$) in the vicinity of the wire. If the wire subtends an angle $2\theta_0$ at the centre of the circle (of which it forms an arc) then the tension in the wire is: [JM April 11, 2015]



- (1) $\frac{IBR}{2\sin\theta_0}$
 (2) $\frac{IBR\theta_0}{\sin\theta_0}$
 (3) IBR
 (4) $\frac{IBR}{\sin\theta_0}$

112. A conductor lies along the z -axis at $-1.5\text{m} \leq z < 1.5\text{m}$ and carries a fixed current of 10.0 A in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x} \hat{a}_y \text{ T}$, find the power required to move the conductor at constant speed to $x = 2.0\text{ m}$, $y = 0\text{ m}$ in $5 \times 10^{-3}\text{ s}$. Assume parallel motion along the x -axis. [JM 2014]



- (1) 1.57 W (2) 2.97 W
 (3) 14.85 W (4) 29.7 W

113. Currents of a 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is 2 m . The force acting on the conductor B , which is situated at 10 cm distance from A will be [JM May 26, 2012]

- (1) $8 \times 10^{-5}\text{ N}$ (2) $5 \times 10^{-5}\text{ N}$
 (3) $8\pi \times 10^{-7}\text{ N}$ (4) $4\pi \times 10^{-7}\text{ N}$

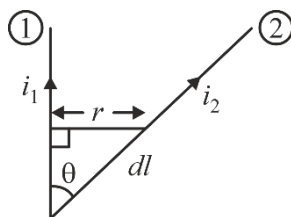
114. Two long conductors, separated by a distance d carry current I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is [JM 2004]

- (1) $-\frac{2F}{3}$ (2) $\frac{F}{3}$
 (3) $-2F$ (4) $-\frac{F}{3}$

115. If a current is passed through a spring then the spring will [JM 2002]

- (1) Expand (2) Compress
 (3) Remains same (4) None of these

116. Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1? [JM 2002]



- (1) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \tan \theta$
- (2) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \sin \theta$
- (3) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \cos \theta$
- (4) $\frac{\mu_0}{4\pi r} i_1 i_2 dl \sin \theta$

MOTION OF CHARGED PARTICLE IN A MAGNETIC FIELD

117. A current carrying solenoid is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along the axis of solenoid. If g is acceleration due to gravity then the acceleration (a) of the charged particle will satisfy:

[JM Jan 21, 2026 (I)]

- (1) $a > g$
- (2) $0 < a < g$
- (1) $a = 0$
- (4) $a = g$

118. A 5 mg particle carrying a charge of $5\pi \times 10^{-6}$ C is moving with velocity of $(3\hat{i} + 2\hat{k}) \times 10^{-2}$ m/s in a region having magnetic field $\vec{B} = 0.1\hat{k}$ Wb/m². It moves a distance of α meter along $\hat{k}$ when it completes 5 revolutions. The value of α is ____.

[JM April 08, 2026 (II)]

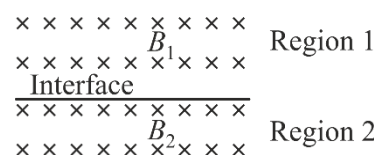
119. The charged particle moving in a uniform magnetic field of $(3\hat{i} + 2\hat{j})$ T has an acceleration $\left(4\hat{i} - \frac{x}{2}\hat{j}\right)$ m/s². The value of x is:

[JM April 5, 2026 (I)]

120. $1\mu\text{C}$ charge moving with velocity $\vec{v} = (\hat{i} - 2\hat{j} + 3\hat{k})$ m/s in the region of magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 5\hat{k})$ T. The magnitude of force acting on it is $\sqrt{\alpha} \times 10^{-6}$ N. The value of α is ____.

[JM April 02, 2026 (I)]

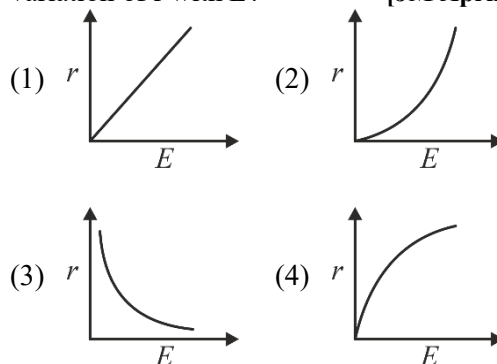
121. Uniform magnetic fields of different strengths (B_1 and B_2), both normal to the plane of the paper exist as shown in the figure. A charged particle of mass m and charge q , at the interface at an instant, moves into the region 2 with velocity v and returns to the interface. It continues to move into region 1 and finally reaches the interface. What is the displacement of the particle during this movement along the interface? [JM April 7, 2025 (I)]



(Consider the velocity of the particle to be normal to the magnetic field and $B_2 > B_1$).

- (1) $\frac{mv}{qB_1} \left(1 - \frac{B_2}{B_1}\right) \times 2$
- (2) $\frac{mv}{qB_1} \left(1 - \frac{B_1}{B_2}\right)$
- (3) $\frac{mv}{qB_1} \left(1 - \frac{B_2}{B_1}\right)$
- (4) $\frac{mv}{qB_1} \left(1 - \frac{B_1}{B_2}\right) \times 2$

122. A particle of charge q , mass m and kinetic energy E enters in magnetic field perpendicular to its velocity and undergoes a circular arc of radius (r). Which of the following curves represents the variation of r with E ? [JM April 7, 2025 (I)]





123. A particle of charge $1.6 \mu\text{C}$ and mass $16 \mu\text{g}$ is present in a strong magnetic field of 6.28 T . The particle is then fired perpendicular to magnetic field. The time required for the particle to return to original location for the first time is _____ s. ($\pi = 3.14$)
[JM April 4, 2025 (II)]
124. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: If oxygen ion (O^{2-}) and Hydrogen ion (H^+) enter normal to the magnetic field with equal momentum, then the path of O^{2-} ion has a smaller curvature than that of H^+ .
Reason R: A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.
In the light of the above statement, choose the correct answer from the options given below:
[JM April 3, 2025 (II)]
- (1) A is true but R is false.
 - (2) Both A and R are true but R is NOT the correct explanation of A.
 - (3) A is false but R is true.
 - (4) Both A and R are true and R is the correct explanation of A.
125. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion (A): A electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path.
Reason (R): The magnetic field in that region is along the direction of velocity of the electron.
In the light of the above statements, choose the correct answer from the options given below:
[JM Jan. 24, 2025 (II)]
- (1) A is true but R is false.
 - (2) Both A and R are true but R is NOT the correct explanation of A.
 - (3) A is false but R is true.
 - (4) Both A and R are true and R is the correct explanation of A.
126. A proton and a neutron ($q = +e$, $m = 2.0 u$) having same kinetic energies enter a region of uniform magnetic field $\vec{B}$, moving perpendicular to $\vec{B}$. The ratio of the radius r_d of neutron path to the radius r_n of the proton path is:
[JM April 9, 2024 (II)]
- (1) 1 : 1
 - (2) $1 : \sqrt{2}$
 - (3) $\sqrt{2} : 1$
 - (4) 1 : 2
127. An electron is projected with uniform velocity along the axis inside a current carrying long solenoid. Then:
[JM April 4, 2024 (I)]
- (1) the electron will be accelerated along the axis.
 - (2) the electron will continue to move with uniform velocity along the axis of the solenoid.
 - (3) the electron path will be circular about the axis.
 - (4) the electron will experience a force at 45° to the axis and execute a helical path.
128. An electron moves through a uniform magnetic field $\vec{B} = (B_0\hat{i} + 2B_0\hat{j})T$. At a particular instant of time, the velocity of electron is $\vec{u} = (3\hat{i} + 5\hat{j}) \text{ m/s}$. If the magnetic force acting on electron is $\vec{F} = 5e\hat{k} \text{ N}$, where e is the charge of electron, then the value of B_0 is _____ T.
[JM Jan. 31, 2024 (I)]
129. Two particles X and Y having equal charges are being accelerated through the same potential difference. There after they enter normally in a region of uniform magnetic field and describes circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is:
[JM Jan. 29, 2024 (II)]
- (1) $\left(\frac{R_1}{R_2}\right)$
 - (2) $\left(\frac{R_2}{R_1}\right)$
 - (3) $\left(\frac{R_2}{R_1}\right)^2$
 - (4) $\left(\frac{R_1}{R_2}\right)^2$
130. A charge of $4.0 \mu\text{C}$ is moving with a velocity of $4.0 \times 10^6 \text{ ms}^{-1}$ along the positive y -axis under a magnetic field $\vec{B}$ of strength $(2\hat{k})T$. The force acting on the charge is $x\hat{i}N$. The value of x is _____.
[JM Jan. 29, 2024 (II)]

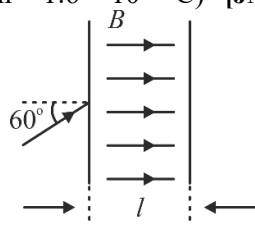


131. An electron is moving along the positive x -axis. If the uniform magnetic field is applied parallel to the negative z -axis. Then
(A) The electron will experience magnetic force along positive y -axis
(B) The electron will experience magnetic force along negative y -axis
(C) The electron will not experience any force in magnetic field
(D) The electron will continue to move along the positive x -axis
(E) The electron will move along circular path in magnetic field
Choose the correct answer from the options given below: [JM April 13, 2023 (II)]
(1) B and E only (2) A and E only
(3) C and D only (4) B and D only
132. An electron is allowed to move with constant velocity along the axis of current carrying straight solenoid.
[JM April 11, 2023 (II)]
(A) The electron will experience magnetic force along the axis of the solenoid.
(B) The electron will not experience magnetic force.
(C) The electron will continue to move along the axis of the solenoid.
(D) The electron will be accelerated along the axis of the solenoid.
(E) The electron will follow parabolic path—inside the solenoid.
Choose the correct answer from the options given below: [JM April 13, 2023 (II)]
(1) B, C and D only (2) B and C only
(3) A and D only (4) B and E only
133. A charge particle moving in magnetic field B , has the components of velocity along B as well as perpendicular to B . The path of the charge particle will be [JM April 8, 2023 (I)]
(1) helical path with the axis perpendicular to the direction of magnetic field B
(2) straight along the direction of magnetic field B
(3) helical path with the axis along magnetic field B
(4) circular path
134. A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3} T$. The angle between the direction of magnetic field and velocity of proton is 60° . The pitch of the helical path taken by the proton is _____ cm. (Take, mass of proton = $1.6 \times 10^{-27} \text{ kg}$ and charge on proton = $1.6 \times 10^{-19} \text{ C}$). [JM April 6, 2023 (II)]
135. A charge particle of $2 \mu\text{C}$ accelerated by a potential difference of 100V enters a region of uniform magnetic field of magnitude 4 mT at right angle to the direction of field. The charge particle completes semicircle of radius 3 cm inside magnetic field. The mass of the charge particle is _____ $\times 10^{-18} \text{ kg}$. [JM Feb. 1, 2023 (I)]
136. A charge particle is moving in a uniform magnetic field $(2\hat{i} + 3\hat{j}) \text{ T}$. If it has an acceleration of $(\alpha\hat{i} - 4\hat{j}) \text{ m/s}^2$, then the value of α will be: [JM July 26, 2022 (I)]
(1) 3 (2) 6
(3) 12 (4) 2
137. Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5 and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be: [JM July 25, 2022 (I)]
(1) 8 : 5 (2) 5 : 4
(3) 5 : 3 (4) 8 : 7
138. An electron with energy 0.1 keV moves at right angle to the magnetic field of $1 \times 10^{-4} \text{ Wbm}^{-2}$. The frequency of revolution of the electron will be (Take mass of electron = $9.0 \times 10^{-31} \text{ kg}$) [JM July 25, 2022 (II)]
(1) $1.6 \times 10^5 \text{ Hz}$ (2) $5.6 \times 10^5 \text{ Hz}$
(3) $2.8 \times 10^6 \text{ Hz}$ (4) $1.8 \times 10^6 \text{ Hz}$
139. A singly ionized magnesium atom ($A = 24$) ion is accelerated to kinetic energy 5 keV and is projected perpendicularly into a magnetic field B of the magnitude 0.5 T. The radius of path formed will be _____ cm. [JM June 28, 2022 (I)]
140. A deuteron and a proton moving with equal kinetic energy enter into a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x} : 1$ where x is _____. [JM June 27, 2022 (II)]



141. A proton and an alpha particle of the same velocity enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the circular paths described by the alpha particle and proton is: [JM June 26, 2022 (I)]
(1) 1 : 4 (2) 4 : 1
(3) 2 : 1 (4) 1 : 2
142. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In an uniform magnetic field, speed and energy remains the same for a moving charged particle.
Reason (R): Moving charged particle experiences magnetic force perpendicular to its direction of motion. [JM June 24, 2022 (I)]
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true.
143. A proton, a deuteron and an α -particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field.
The ratio of the radii of their respective circular paths is: [JM June 24, 2022 (II)]
(1) $1:\sqrt{2}:\sqrt{2}$ (2) $1:1:\sqrt{2}$
(3) $\sqrt{2}:1:1$ (4) $1:\sqrt{2}:1$
144. Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in a uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is: [JM July 25, 2021 (II)]
(1) 1 : 4 (2) 4 : 3
(3) 3 : 1 (4) 2 : 3
145. A deuteron and an alpha particle having equal kinetic energy enter perpendicular into a magnetic field. Let r_d and r_α be their respective radii of circular path. The value of $\frac{r_d}{r_\alpha}$ is equal to: [JM July 20, 2021 (I)]
(1) $\frac{1}{\sqrt{2}}$ (2) $\sqrt{2}$
(3) 1 (4) 2
146. A proton and an α -particle, having kinetic energies K_p and K_α respectively, enter into a magnetic field at right angles. The ratio of the radii of trajectory of proton to that of α -particle is 2 : 1. The ratio of $K_p : K_\alpha$ is: [JM March 18, 2021 (II)]
(1) 1 : 8 (2) 8 : 1
(3) 1 : 4 (4) 4 : 1
147. A charge Q is moving $\overline{dl}$ distance in the magnetic field $\vec{B}$. Find the value of work done by $\vec{B}$. [JM March 16, 2021 (II)]
(1) -1 (2) Zero
(3) 1 (4) Infinte
148. A particle of charge q and mass m is moving with a velocity $-v\hat{i}$ ($v \neq 0$) towards a large screen placed in the Y-Z plane at a distance d . If there is a magnetic field $\vec{B} = B_0\hat{k}$, the minimum value of v for which the particle will not hit the screen is: [JM Sep. 06, 2020 (I)]
(1) $\frac{qdB_0}{3m}$ (2) $\frac{2qdB_0}{m}$
(3) $\frac{qdB_0}{m}$ (4) $\frac{qdB_0}{2m}$
149. A charged particle carrying $1 \mu\text{C}$ is moving with velocity $(2\hat{i} + 3\hat{j} + 4\hat{k}) \text{ ms}^{-1}$. If an external magnetic field of $(5\hat{i} + 3\hat{j} - 6\hat{k}) \times 10^{-3} \text{ T}$ exists in the region where the particle is moving then the force on the particle is $\vec{F} \times 10^{-9} \text{ N}$. The vector $\vec{F}$ is: [JM Sep. 03, 2020 (I)]
(1) $-0.30\hat{i} + 0.32\hat{j} - 0.09\hat{k}$
(2) $-30\hat{i} + 32\hat{j} - 9\hat{k}$
(3) $-300\hat{i} + 320\hat{j} - 90\hat{k}$
(4) $-3.0\hat{i} + 3.2\hat{j} - 0.9\hat{k}$
150. A beam of protons with speed $4 \times 10^5 \text{ ms}^{-1}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to:
(Mass of the proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.69 \times 10^{-19} \text{ C}$) [JM Sep. 02, 2020 (I)]
(1) 2 cm
(2) 5 cm
(3) 12 cm
(4) 4 cm

151. The figure shows a region of length ' l ' with a uniform magnetic field of 0.3 T in it and a proton entering the region with velocity $4 \times 10^5 \text{ ms}^{-1}$ making an angle 60° with the field. If the proton completes 10 revolution by the time it cross is the region shown, ' l ' is close to (mass of proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.6 \times 10^{-19} \text{ C}$) [JM Sep. 02, 2020 (II)]



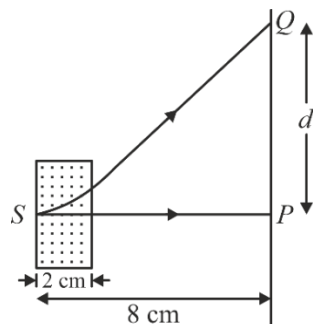
- (1) 0.11 m (2) 0.88 m
(3) 0.44 m (4) 0.22 m
152. Proton with kinetic energy of 1 MeV moves from south to north. It gets an acceleration of 10^{12} m/s^2 by an applied magnetic field (west to east). The value of magnetic field:

(Rest mass of proton is $1.6 \times 10^{-27} \text{ kg}$).

[JM 8 Jan 2020 (I), Online 23 April 2013]

- (1) 0.71 mT (2) 7.1 mT
(3) 0.071 mT (4) 71 mT
153. An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3} \text{ T})\hat{k}$ at S (see figure). The field extends between $x = 0$ and $x = 2 \text{ cm}$. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is: (Electron's charge = $1.6 \times 10^{-19} \text{ C}$, Mass of electron = $9.1 \times 10^{-31} \text{ kg}$).

[JM 12 April 2019 (II)]



- (1) 11.65 cm (2) 12.87 cm
(3) 1.22 cm (4) 2.25 cm

154. A proton, an electron, and a Helium nucleus, have the same energy. They are in the circular orbits in a plane due to magnetic field perpendicular to the plane. Let r_p , r_e and r_{He} be their respective radii, then,

[JM 10 April 2019 (I), 2018, 2012]

- (1) $r_e > r_p = r_{\text{He}}$ (2) $r_e < r_p = r_{\text{He}}$
(3) $r_e < r_p < r_{\text{He}}$ (4) $r_e > r_p > r_{\text{He}}$

155. A proton and an α -particle (with their masses in the ratio of 1 : 4 and charges in the ratio 1 : 2) are accelerated from rest through a potential difference V. If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii $r_p : r_\alpha$ of the circular paths described by them will be:

[JM 12 Jan 2019, (I)]

- (1) $1 : \sqrt{2}$ (2) $1 : 2$
(3) $1 : 3$ (4) $1 : \sqrt{3}$

156. In an experiment, electrons are accelerated, from rest, by applying a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied. [Charge of the electron = $1.6 \times 10^{-19} \text{ C}$, Mass of the electron = $9.1 \times 10^{-31} \text{ kg}$]

[JM 11 Jan 2019, (I)]

- (1) $7.5 \times 10^{-3} \text{ m}$ (2) $7.5 \times 10^{-2} \text{ m}$
(3) 7.5 m (4) $7.5 \times 10^{-4} \text{ m}$

157. The region between $y = 0$ and $y = d$ contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$. If $d = \frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is:

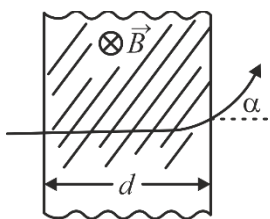
[JM 11 Jan 2019, (II)]

- (1) $\frac{qvB}{m} \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j} \right)$
(2) $\frac{qvB}{m} \left(\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j} \right)$
(3) $\frac{qvB}{m} \left(\frac{-\hat{j} + \hat{i}}{\sqrt{2}} \right)$
(4) $\frac{qvB}{m} \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$

158. A negative test charge is moving near a long straight wire carrying a current. The force acting on the test charge is parallel to the direction of the current. The motion of the charge is: [JM Online April 9, 2017]

- (1) away from the wire
- (2) towards the wire
- (3) parallel to the wire along the current
- (4) parallel to the wire opposite to the current

159. A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B . The field occupies a region of space by width ' d '. If α be the angle of deviation of proton from initial direction of motion (see figure), the value of $\sin \alpha$ will be: [JM Online April 10, 2015]



- (1) $qV\sqrt{\frac{Bd}{2m}}$
- (2) $\frac{B}{2}\sqrt{\frac{qd}{mV}}$
- (3) $\frac{B}{d}\sqrt{\frac{q}{2mV}}$
- (4) $Bd\sqrt{\frac{q}{2mV}}$

160. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: A charged particle is moving at right angle to a static magnetic field. During the motion the kinetic energy of the charge remains unchanged.

Statement 2: Static magnetic field exert force on a moving charge in the direction perpendicular to the magnetic field. [JM Online May 26, 2012]

- (1) Statement 1 is false, Statement 2 is true.
- (2) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
- (3) Statement 1 is true, Statement 2 is false.
- (4) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

161. A charged particle moves through a magnetic field perpendicular to its direction. Then [JM 2007]

- (1) kinetic energy changes but the momentum is constant
- (2) the momentum changes but the kinetic energy is constant
- (3) both momentum and kinetic energy of the particle are not constant
- (4) both momentum and kinetic energy of the particle are constant

162. A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B . The time taken by the particle to complete one revolution is: [JM 2005]

- (1) $\frac{2\pi q^2 B}{m}$
- (2) $\frac{2\pi m q}{B}$
- (3) $\frac{2\pi m}{qB}$
- (4) $\frac{2\pi q B}{m}$

163. A particle of mass M and charge Q moving with velocity describe a circular path of radius R when subjected to a uniform transverse magnetic field of induction B . The work done by the field when the particle completes one full circle is: [JM 2003]

- (1) $\left(\frac{Mv^2}{R}\right)2\pi R$
- (2) Zero
- (3) $BQ2\pi R$
- (4) $BQv2\pi R$

164. If an electron and a proton having same momenta enter perpendicular to a magnetic field, then [JM 2002]

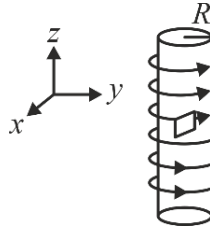
- (1) curved path of electron and proton will be same (ignoring the sense of revolution)
- (2) they will move undeflected
- (3) curved path of electron is more curved than that of the proton
- (4) path of proton is more curved.

165. The time period of a charged particle undergoing a circular motion in a uniform magnetic field is independent of its [JM 2002]

- (1) speed
- (2) mass
- (3) charge
- (4) magnetic induction

166. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I . The electron gun shoots an electron along the radius of the solenoid with speed v . If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning):

[JM 9 Jan 2020, (II)]



- (1) $\frac{e\mu_0 nIR}{m}$
- (2) $\frac{e\mu_0 nIR}{2m}$
- (3) $\frac{e\mu_0 nIR}{4m}$
- (4) $\frac{2e\mu_0 nIR}{m}$

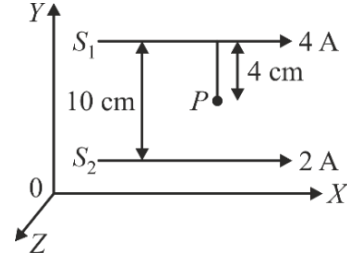
167. Consider a long thin conducting wire carrying a uniform current I . A particle having mass " m " and charge " q " is released at a distance " a " from the wire with a speed V_0 along the direction of current in the wire.

The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability]

[JM Jan. 28, 2025 (I)]

- (1) $a \left[1 - \frac{mv_0}{2q\mu_0 I} \right]$
- (2) $\frac{a}{2}$
- (3) $a \left[1 - \frac{mv_0}{q\mu_0 I} \right]$
- (4) $ae^{\frac{-4\pi mv_0}{q\mu_0 I}}$

168. Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of 4A and 2A respectively. The conductors are placed along x -axis in X - Y plane. There is a point P located between the conductors (as shown in figure).



A charge particle of 3π coulomb is passing through the point P with velocity.

$\vec{v} = (2\hat{i} + 3\hat{j})$ m/s; $\hat{i}$ and $\hat{j}$ represent unit vector along x and y axis respectively. The force acting on the charge particle is $4\pi \times 10^{-5} (-x\hat{i} + 2\hat{j})$ N. The value of x is:

[JM June 27, 2022 (11)]

- (1) 2
- (2) 1
- (3) 3
- (4) -3

MOTION IN COMBINED ELECTRIC AND MAGNETIC FIELD

169. A particle of charge q and mass m is projected from origin with an initial velocity $\vec{v} = \left(\frac{v_0}{\sqrt{2}} \hat{x} + \frac{v_0}{\sqrt{2}} \hat{y} \right)$.

There exists a uniform magnetic field $\vec{B} = B_0 \hat{z}$ and a space varying electric field $\vec{E} = E_0 e^{-\lambda x} \hat{x}$ within the region $0 \leq x \leq L$. After travelling a distance such that x -coordinate has changed from $x = 0$ to $x = L$, the change in the kinetic energy is _____.

[JM April 05, 2026 (II)]

- (1) $\frac{qE_0}{\lambda} [1 - e^{-\lambda L}]$
- (2) $\left(\frac{v_0 q B_0}{2\lambda} \right) [2 - e^{-2\lambda L}]$
- (3) $\frac{qE_0}{\lambda} [1 + e^{-\lambda L}]$
- (4) $q \left(\frac{E_0 + v_0 B_0}{\lambda} \right) [1 - e^{-\lambda L/2}]$



170. A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of $2 \times 10^5 \text{ ms}^{-1}$. When the electric field is switched off, the proton moves along a circular path of radius 2 cm. The magnitude of electric field is $x \times 10^4 \text{ N/C}$. The value of x is _____.
(Take the mass of the proton = $1.6 \times 10^{-27} \text{ kg}$.)
[JM Jan. 22, 2025 (II)]
171. An electron with kinetic energy 5 eV enters a region of uniform magnetic field of $3 \mu\text{T}$ perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E , so that electron moves along the same path, is _____ NC^{-1} .
(Given, mass of electron = $9 \times 10^{-31} \text{ kg}$, electric charge = $1.6 \times 10^{-19} \text{ C}$)
[JM April 8, 2024 (I)]
172. The electrostatic force ($\vec{F}_1$) and magnetic force ($\vec{F}_2$) acting on a charge q moving with velocity $\vec{v}$ can be written: [JM April 5, 2024 (II)]
(1) $\vec{F}_1 = q\vec{v} \cdot \vec{E}$, $\vec{F}_2 = q(\vec{B} \cdot \vec{v})$
(2) $\vec{F}_1 = q\vec{B}$, $\vec{F}_2 = q(\vec{B} \times \vec{v})$
(3) $\vec{F}_1 = q\vec{E}$, $\vec{F}_2 = q(\vec{v} \times \vec{B})$
(4) $\vec{F}_1 = q\vec{E}$, $\vec{F}_2 = q(\vec{B} \times \vec{v})$
173. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then the region of space may have:
(A) $E = 0$, $B = 0$ (B) $E = 0$, $B \neq 0$
(C) $E \neq 0$, $B = 0$ (D) $E \neq 0$, $B \neq 0$
Choose the most appropriate answer from the options given below: [JM Jan. 27, 2024 (I)]
(1) (A), (B) and (C) only
(2) (A), (C) and (D) only
(3) (A), (B) and (D) only
(4) (B), (C) and (D) only
174. A velocity selector consists of electric field $\vec{E} = E\hat{k}$ and magnetic field $\vec{B} = B\hat{j}$ with $B = 12 \text{ mT}$. The value E required for an electron of energy 728 eV moving along the positive x -axis to pass undeflected is: (Given, mass of electron = $9.1 \times 10^{-31} \text{ kg}$)
[JM July 26, 2022 (II)]
(1) 192 kV m^{-1} (2) 192 mV m^{-1}
(3) 9600 kV m^{-1} (4) 16 kV m^{-1}
175. Given below are two statements:
Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy; whereas the magnetic force does not change the kinetic energy of the charged particle.
Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particle along the direction of magnetic field. In the light of the above statements, choose the most appropriate answer from the options given below: [JM June 29, 2022 (II)]
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.
176. An electron is moving along $+x$ direction with a velocity of $6 \times 10^6 \text{ ms}^{-1}$. It enters a region of uniform electric field of 300 V/cm pointing along $+y$ direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be: [JM Sep. 06, 2020 (I)]
(1) $3 \times 10^{-4} \text{ T}$, along $+z$ direction
(2) $5 \times 10^{-3} \text{ T}$, along $-z$ direction
(3) $5 \times 10^{-3} \text{ T}$, along $+z$ direction
(4) $3 \times 10^{-4} \text{ T}$, along $-z$ direction
177. A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5T. If an electric field of 100 V/m makes it to move in a straight path then the mass of the particle is (Given charge of electron = $1.6 \times 10^{-19} \text{ C}$) [JM 12 April 2019 (I)]
(1) $9.1 \times 10^{-31} \text{ kg}$
(2) $1.6 \times 10^{-27} \text{ kg}$
(3) $1.6 \times 10^{-19} \text{ kg}$
(4) $2.0 \times 10^{-24} \text{ kg}$
178. The velocity of certain ions that pass undeflected through crossed electric field $E = 7.7 \text{ k V/m}$ and magnetic field $B = 0.14 \text{ T}$ is [JM Online May 7, 2012]
(1) 18 km/s (2) 77 km/s
(3) 55 km/s (4) 1078 km/s



179. An electric charge $+q$ moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$ in an electromagnetic field given $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$.

The y-component of the force experienced by $+q$ is:

[JM 2011 RS]

- (1) $11q$ (2) $5q$
(3) $3q$ (4) $2q$

180. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal field $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then [JM 2007]

- (1) $\vec{v} = \vec{B} \times \vec{E} / E^2$ (2) $\vec{v} = \vec{E} \times \vec{B} / B^2$
(3) $\vec{v} = \vec{B} \times \vec{E} / B^2$ (4) $\vec{v} = \vec{E} \times \vec{B} / E^2$

181. In a region, steady and Uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a

[JM 2006]

- (1) helix (2) straight line
(3) ellipse (4) circle

182. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected along the direction of the fields with a certain velocity then [JM 2005]

- (1) its velocity will increase
(2) its velocity will decrease
(3) it will turn towards left of direction of motion
(4) it will turn towards right of direction of motion

CURRENT LOOP AS A MAGNETIC DIPOLE

183. A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{\hat{k} + \hat{i}}{\sqrt{2}}$ is placed in a magnetic field,

$\vec{B} = B_0(3\hat{i} + 2\hat{k})$. If $B_0 = 4 \times 10^{-3}$ T and current

$I = 100\sqrt{2}$ A, the torque experienced by the loop is _____ Wb.A. ($\pi = 3.14$). [JM April 8, 2026 (II)]

- (1) $16 \times 10^{-5} \hat{k}$
(2) $5024 \times 10^{-7} \hat{k}$
(3) $5024 \times 10^{-7} \hat{i}$
(4) $5024 \times 10^{-7} \hat{j}$

184. An insulated wire is wound so that it forms a flat coil with $N = 200$ turns. The radius of the innermost turn is $r_1 = 3$ cm, and of the outermost turn $r_2 = 6$ cm. If 20 mA current flows in it then the magnetic moment will be $\alpha \times 10^{-2}$ A.m². The value of α is ____.

[JM April 04, 2026 (I)]

- (1) 4.4 (2) 2.64
(3) 3.25 (4) 1.2

185. A particle having charge 10^{-9} C moving in x - y plane in fields of $0.4\hat{j}$ N/C and $4 \times 10^{-3}\hat{k}$ T experiences a force of $(4\hat{i} + 2\hat{j}) \times 10^{-10}$ N. The velocity of the particle at that instant is _____ m/s.

[JM April 2, 2026 (II)]

- (1) $50\hat{i} + 100\hat{j}$ (2) $100\hat{i} + 50\hat{j}$
(3) $-50\hat{i} + 100\hat{j}$ (4) $50\hat{i} - 100\hat{j}$

186. A circular coil of radius 2 cm and 125 turns carries a current of 1 A. The coil is placed in a uniform magnetic field of magnitude 0.4 T. The axis of the coil makes an angle of 30° with the direction of the magnetic field. The torque acting on the coil is $\alpha \times 10^{-4}$ N.m. The value of α is ____ ($\pi = 3.14$)

[JM April 4, 2026 (II)]

187. The magnetic moments associated with two closely wound circular coils A and B of radius $r_A = 10$ cm and 20 cm respectively are equal if: (Where N_A, I_A and N_B, I_B are number of turn and current of A and B respectively) [JM Jan. 30, 2023 (I)]

- (1) $2 N_A I_A = N_B I_B$ (2) $N_A = 2 N_B$
(3) $N_A I_A = 4 N_B I_B$ (4) $4 N_A I_A = N_B I_B$

188. A wire of length 314 cm carrying current of 14 A is bent to form a circle. The magnetic moment of the coil is A.m². [Given $\pi = 3.14$] [JM July 29, 2022 (I)]

189. A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle:

[JM Sep. 06, 2020 (II)]

- (1) $\frac{mv^2 \vec{B}}{2B^2}$ (2) $\frac{mv^2 \vec{B}}{2\pi B^2}$
(3) $-\frac{mv^2 \vec{B}}{B^2}$ (4) $-\frac{mv^2 \vec{B}}{2B^2}$

190. A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m . If this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be:

[JM 10 April 2019, II]

- (1) $\frac{m}{\pi}$ (2) $\frac{3m}{\pi}$
(3) $\frac{2m}{\pi}$ (4) $\frac{4m}{\pi}$

191. The dipole moment of a circular loop carrying a current I , is m and the magnetic field at the centre of the loop is B_1 . When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of the loop is B_2 . The ratio B_1 / B_2 is:

[JM 2018]

- (1) 2 (2) $\sqrt{3}$
(3) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

192. A magnetic dipole experiences a torque of $80\sqrt{3}$ Nm when placed in uniform magnetic field in such a way that dipole moment makes angle of 60° with magnetic field. The potential energy of the dipole is:

[JM April 3, 2025 (II)]

- (1) 80 J
(2) $-40\sqrt{3}$ J
(3) -60 J
(4) -80 J

193. A uniform magnetic field of 2×10^{-3} T acts along positive Y -direction. A rectangular loop of sides 20 cm and 10 cm with current of 5 A is Y - Z plane. The current is in anticlockwise sense with reference to negative X axis. Magnitude and direction of the torque is:

[JM Jan. 31, 2024 (II)]

- (1) 2×10^{-4} N - m along positive Z -direction
(2) 2×10^{-4} N - m along negative Z -direction
(3) 2×10^{-4} N - m along positive X -direction
(4) 2×10^{-4} N - m positive Y -direction

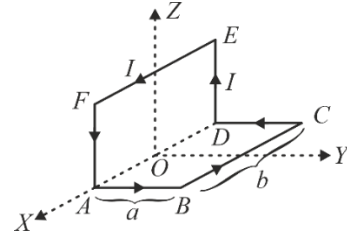
194. A square loop of side $2a$ and carrying current I is kept in xz plane with its centre at origin. A long wire carrying the same current I is placed parallel to z -axis and passing through point $(0, b, 0)$, ($b \gg a$). The magnitude of torque on the loop about z -axis will be:

[JM Sep. 06, 2020 (II)]

- (1) $\frac{2\mu_0 I^2 a^2}{\pi b}$ (2) $\frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$
(3) $\frac{\mu_0 I^2 a^2 b}{2\pi(a^2 + b^2)}$ (4) $\frac{\mu_0 I^2 a^2}{2\pi b}$

195. A wire carrying current I is bent in the shape $ABCDEF$ as shown, where rectangle $ABCD$ and $ADEFA$ are perpendicular to each other. If the sides of the rectangles are of lengths a and b , then the magnitude and direction of magnetic moment of the loop $ABCDEF$ is:

[JM Sep. 02, 2020 (II)]



- (1) abI , along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
(2) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
(3) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$
(4) abI , along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$

196. A small circular loop of conducting wire has radius a and carries current I . It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T . If the mass of the loop is m then:

[9 Jan 2020, II]

- (1) $T = \sqrt{\frac{2m}{IB}}$ (2) $T = \sqrt{\frac{\pi m}{2IB}}$
(3) $T = \sqrt{\frac{2\pi m}{IB}}$ (4) $T = \sqrt{\frac{\pi m}{IB}}$

197. A rectangular coil (Dimension $5 \text{ cm} \times 2.5 \text{ cm}$) with 100 turns, carrying a current of 3 A in the clock-wise direction, is kept centered at the origin and in the X - Z plane. A magnetic field of 1 T is applied along X -axis. If the coil is tilted through 45° about Z -axis, then the torque on the coil is: [JM 9 April 2019 I]

- (1) 0.38 Nm (2) 0.55 Nm
(3) 0.42 Nm (4) 0.27 Nm

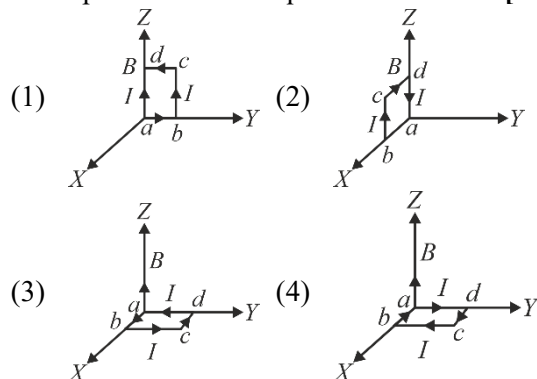
198. A circular coil having N turns and radius r carries a current I . It is held in the XZ plane in a magnetic field B . The torque on the coil due to the magnetic field is: [JM 8 April 2019]

- (1) $\frac{Br^2 I}{\pi N}$ (2) $B\pi r^2 IN$
(3) $\frac{Br^2 I}{N}$ (4) Zero

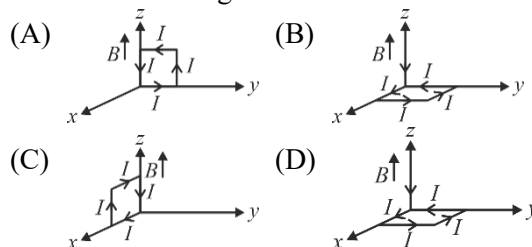
199. A charge q is spread uniformly over an insulated loop of radius r . If it is rotated with an angular velocity and with respect to normal axis then the magnetic moment of the loop is: [JM April 16, 2018]

- (1) $\frac{1}{2} q\omega r^2$ (2) $\frac{4}{3} q\omega r^2$
(3) $\frac{3}{2} q\omega r^2$ (4) $q\omega r^2$

200. A uniform magnetic field B of 0.3 T is along the positive Z -direction. A rectangular loop (abcd) of sides $10 \text{ cm} \times 5 \text{ cm}$ carries a current I of 12 A. Out of the following different orientations which one corresponds to stable equilibrium? [JM 2017]



201. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below:



If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium? [JM 2015]

- (1) (B) and (D), respectively
(2) (B) and (C), respectively
(3) (A) and (B), respectively
(4) (A) and (C), respectively

THE MOVING COIL GALVANOMETER - ITS CONVERSION INTO AMMETER AND VOLTMETER

202. A moving coil of galvanometer when shunted with 2Ω resistance gives a full scale deflection for a current of 500 mA. When a resistance of 470Ω is connected in series it gives a full scale deflection for 10 V potential applied on it. The value of resistance of galvanometer coil is $_____\Omega$.

[JM April 6, 2026 (II)]

203. In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars:

$$R_1 = 5\Omega, N_1 = 15, A_1 = 3.6 \times 10^{-3} \text{ m}^2, B_1 = 0.25 \text{ T}$$

$$R_2 = 7\Omega, N_2 = 21, A_2 = 1.8 \times 10^{-3} \text{ m}^2, B_2 = 0.50 \text{ T}$$

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ? [JM April 2, 2025 (II)]

- (1) 1:1 (2) 1:4
(3) 1:3 (4) 1:2

204. A galvanometer having a coil of resistance 30Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}\Omega$ where X is

[JM Jan. 23, 2025 (II)]

- (1) 447 (2) 298
(3) 149 (4) 596



205. Consider a moving coil galvanometer (MCG):

[JM Jan. 23, 2025 (I)]

- A: The torsional constant in moving coil galvanometer has dimensions $[ML^2T^{-2}]$
B: Increasing the current sensitivity may not necessarily increase the voltage sensitivity.
C: If we increase number of turns (N) to its double ($2N$), then the voltage sensitivity doubles.
D: MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.
E: Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from options given below:

- (1) A, B only (2) A, D, only
(3) B, D, E only (4) A, B, E only

206. A galvanometer has a coil of resistance 200Ω with a full-scale deflection at $20\mu A$. The value of resistance to be added to use it as an ammeter of range (0-20) mA is:

[JM April 9, 2024 (I)]

- (1) 0.40Ω (2) 0.20Ω
(3) 0.50Ω (4) 0.10Ω

207. A galvanometer of resistance 100Ω when connected in series with 400Ω measures a voltage of upto 10 V. The value of resistance required to convert the galvanometer into ammeter to read upto 10 A is $x \times 10^{-2}\Omega$. The value of x is:

[JM April 5, 2024 (II)]

- (1) 2 (2) 800
(3) 20 (4) 200

208. A galvanometer has a resistance of 50Ω and it allows maximum current of 5 mA. It can be converted into voltmeter to measure upto 100V by connected in series a resistor of resistance:

[JM Feb. 1, 2024 (I)]

- (1) 5975Ω (2) 20050Ω
(3) 19950Ω (4) 19500Ω

209. In an ammeter, 5% of the main current passes through the galvanometer. If resistance of the galvanometer is G , the resistance of ammeter will be:

[JM Feb. 1, 2024 (II)]

- (1) $199G$ (2) $200G$
(3) $G/200$ (4) $G/19$

210. A moving coil galvanometer has 100 turns and each turn has an area of 2.0 cm^2 . The magnetic field produced by the magnet is 0.01 T and the deflection in the coil is 0.05 radian when a current of 10 mA is passed through it. The torsional constant of the suspension wire is $x \times 10^{-5}\text{ N-m/rad}$. The value of x is _____.
[JM Feb. 1, 2024 (II)]

211. A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of 3mA. For it to measure a current of 8A, the value of the shunt should be:

[JM Jan. 29, 2024 (I)]

- (1) $4.85 \times 10^{-3}\Omega$ (2) $3 \times 10^{-3}\Omega$
(3) $3.75 \times 10^{-3}\Omega$ (4) $2.75 \times 10^{-3}\Omega$

212. The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 242 is applied. The resistance of galvanometer coil will be:

[JM Jan. 29, 2024 (I)]

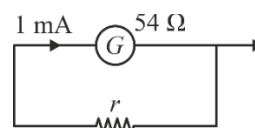
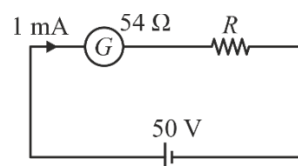
- (1) 100Ω (2) 48Ω
(3) 12Ω (4) 96Ω

213. A current of $200\mu A$ deflects the coil of a moving coil galvanometer through 60° . The current to cause deflection through $\pi/10$ radian is:

[JM Jan. 27, 2024 (II)]

- (1) $30\mu A$ (2) $120\mu A$
(3) $60\mu A$ (4) $180\mu A$

214. For designing a voltmeter of range 50 V and an ammeter of range 10 mA using a galvanometer which has a coil of resistance 54Ω showing a full scale deflection for 1 mA as in figure.



- (A) for voltmeter $R \approx 50\text{ k}\Omega$
(B) for ammeter $r = 0.2\Omega$
(C) for ammeter $r = 6\Omega$
(D) for voltmeter $R = 5\text{ k}\Omega$
(E) for voltmeter $R \approx 500\text{ k}\Omega$

Choose the correct answer from the options given below:
[JM April 15, 2023 (I)]

- (1) (C) and (E) (2) (C) and (D)
(3) (A) and (C) (4) (A) and (B)



215. When a resistance of $5\ \Omega$ is shunted with a moving coil galvanometer, it shows a full scale deflection for a current of 250 mA, however when $1050\ \Omega$ resistance is connected with it in series, it gives full scale deflection for 25 volt. The resistance of galvanometer is ____ Ω . [JM April 13, 2023 (I)]
216. The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be:
[JM April 11, 2023 (I)]
- (1) +25% (2) -50%
(3) Zero (4) -25%
217. Given below are two statement:
Statement I: If the number of turns in the coil of a moving coil galvanometer is doubled then the current sensitivity becomes double.
Statement II: Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio:
In the light of the above statement, choose the correct answer from the options given below:
[JM April 10, 2023 (I)]
- (1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Both Statement I and Statement II are false
(4) Statement I is true but Statement II is false
218. Certain galvanometers have a fixed core made of non-magnetic metallic material. The function of this metallic material is [JM April 8, 2023 (I)]
- (1) To oscillate the coil in magnetic field for longer period of time
(2) To bring the coil to rest quickly
(3) To produce large deflecting torque on the coil
(4) To make the magnetic field radial
219. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: For measuring the potential difference across a resistance of $600\ \Omega$, the voltmeter with resistance $1000\ \Omega$ will be preferred over voltmeter with resistance $4000\ \Omega$.
Reason R: Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.
In the light of the above statements, choose the most appropriate answer from the options given below:
[JM Feb. 1, 2023 (II)]
- (1) A is not correct but R is correct
(2) Both A and R are correct and R is the correct explanation of A
(3) Both A and R are correct but R is not the correct explanation of A
(4) A is correct R is not correct
220. The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be:
[JM Jan. 31, 2023 (II)]
- (1) 100% (2) 50%
(3) 75% (4) 0%
221. For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10 mA is passed through it. If the torsional constant of suspension wire is $4.0 \times 10^{-5}\ \text{Nm rad}^{-1}$, the magnetic field is 0.01 T and the number of turns in the coil is 200 the area of each turn (in cm^2) is: [JM Jan. 25, 2023 (II)]
- (1) 2.0 (2) 1.0
(3) 1.5 (4) 0.5
222. The current sensitivity of a galvanometer can be increased by:
(A) decreasing the number of turns
(B) increasing the magnetic field
(C) decreasing the area of the coil
(D) decreasing the torsional constant of the spring
Choose the most appropriate answer from the options given below: [JM July 28, 2022 (I)]
- (1) (B) and (C) only
(2) (C) and (D) only
(3) (A) and (C) only
(4) (B) and (D) only

223. A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is:

[JM June 27, 2022 (I)]

- (1) 0.1% (2) 10%
(3) 25% (4) 0.25%

224. If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S . Then the total current I when its figure of merit is K will be:

[JM June 25, 2022 (II)]

- (1) $\frac{KS}{(S+G)}$ (2) $\frac{(G+S)}{nKS}$
(3) $\frac{nKS}{(G+S)}$ (4) $\frac{nK(G+S)}{S}$

225. Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer?

[JM Aug. 31, 2021 (I)]

- (1) 300Ω (2) 344Ω
(3) 245Ω (4) 226Ω

226. For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be:

[JM Aug. 27, 2021 (II)]

- (1) 1Ω (2) 5Ω
(3) 4Ω (4) 2Ω

227. A galvanometer of resistance G is converted into a volt-meter of range 0 - 1V by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to 0 - 2V will be:

[JM Sep. 05, 2025 (I)]

- (1) G (2) R_1
(3) $R_1 - G$ (4) $R_1 + G$

228. A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6 mA it produces a deflection of 2° , its figure of merit is close to:

[JM Sep. 05, 2020 (II)]

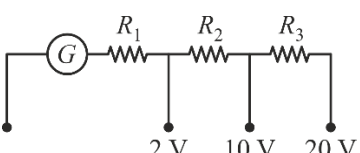
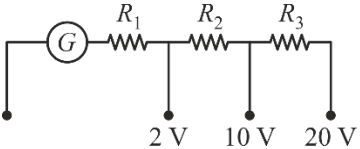
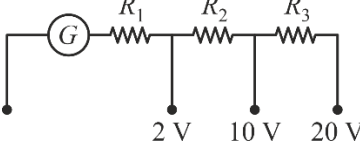
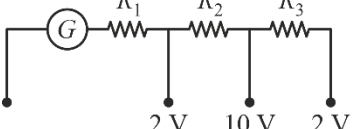
- (1) 333° A/div.
(2) $6 \times 10^{-3} \text{ A/div.}$
(3) 666° A/div.
(4) $3 \times 10^{-3} \text{ A/div.}$

229. A galvanometer coil has 500 turns and each turn has an average area of $3 \times 10^{-4} \text{ m}^2$. If a torque of 1.5 Nm is required to keep this coil parallel to a magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is _____.

[JM Sep. 03, 2020 (II)]

230. A galvanometer of resistance 100Ω has 50 divisions on its scale and has sensitivity of $20 \mu\text{A/division}$. It is to be converted to a voltmeter with three ranges, of 0 - 2V, 0 - 10V and 0 - 20V. The appropriate circuit to do so is

[JM 12 April 2019, (I)]

- (1)  $R_1 = 2000\Omega$
 $R_2 = 8000\Omega$
 $R_3 = 10000\Omega$
- (2)  $R_1 = 1900\Omega$
 $R_2 = 9900\Omega$
 $R_3 = 19900\Omega$
- (3)  $R_1 = 1900\Omega$
 $R_2 = 8000\Omega$
 $R_3 = 10000\Omega$
- (4)  $R_1 = 19900\Omega$
 $R_2 = 9900\Omega$
 $R_3 = 1900\Omega$

- 231.** A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into
- an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and
 - into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then,

[JM 12 April 2019, (II)]

$$(1) \quad R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g} \right) \text{ and } \frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$$

$$(2) \quad R_A R_V = G^2 \text{ and } \frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$$

$$(3) \quad R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g} \right) \text{ and } \frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$$

$$(4) \quad R_A R_V = G^2 \text{ and } \frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$$

- 232.** A moving coil galvanometer allows a full scale current of 10^{-4} A. A series resistance of $2 \text{ M}\Omega$ is required to convert the above galvanometer into a voltmeter of range 0 - 5V.

Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0-10 mA is:

[JM 10 April 2019 (I)]

- 500 Ω
- 100 Ω
- 200 Ω
- 10 Ω

- 233.** A moving coil galvanometer has resistance 50Ω and it indicates full deflection at 4 mA current. A voltmeter is made using this galvanometer and a $5 \text{ k}\Omega$ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to:

[JM 9 April 2019 (I)]

- 40 V
- 15 V
- 20 V
- 10 V

- 234.** A moving coil galvanometer has a coil with 175 turns and area 1 cm^2 . It uses a torsion band of torsion constant 10^{-6} N-m/rad . The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1 mA. The value of B (in tesla) is approximately:

[JM 9 April 2019, (II)]

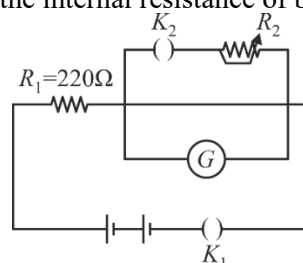
- 10^{-4}
- 10^{-2}
- 10^{-1}
- 10^{-3}

- 235.** The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A . What resistance must be connected to it in order to convert it into an ammeter of range 0 - 0.5 A ?

[JM 9 April 2019, (II)]

- 0.5 ohm
- 0.002 ohm
- 0.02 ohm
- 0.2 ohm

- 236.** The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω , the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [JM 12 Jan 2019, (I)] [Neglect the internal resistance of battery]:



- 5 Ω
- 22 Ω
- 25 Ω
- 12 Ω

- 237.** A galvanometer, resistance is 50 ohm , has 25 divisions in it. When a current of $4 \times 10^{-4} \text{ A}$ passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5 V , it should be connected to a resistance of:

[JM 12 Jan 2019, (II)]

- 250 ohm
- 200 ohm
- 6200 ohm
- 6250 ohm

- 238.** A galvanometer having a resistance of 20Ω and 30 division on both sides has figure of merit $0.005 \text{ ampere/division}$. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt , is:

[JM 11 Jan 2019, (II)]

- 100 Ω
- 120 Ω
- 80 Ω
- 125 Ω

- 239.** A galvanometer having a coil resistance 100Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V ?

[JM 8 Jan 2019, (II)]

- 10 k Ω
- 8.9 k Ω
- 7.9 k Ω
- 9.9 k Ω



- 240.** In a circuit for finding the resistance of a galvanometer by half deflection method, a 6V battery and a high resistance of $11\text{ k}\Omega$ are used. The figure of merit of the galvanometer $60\text{ }\mu\text{A/division}$. In the absence of shunt resistance, the galvanometer produces a deflection of $\theta = 9$ divisions when current flows in the circuit. The value of the shunt resistance that can cause the deflection of $\theta/2$, is closest
[JM April 16, 2018]
(1) $55\text{ }\Omega$ (2) $110\text{ }\Omega$
(3) $220\text{ }\Omega$ (4) $550\text{ }\Omega$
- 241.** A galvanometer with its coil resistance $25\text{ }\Omega$ requires a current of 1 mA for its full deflection. In order to construct an ammeter to read up to a current of 2 A , the approximate value of the shunt resistance should be
[JM April 16, 2018]
(1) $2.5 \times 10^{-2}\text{ }\Omega$ (2) $1.25 \times 10^{-3}\text{ }\Omega$
(3) $2.5 \times 10^{-3}\text{ }\Omega$ (4) $1.25 \times 10^{-2}\text{ }\Omega$
- 242.** When a current of 5 mA is passed through a galvanometer having a coil of resistance $15\text{ }\Omega$, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into voltmeter of range $0 - 10\text{ V}$ is
[JM 2017]
(1) $2.535 \times 10^3\text{ }\Omega$ (2) $4.005 \times 10^5\text{ }\Omega$
(3) $1.985 \times 10^3\text{ }\Omega$ (4) $2.045 \times 10^3\text{ }\Omega$
- 243.** A galvanometer having a coil resistance of $100\text{ }\Omega$ gives a full-scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A , is:
[JM 2016]
(1) $0.1\text{ }\Omega$ (2) $3\text{ }\Omega$
(3) $0.01\text{ }\Omega$ (4) $2\text{ }\Omega$
- 244.** A $50\text{ }\Omega$ resistance is connected to a battery of 5 V . A galvanometer of resistance $100\text{ }\Omega$ is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit?
[JM Online April 9, 2016]
(1) $r_s = 0.5\text{ }\Omega$ in series with the galvanometer
(2) $r_s = 1\text{ }\Omega$ in series with galvanometer
(3) $r_s = 1\text{ }\Omega$ in parallel with galvanometer
(4) $r_s = 0.5\text{ }\Omega$ in parallel with the galvanometer.
- 245.** To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ . If a shunt of resistance S is needed to get half deflection then G , R and S related by the equation:
[JM Online April 9, 2016]
(1) $S(R + G) = RG$ (2) $2S(R + G) = RG$
(3) $2G = S$ (4) $2S = G$
- 246.** The AC voltage across a resistance can be measured using a:
[JM Online April 11, 2015]
(1) hot wire voltmeter
(2) moving coil galvanometer
(3) potential coil galvanometer
(4) moving magnet galvanometer
- 247.** This questions has Statement I and Statement II. Of the four choices given after the Statements, choose the one that best describes into two Statements.
Statement-I: Higher the range, greater is the resistance of ammeter.
Statement-II: To increase the range of ammeter, additional shunt needs to be used across it. [JM 2013]
(1) Statement-I is true, Statement-II is true, Statement-II is the correct explanation of Statement-I.
(2) Statement-I is true, Statement-II is true, Statement-II is not the correct explanation of Statement-I.
(3) Statement-I is true, Statement-II is false.
(4) Statement-I is false, Statement-II is true.
- 248.** A shunt of resistance $1\text{ }\Omega$ is connected across a galvanometer of $120\text{ }\Omega$ resistance. A current of 5.5 ampere gives full scale deflection in the galvanometer. The current that will give full scale deflection in the absence of the shunt is nearly:
[JM Online April 9, 2013]
(1) 5.5 ampere (2) 0.5 ampere
(3) 0.004 ampere (4) 0.045 ampere
- 249.** A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 -divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt , the resistance in ohms needed to be connected in series with the coil will be:
[JM 2005]
(1) 10^5 (2) 10^3
(3) 9995 (4) 99995

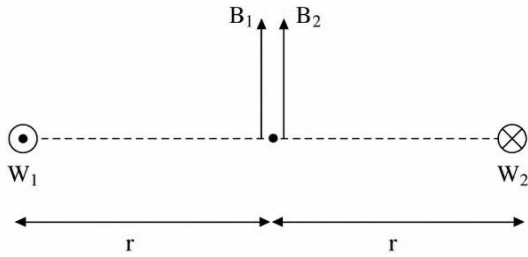


Answer Key

1. (2)	43. (3)	85. (3)	127. (2)	169. (1)	211. (3)
2. (1)	44. (1)	86. (2)	128. (5)	170. (2)	212. (4)
3. (2)	45. (1)	87. (1)	129. (4)	171. (4)	213. (3)
4. (2)	46. (4)	88. (4)	130. (32)	172. (3)	214. (3)
5. (1)	47. (3)	89. (2)	131. (1)	173. (3)	215. (50)
6. (2)	48. (4)	90. (2)	132. (2)	174. (1)	216. (1)
7. (3)	49. (3)	91. (48)	133. (3)	175. (3)	217. (4)
8. (1)	50. (1)	92. (5)	134. (40)	176. (3)	218. (2)
9. (3)	51. (4)	93. (50)	135. (144)	177. (4)	219. (1)
10. (8)	52. (3)	94. (1)	136. (2)	178. (3)	220. (4)
11. (2)	53. (1)	95. (35)	137. (2)	179. (1)	221. (2)
12. (2)	54. (1)	96. (2)	138. (3)	180. (2)	222. (4)
13. (250)	55. (1)	97. (4)	139. (10)	181. (2)	223. (2)
14. (8)	56. (2)	98. (2)	140. (2)	182. (2)	224. (4)
15. (3)	57. (2)	99. (1)	141. (3)	183. (4)	225. (3)
16. (500)	58. (1)	100. (9)	142. (1)	184. (2)	226. (4)
17. (5)	59. (3)	101. (1)	143. (4)	185. (1)	227. (4)
18. (72)	60. (4)	102. (1)	144. (2)	186. (314)	228. (4)
19. (20)	61. (4)	103. (3)	145. (2)	187. (3)	229. (20)
20. (3)	62. (2)	104. (3)	146. (4)	188. (11)	230. (3)
21. (40)	63. (2)	105. (5)	147. (2)	189. (4)	231. (2)
22. (3)	64. (2)	106. (4)	148. (3)	190. (4)	232. (None)
23. (2)	65. (1)	107. (3)	149. (2)	191. (3)	233. (3)
24. (8)	66. (2)	108. (4)	150. (4)	192. (4)	234. (4)
25. (628)	67. (1)	109. (3)	151. (3)	193. (2)	235. (4)
26. (1)	68. (4)	110. (1)	152. (1)	194. (1)	236. (2)
27. (3)	69. (1)	111. (3)	153. (2)	195. (2)	237. (2)
28. (3)	70. (3)	112. (2)	154. (2)	196. (3)	238. (3)
29. (4)	71. (3)	113. (1)	155. (1)	197. (4)	239. (4)
30. (3)	72. (1)	114. (1)	156. (4)	198. (2)	240. (2)
31. (4)	73. (2)	115. (2)	157. (None)	199. (1)	241. (4)
32. (3)	74. (3)	116. (3)	158. (2)	200. (3)	242. (3)
33. (3)	75. (1)	117. (4)	159. (4)	201. (1)	243. (3)
34. (2)	76. (1)	118. (2)	160. (4)	202. (50)	244. (4)
35. (3)	77. (160)	119. (12)	161. (2)	203. (1)	245. (1)
36. (3)	78. (68)	120. (171)	162. (3)	204. (3)	246. (2)
37. (2)	79. (2)	121. (4)	163. (2)	205. (1)	247. (4)
38. (2)	80. (1)	122. (4)	164. (1)	206. (2)	248. (4)
39. (2)	81. (4)	123. (0 or 10)	165. (1)	207. (3)	249. (3)
40. (4)	82. (4)	124. (1)	166. (2)	208. (3)	
41. (4)	83. (3)	125. (2)	167. (4)	209. (4)	
42. (1)	84. (1)	126. (3)	168. (3)	210. (4)	

Solution

1. (2)



Since both fields point in the same direction, the net magnetic field $B_{\text{net}} = B_1 + B_2$.

The current in the wires are given as $I_1 = I_2 = 30 \text{ A}$ and $r = 0.04 \text{ m}$:

$$B_1 = B_2 = \frac{\mu_0 I}{2\pi r} = 2 \times \left(\frac{\mu_0}{4\pi} \right) \frac{I}{r}$$

$$\Rightarrow B_1 = B_2 = 2 \times 10^{-7} \times \frac{30}{0.04} = 2 \times 10^{-7} \times 750$$

$$= 15 \times 10^{-5} \text{ T}$$

$$\Rightarrow B_{\text{net}} = 15 \times 10^{-5} + 15 \times 10^{-5} = 30 \times 10^{-5} \text{ T}$$

$$\Rightarrow B_{\text{net}} = 300 \times 10^{-6} \text{ T} = 300 \mu\text{T}$$

Therefore, the magnitude of magnetic field at the midpoint is $300 \mu\text{T}$.

Thus, the correct option is (2).

2. (1)

The correct option is (1) $\frac{2 + \pi}{1 + \pi}$

$$B_1 = \frac{\mu_0 I}{4\pi r} \times 2 + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r} \left(\frac{2}{\pi} + 1 \right)$$

$$B_2 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r} \left(\frac{1}{\pi} + 1 \right)$$

$$\frac{B_1}{B_2} = \frac{2 + \pi}{1 + \pi}$$

3. (2)

The magnetic field at the centre of a circular loop of radius R carrying current i is given by:

$$B_c = \frac{\mu_0 i}{2R}$$

We are given that $B_c = 16 \mu\text{T}$.

The magnetic field at a point on the axis at a distance x from the centre is given by:

$$B_a = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{\frac{3}{2}}}$$

The ratio of magnetic field at the centre to the axis,

$$\frac{B_c}{B_a} = \frac{\left(\frac{\mu_0 i}{2R} \right)}{\left(\frac{\mu_0 R^2 i}{2(R^2 + x^2)^{\frac{3}{2}}} \right)}$$

$$\Rightarrow B_a = B_c \left[\frac{R^3}{(R^2 + x^2)^{\frac{3}{2}}} \right]$$

Given the distance $x = \sqrt{3}R$,

$$\Rightarrow B_a = B_c \left[\frac{R^3}{(R^2 + (\sqrt{3}R)^2)^{\frac{3}{2}}} \right] = \frac{B_c}{8}$$

$$\Rightarrow B_a = \frac{16 \mu\text{T}}{8} = 2 \mu\text{T}$$

Hence, the correct option is (2).

4. (2)

The magnetic field (2) on the axis of a circular loop of radius r at a distance x from its center is given by:

$$B = \frac{\mu_0 I r^2}{2(r^2 + x^2)^{\frac{3}{2}}}$$

The point O is at a distance $x = r$ from the center of both loops P and Q .

For loop P :

Current is $i = I$.

The magnetic field B_p at O due to loop P is:

$$B_p = \frac{\mu_0 I r^2}{2(r^2 + r^2)^{\frac{3}{2}}} = \frac{\mu_0 I}{4\sqrt{2}r}$$

Using the right-hand thumb rule, the current is clockwise as seen from O , so the field B_p due to loop P points towards P .

For loop Q :

Current is $i = 4I$.

The magnetic field B_Q at O due to loop Q is:

$$B_Q = \frac{\mu_0 (4I) r^2}{2(r^2 + r^2)^{3/2}} = \frac{4\mu_0 I}{4\sqrt{2}r}$$

Using the right-hand thumb rule, the current is clockwise as seen from O, so the field B_Q due to loop Q points towards Q.

The net magnetic field (B_{net}) at O is the vector sum of these opposing fields. Since $B_Q > B_P$, the net field will be in the direction of B_Q (towards Q):

$$B_{net} = B_Q - B_P = \frac{4\mu_0 I}{4\sqrt{2}r} - \frac{\mu_0 I}{4\sqrt{2}r} = \frac{3\mu_0 I}{4\sqrt{2}r}$$

Therefore, the resulting magnetic field at the midpoint is $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q.

Hence, the correct option is (2).

5. (1)

The magnetic field B_1 due to a straight wire of finite length at a distance r is given by:

$$B_1 = \frac{\mu_0 I}{4\pi r} (\sin\theta_1 + \sin\theta_2)$$

Here, $\theta_1 = \theta_2 = 60^\circ$

$$B_1 = \frac{\mu_0 I}{4\pi r} (\sin 60^\circ + \sin 60^\circ) = \frac{\mu_0 I}{4\pi r} (2\sin 60^\circ)$$

$$\Rightarrow B_1 = \frac{\mu_0 I}{4\pi r} \left(2 \times \frac{\sqrt{3}}{2} \right) = \frac{\mu_0 I \sqrt{3}}{4\pi r}$$

Since there are three sides and the current flows in a way that all three fields at the centroid point in the same direction,

$$B_{net} = 3 \times B_1 = \frac{3\sqrt{3}\mu_0 I}{4\pi r}$$

$$B_{net} = \frac{3\sqrt{3} \times (4\pi \times 10^{-7}) \times 2}{4\pi \times (2 \times 10^{-2})} \text{ T}$$

$$\Rightarrow B_{net} = \frac{3\sqrt{3} \times 10^{-7} \times 2}{2 \times 10^{-2}} \text{ T}$$

$$\Rightarrow B_{net} = 3\sqrt{3} \times 10^{-5} \text{ T} = \alpha \times 10^{-5} \text{ T}$$

$$\Rightarrow \alpha = 3\sqrt{3}$$

6. (2)

The segments AB and DE together form an infinitely long straight wire.

For an infinite wire the magnitude of the magnetic field at distance r is given as

$$B_{\text{straight}} = \frac{\mu_0 I}{2\pi r}$$

Using the Right-Hand Thumb Rule, if the thumb points in the direction of current I (along +y), the fingers curl out of the page at point O, i.e., in the x direction.

$$\vec{B}_{\text{straight}} = \frac{\mu_0 I}{2\pi r} \hat{i}$$

The circular part is a nearly complete loop of radius r . For a circular loop at its centre the magnitude of the magnetic field is given as,

$$B_{\text{loop}} = \frac{\mu_0 I}{2r}$$

Using the Right-Hand Rule for loops, curling the fingers in the direction of the current (clockwise in the yz -plane) the thumb points into the page, which is the $-x$ direction.

$$\vec{B}_{\text{loop}} = -\frac{\mu_0 I}{2r} \hat{i}$$

The total field is the vector sum of both parts :

$$\vec{B}_{\text{total}} = \vec{B}_{\text{straight}} + \vec{B}_{\text{loop}}$$

$$\Rightarrow \vec{B}_{\text{total}} = \left(\frac{\mu_0 I}{2\pi r} \right) \hat{i} + \left(-\frac{\mu_0 I}{2r} \right) \hat{i}$$

$$\Rightarrow \vec{B}_{\text{total}} = \frac{\mu_0 I}{2r} \left[\frac{1}{\pi} - 1 \right] \hat{i} = \frac{\mu_0 I}{2\pi r} [1 - \pi] \hat{i}$$

$$= -\frac{\mu_0 I}{2\pi r} [\pi - 1] \hat{i}$$

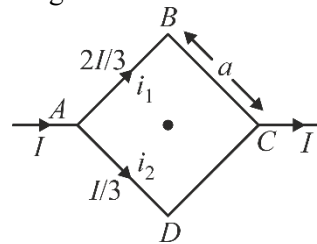
So, the magnetic field at the centre O of the circular

$$\text{loop is } = -\frac{\mu_0 I}{2\pi r} [\pi - 1] \hat{i}.$$

Hence, the correct option is (2).

7. (3)

Resultant magnetic field at the centre of square loop



$$\vec{B} = \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CD} + \vec{B}_{DA}$$

$$\vec{B} = \left[\begin{aligned} & \frac{-\mu_0(2I/3)}{4\pi(a/2)}\sqrt{2} - \frac{\mu_0(2I/3)}{4\pi(a/2)}\sqrt{2} \\ & + \frac{\mu_0(I/3)}{4\pi(a/2)}\sqrt{2} + \frac{\mu_0(I/3)}{4\pi(a/2)}\sqrt{2} \end{aligned} \right] \hat{k}$$

$$\Rightarrow \vec{B} = \left[\frac{-2\sqrt{2}\mu_0 I}{3\pi a} + \frac{\sqrt{2}\mu_0 I}{3\pi a} \right] \hat{k} \therefore \vec{B} = \frac{-\sqrt{2}\mu_0 I}{3\pi a} \hat{k}$$

8. (1)

Magnetic field due to wires AB & $CD = 0$

$$B_0 = B_{BC} - B_{AD} = \frac{\mu_0 I}{4R_2} - \frac{\mu_0 I}{4R_1} = \frac{\mu_0 I}{4} \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$$

Here, $R_1 = 6\pi m$, $R_2 = 4\pi m$

$$\therefore B_0 = \frac{4\pi \times 10^{-7} \times 12}{4} \left(\frac{1}{4\pi} - \frac{1}{6\pi} \right)$$

$$= 12\pi \times 10^{-7} \left(\frac{1}{12\pi} \right)$$

$$= 1 \times 10^{-7} \text{ T} = k \times 10^{-7} \text{ (given)}$$

$$\therefore K = 1$$

9. (3)

Magnetic field (i) at the centre of the loop

$$B_1 = \frac{\mu_0 I}{2R} \text{ and (ii) at the axis at point } P$$

$$B_2 = \frac{\mu_0 I R^2}{2x^3}$$

$$\therefore \frac{B_2}{B_1} = \frac{\mu_0 I R^2}{2x^3} \cdot \frac{2R}{\mu_0 I} = \frac{R^3}{x^3} = \left(\frac{R}{x} \right)^3 = \left(\frac{4}{5} \right)^3 = \frac{64}{125}$$

10. (8)

Magnetic field, $B = 2.9 \times 10^{-4} \text{ T}$

Radius, $r = 10 \text{ cm}$

Number of turns, $N = 200$

Current, $I = 0.29 \text{ A}$

The expression of magnetic field of solenoid is

$$B = \mu_0 \left(\frac{N}{l} \right) I$$

$$\Rightarrow l = \frac{\mu_0 N I}{B} = \frac{(4\pi \times 10^{-7})(200)(0.29)}{2.9 \times 10^{-4}} \text{ m} = 8\pi \text{ cm}$$

11. (2)

For long straight current carrying wire, Maximum possible magnetic field is

$$B_{\max} = B_{\text{surface}} = \frac{\mu_0 I}{2\pi a}$$

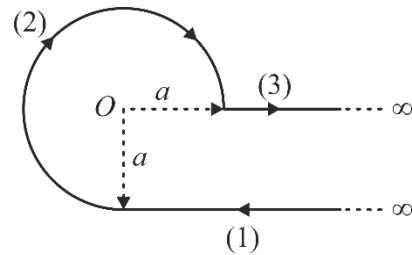
$$\text{For inside, } B_{\text{inside}} = \frac{B_{\max}}{2}$$

$$\frac{\mu_0 I r}{2\pi a^2} = \frac{\mu_0 I}{4\pi a} \Rightarrow r = \frac{a}{2}$$

$$\text{For outside, } B_{\text{outside}} = \frac{B_{\max}}{2} \Rightarrow \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{4\pi a} \Rightarrow r = 2a$$

The distance from the centre of wire is $\left[\frac{a}{2}, 2a \right]$

12. (2)



Magnetic field

$$B_1 = \frac{\mu_0 I}{4\pi a} \otimes$$

$$B_2 = \frac{\mu_0 I}{4\pi a} \left(\frac{3\pi}{2} \right) \otimes$$

$$B_3 = 0$$

$$\therefore B_0 = B_1 + B_2 + B_3 = \frac{\mu_0 I}{4\pi a} \left(1 + \frac{3\pi}{2} \right) \otimes$$

13. (250)

Given, Current, $I = 1.5 \text{ A}$

Time period of revolution of electron, $T = 75 \text{ ns}$

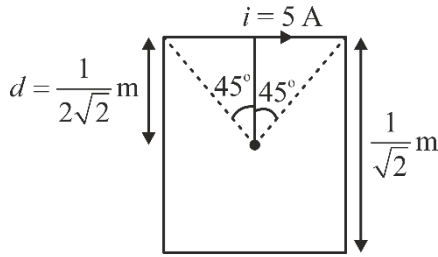
For charged particle moving on circle in magnetic field,

$$T = \frac{2\pi m}{qB} = \frac{2\pi m}{q(\mu_0 n I)}$$

$$\Rightarrow 75 \times 10^{-9} = \frac{2\pi(9 \times 10^{-31})}{1.6 \times 10^{-19} \times 4 \times 10^{-7} \times n \times 1.5}$$

$$\therefore n = 250.$$

14. (8)



$$d = \frac{1}{2\sqrt{2}} \text{ m}, \quad i = 5 \text{ A}$$

The net magnetic field at the centre of square loop is

$$\begin{aligned} B_{\text{net}} &= 4B_{\text{single side}} = \frac{\mu_0 i}{4\pi d} (\sin \theta_1 + \sin \theta_2) \\ &= \frac{4 \times 10^{-7} \times 5 \times 2}{\frac{1}{2\sqrt{2}}} \times \frac{1}{\sqrt{2}} = 8 \times 10^{-6} \\ &= p \times 10^{-6} (\text{given}) \quad \therefore p = 8 \end{aligned}$$

15. (3)

Magnetic field due to straight wire, $B = \frac{\mu_0 I}{2\pi r}$

Magnetic field at $2a$ is,

$$B_{2a} = \frac{\mu_0 I}{2\pi(2a)} = \frac{\mu_0 I}{4\pi a}$$

Inside the wire, $B_{a/2} = \left(\frac{\mu_0 I}{2\pi a^2} \right) \cdot \frac{a}{2} = \frac{\mu_0 I}{4\pi a}$

$$\therefore \frac{B_{a/2}}{B_{2a}} = \frac{1}{1} = 1:1$$

16. (500)

Magnetic field inside a solenoid, $B = \mu_0 n i$

$$\Rightarrow \mu_0 \left(\frac{m}{l} \right) i = B \quad \left[\because n = \frac{m}{l} \right]$$

$$\Rightarrow m = \frac{Bl}{\mu_0 i} = \frac{6.28 \times 10^{-3} \times 0.5}{12.56 \times 10^{-7} \times 5} \quad \therefore m = 500$$

17. (5)

Magnetic field at point A

$$B_A = \frac{\mu_0 i}{2\pi r} + \frac{\mu_0 (2i)}{2\pi(3r)} = \frac{5\mu_0 i}{6\pi r}$$

Magnetic field at point C

$$B_C = \frac{\mu_0 (2i)}{2\pi r} + \frac{\mu_0 i}{2\pi(3r)} = \frac{7\mu_0 i}{6\pi r}$$

$$\therefore \frac{B_A}{B_C} = \frac{5}{7} = \frac{x}{7} \quad \therefore x = 5$$

18. (72)

Length of each side, $a = \frac{4\pi}{6}$

Distance from centre, $r = \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{2} \times \frac{4\pi}{6} = \frac{\pi}{\sqrt{3}}$

Magnetic field at centre,

$$B = 6 \left(\frac{\mu_0 I}{4\pi r} \right) (\sin \theta_1 + \sin \theta_2)$$

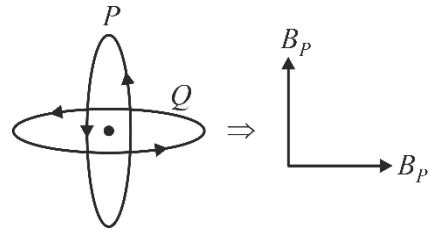
$$\Rightarrow B = \frac{6 \times 6\pi \times 10^{-7} \times 4\pi \sqrt{3}}{4\pi \times \left(\frac{\pi}{\sqrt{3}} \right)} (\sin 30^\circ + \sin 30^\circ)$$

$$\Rightarrow B = 6 \times 4 \times 3 \times 10^{-7} = 72 \times 10^{-7} \text{ T}$$

$$= x \times 10^{-7} \text{ T (given)}$$

$$\therefore x = 72$$

19. (20)



Magnetic field due to coil P

$$B_P = \frac{\mu_0 N i_1}{2r} = \frac{\mu_0 \times 1 \times 100}{2\pi} = 2 \times 10^{-3} \text{ T}$$

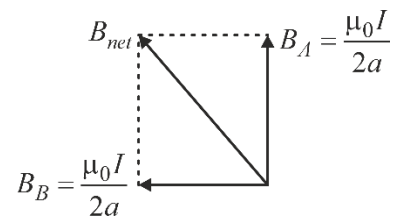
Magnetic field due to coil Q

$$B_Q = \frac{\mu_0 N i_2}{2r} = \frac{\mu_0 \times 2 \times 100}{2\pi} = 4 \times 10^{-3} \text{ T}$$

$$\begin{aligned} B_{\text{net}} &= \sqrt{B_P^2 + B_Q^2 + 2B_P B_Q \cos 90^\circ} = \sqrt{B_P^2 + B_Q^2} \\ &= \sqrt{(2 \times 10^{-3})^2 + (4 \times 10^{-3})^2} = \sqrt{20} \times 10^{-3} \text{ T} \\ &= \sqrt{20} \text{ mT} = \sqrt{x} \text{ mT (given)} \end{aligned}$$

$$\therefore x = 20.$$

20. (3)



$$B_{\text{net}} = \sqrt{B_A^2 + B_B^2} = \sqrt{2} B_A$$

$$\therefore B_{\text{net}} = \frac{\sqrt{2}\mu_0 I}{2a} = \frac{\mu_0 I}{\sqrt{2}a}$$

21. (40)

The magnetic field at the centre is

$$B = 4 \times \frac{\mu_0 i}{4\pi r} (\sin 45^\circ + \sin 45^\circ) \left(\because r = \frac{1}{2}m \right)$$

$$B = 4 \times \frac{\mu_0 i}{4\pi \left(\frac{1}{2}\right)} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right)$$

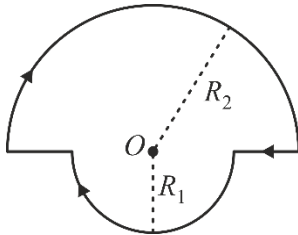
$$= 4 \times 10^{-7} \times 5 \times 2 \times \sqrt{2} = 40\sqrt{2} \times 10^{-7} \text{ T}$$

$$= x\sqrt{2} \times 10^{-7} \text{ T (given)}$$

$$\therefore x = 40$$

22. (3)

Here magnetic field at the centre (O) of a wire loop formed by two semicircular wires of radii R_1 and R_2



$$\vec{B}_{\text{centre}} = \frac{\mu_0 i}{2R_2} \left(\frac{\pi}{2\pi} \right) \otimes + \frac{\mu_0 i}{2R_1} \left(\frac{\pi}{2\pi} \right) \odot$$

$$= \left(\frac{\mu_0 i}{4R_2} + \frac{\mu_0 i}{4R_1} \right) \otimes$$

$$= \frac{4\pi \times 10^{-7} \times 4}{4 \times 4\pi} + \frac{4\pi \times 10^{-7} \times 4}{4 \times 2\pi}$$

$$= 3 \times 10^{-7} \text{ T} = \alpha \times 10^{-7} \text{ T (given)}$$

$$\therefore \alpha = 3$$

23. (2)

Given,

Current, $I = 14 \text{ A}$

Radius of semicircular arc,

$$R = 2.2 \text{ cm} = 2.2 \times 10^{-2} \text{ m}$$

Magnetic field at the centre of semicircular arc

$$B = \frac{\mu_0 I}{4R} = \frac{4\pi \times 10^{-7} \times 14}{4 \times 2.2 \times 10^{-2}}$$

$$= \frac{4 \times 22 \times 10^{-7} \times 14}{4 \times 7 \times 2.2 \times 10^{-2}} = 2 \times 10^{-4} \text{ T}$$

24. (8)

$$\text{Magnetic field at centre } (B_1) = \frac{\mu_0 I}{2r}$$

$$\text{Magnetic field on axis } (B_2) = \frac{\mu_0 I r^2}{2(x^2 + r^2)^{3/2}}$$

Value of $x = r$ (given)

$$B_2 = \frac{\mu_0 I}{4\sqrt{2}r}$$

$$\frac{B_1}{B_2} = \frac{\mu_0 I}{2r} \times \frac{4\sqrt{2}r}{\mu_0 I} = \frac{2\sqrt{2}}{1} = \frac{\sqrt{8}}{1} = \frac{\sqrt{x}}{1} \text{ (given)}$$

$$\therefore x = 8$$

25. (628)

Given,

Radius of circular wires, $r = 20 \text{ cm} = 0.2 \text{ m}$

Current, $i = \sqrt{2} \text{ A}$

Magnetic field due to circular wire X = magnetic field due to circular wire Y

$$\therefore B_x = B_y = \frac{\mu_0 i}{2r}$$

$$= \frac{4\pi \times 10^{-7}}{2 \times 0.2} \times \sqrt{2} \text{ T} \quad \left[\because \mu_0 = 4\pi \times 10^{-7} \right]$$

Net magnetic field is

$$B_N = \sqrt{B_x^2 + B_y^2}$$

$$= \frac{4\pi \times 10^{-7} \times \sqrt{2}}{2 \times 0.2} \times \sqrt{2} \text{ T} = 20\pi \times 10^{-6} \text{ T}$$

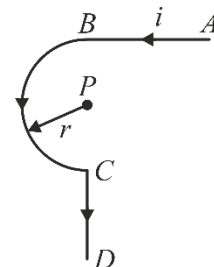
$$= 200\pi \times 10^{-8} \text{ T} = 2 \times 314 \times 10^{-8} \text{ T} = 628 \times 10^{-8} \text{ T}$$

26. (1)

The magnetic field at the point P is

$$B_P = B_{AB} + B_{BC} + B_{CD}$$

$$B_{AB} = \frac{\mu_0 i}{4\pi r}$$

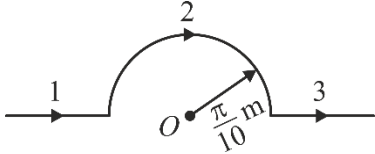


$$B_{BC} = \frac{\mu_0 i}{4\pi r} \times \pi = \frac{\mu_0 i}{4r}$$

$$B_{CD} = 0$$

$$B_P = \left(\frac{\mu_0 i}{4r} + \frac{\mu_0 i}{4\pi r} \right) = \frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{2\pi} \right)$$

27. (3)



$$B_{net} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = \left(0 + \frac{\mu_0 I}{4r} + 0 \right) \otimes$$

$$= \left(\frac{\mu_0 \times 3 \times 10}{4 \times \pi} \right) \otimes = 10^{-7} \times 3 \times 10 \otimes = 3 \mu T$$

28. (3)

$$\text{We have, } B_1 = \frac{N\mu_0 I}{2r_1} \text{ and } B_2 = \frac{n\mu_0 I}{2r_2}$$

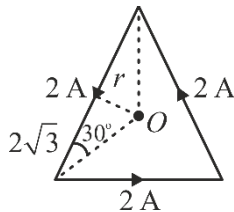
$$\text{Now, } N \times 2\pi r_1 = n \times 2\pi r_2 \Rightarrow Nr_1 = nr_2 \Rightarrow \frac{r_2}{r_1} = \frac{N}{n}$$

$$\text{So, } \frac{B_1}{B_2} = \frac{Nr_2}{nr_1} = \frac{N}{n} \left(\frac{r_2}{r_1} \right) = \frac{N}{n} \frac{N}{n} = \frac{N^2}{n^2}$$

29. (4)

From Figure

$$\Rightarrow \frac{r}{2\sqrt{3}} = \tan 30^\circ \Rightarrow r = 2 \text{ cm}$$

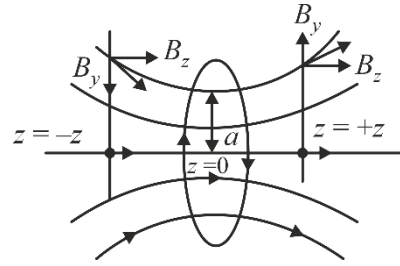


Magnetic field at the centroid of triangle,

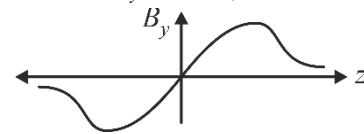
$$B = 3 \times \frac{\mu_0 i}{4\pi r} (\sin 60^\circ + \sin 60^\circ) \Rightarrow B = 3 \times \frac{\mu_0 i}{2\pi r} \sin 60^\circ$$

$$= 3 \times \frac{2 \times 10^{-7} \times 2}{2 \times 10^{-2}} \times \frac{\sqrt{3}}{2} = 3\sqrt{3} \times 10^{-5} \text{ T}$$

30. (3)



We can see from magnetic field lines drawing that B_y direction is equal and opposite on $z = +z$ and $z = -z$. And, at $z = 0 \Rightarrow B_y = 0$. So, curves will look like:



31. (4)

Magnetic field due to wire AB and wire ED cancel each other. So, magnetic field due to BC and ET will be

$$B_1 = \frac{\mu_0 I}{4\pi(a/2)} \cdot \left(\sin \frac{\pi}{2} + \sin 0^\circ \right) = \frac{\mu_0 I}{2\pi a}$$

$$B_2 = \frac{\mu_0 I}{4\pi(a/2)} \cdot \left(\sin \frac{\pi}{2} + \sin 0^\circ \right) = \frac{\mu_0 I}{2\pi a}$$

$$\text{So, } B_{net} = B_1 + B_2 = \frac{\mu_0 I}{\pi a}$$

32. (3)

Initial length = Final length

$$\Rightarrow 4 \times 2\pi R_1 = 2\pi R_2 \Rightarrow R_2 = 4R_1$$

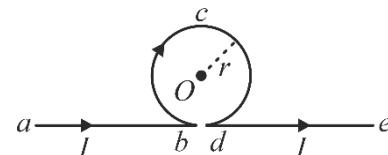
$$\text{As } B = \frac{\mu_0 I}{2R} \times n \Rightarrow B \propto \frac{n}{R}$$

$$\Rightarrow \frac{B_2}{B_1} = \frac{n_2}{n_1} \times \frac{R_1}{R_2} = \frac{1}{1} \times \frac{1}{4} = \frac{1}{16}$$

$$\text{So, } B_2 = \frac{B_1}{16} = \frac{32}{16} = 2 \text{ T}$$

33. (3)

(I)



$$B_{ab} = \frac{\mu_0 I}{4\pi r} \text{ (out of the plane)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (2\pi) \text{ (into the plane)}$$

$$B_{de} = \frac{\mu_0 I}{4\pi r} \text{ (out of the plane)}$$

Hence magnetic field at O is

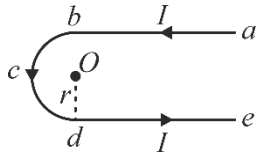
$$B_0 = B_{ab} + B_{bcd} + B_{de}$$

$$B_0 = -\frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} (2\pi) - \frac{\mu_0 I}{4\pi r}$$

$$B_0 = \frac{\mu_0 I}{4\pi r} (2\pi - 2)$$

$$B_0 = \frac{\mu_0 I}{2\pi r} (\pi - 1) \quad \dots(\text{iii})$$

(II)



$$B_{ab} = \frac{\mu_0 I}{4\pi r} \text{ (out of the plane)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (\pi) \text{ (out of the plane)}$$

$$B_{de} = \frac{\mu_0 I}{4\pi r} \text{ (out of the plane)}$$

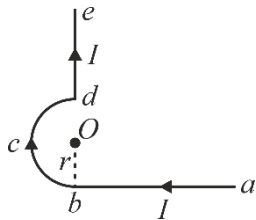
Hence magnetic field of O is

$$B_0 = B_{ab} + B_{bcd} + B_{de}$$

$$B_0 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} (\pi) + \frac{\mu_0 I}{4\pi r}$$

$$B_0 = \frac{\mu_0 I}{4\pi r} (\pi + 2) \quad \dots(\text{i})$$

(III)



$$B_{ab} = \frac{\mu_0 I}{4\pi r} \text{ (into the plane)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (\pi) \text{ (into the plane)}$$

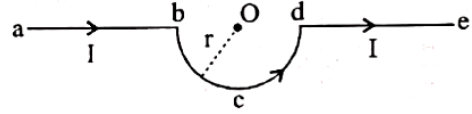
$$B_{de} = 0 \text{ (at the axis)}$$

Hence, magnetic field at O is

$$B_0 = B_{ab} + B_{bcd} + B_{de}$$

$$B_0 = \frac{\mu_0 I}{4\pi r} (1 + \pi) \quad \dots(\text{iv})$$

(IV)



$$B_{ab} = 0 \text{ (at the axis)}$$

$$B_{bcd} = \frac{\mu_0 I}{4\pi r} (\pi) \text{ (out of the plane)}$$

$$B_{de} = 0 \text{ (at the axis)}$$

$$B_0 = B_{ab} + B_{bcd} + B_{de} = 0 + \frac{\mu_0 I}{4r} + 0$$

$$B_0 = \frac{\mu_0 I}{4r} \quad \dots(\text{ii})$$

34. (2)

Magnetic field inside solenoid is given as

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 70 \times 10^2 \times 2 = 56\pi \times 10^{-4} \text{ T} \\ = 176 \times 10^{-4} \text{ T}$$

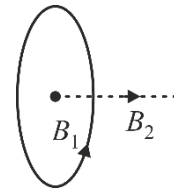
35. (3)

Magnetic field at centre of ring/coil is given as,

$$B_{\text{centre}} = \frac{N\mu_0 I}{2R}$$

$$\Rightarrow 37.68 \times 10^{-4} = \frac{100 \times 4\pi \times 10^{-7} \times I}{2 \times 5 \times 10^{-2}} \Rightarrow I = 3 \text{ A}$$

36. (3)



$$B_1 = \frac{\mu_0 I}{2R} \quad \dots(\text{i})$$

$$B_2 = \frac{\mu_0 I R^2}{2(R^2 + 3R^2)^{3/2}} \quad \dots(\text{ii})$$

$$= \frac{1}{8} \left(\frac{\mu_0 I}{2R} \right) = \frac{B_1}{8}$$

Dividing (i) by (ii), we get

$$\Rightarrow \frac{B_1}{B_2} = \frac{8}{1}$$

37. (2)

The magnetic field at the centre of coil, $B = \frac{\mu_0 NI}{2R}$

Given, for coil x

Number of turns, $N_x = 200$

Radius, $R_x = 20$ cm

$$\Rightarrow B_x = \frac{\mu_0 N_x i}{2R_x} = \frac{200\mu_0 i}{2 \times 20} \quad \dots(i)$$

For coil y

Number of turns, $N_y = 400$

Radius, $R_y = 20$ cm

$$\Rightarrow B_y = \frac{\mu_0 N_y i}{2R_y} = \frac{400\mu_0 i}{2 \times 20} \quad \dots(ii)$$

Divide equation (i) by equation (ii), we have

$$\Rightarrow \frac{B_x}{B_y} = \frac{\frac{200\mu_0 i}{2 \times 20}}{\frac{400\mu_0 i}{2 \times 20}} = \frac{1}{2}$$

38. (2)

From the given figure. Magnetic moment will be,

$\vec{M} = \text{current } (I) \times \text{Area } (A)$

$$\vec{M} = -I\pi(0.5)^2 \hat{k} + I\pi(0.3)^2 \hat{k}$$

$$\vec{M} = -I \times \frac{22}{7} \left(\frac{25}{100} - \frac{9}{100} \right) \hat{k} = -22 \left(\frac{16}{100} \right) \hat{k}$$

$$\vec{M} = -3.52 \hat{k} \text{ Am}^2 = -\frac{7}{2} \hat{k} \text{ Am}^2$$

39. (2)

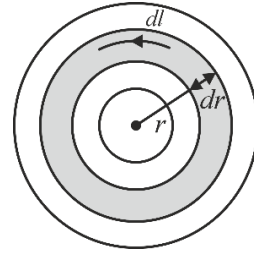
Magnetic field at centre is given as

$$B = \frac{N\mu_0 i}{2R}$$

$$\text{So, } B_1 = \frac{N_1 \mu_0 i}{2R_1} \text{ and, } B_2 = \frac{N_2 \mu_0 i}{2R_2}, R_2 = \frac{N_1 R_1}{N_2}$$

$$\frac{B_2}{B_1} = \frac{N_2}{N_1} \frac{R_1}{R_2} = \frac{N_2}{N_1} \times \frac{N_2}{N_1} \Rightarrow \frac{B_2}{B_1} = \frac{25}{4}$$

40. (4)



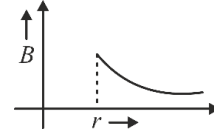
Let us consider a circular strip of length 'dr' at distance 'r' from centre

Magnetic field at centre due to this strip is given as

$$dB = \frac{\mu_0 I}{2r} dn = \frac{\mu_0 I}{2r} \cdot \frac{n}{r_2 - r_1} \cdot dr$$

$$\text{So, } B_{\text{net}} = \int dB = \frac{\mu_0 n I}{2(r_2 - r_1)} \int_{r_1}^{r_2} \frac{dr}{r} = \frac{\mu_0 n I}{2(r_2 - r_1)} \ln \left(\frac{r_2}{r_1} \right)$$

41. (4)



$$\vec{B} = 0, r < R = \frac{\mu_0 I}{2r}, r \geq R$$

So, $B = 0, r < R$

$\propto \frac{1}{r}, r \geq R$. Therefore graph will be as such shown in figure above.

42. (1)

Magnetic field due to solenoid is given by

$$B = \mu_0 n I \Rightarrow B \propto n I \Rightarrow \frac{B_2}{B_1} = \frac{n_2 I_2}{n_1 I_1}$$

$$\Rightarrow B_2 = \frac{n_2 I_2}{n_1 I_1} \times B_1 \Rightarrow B_2 = \frac{1}{2} \times 2 \times B_1 \therefore B_2 = B_1 = B$$

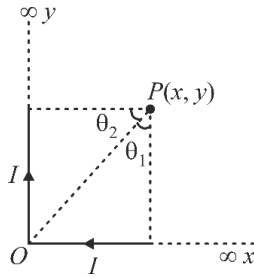
43. (3)

$$B_{\text{centre}} = \frac{\mu_0 n I}{2r}$$

$$B_{r/2} = \frac{\mu_0 I}{2} \frac{r^2}{\left[r^2 + \left(\frac{r}{2} \right)^2 \right]^{3/2}} = \frac{\mu_0 I}{2} \frac{r^2}{\left[\frac{5r^2}{4} \right]^{3/2}}$$

$$= \frac{\mu_0 I}{2r} \left(\frac{4}{5} \right)^{3/2} = B_{\text{centre}} \left(\frac{2}{\sqrt{5}} \right)^3$$

44. (1)



$$B_1 = \frac{\mu_0 I}{4\pi y} [\sin 90^\circ + \sin \theta_1]$$

$$= \frac{\mu_0 I}{4\pi y} \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right) \dots (i)$$

$$B_2 = \frac{\mu_0 I}{4\pi x} (\sin 90^\circ + \sin \theta_2)$$

$$= \frac{\mu_0 I}{4\pi x} \left(1 + \frac{y}{\sqrt{x^2 + y^2}} \right) \dots (ii)$$

$$B_{\text{Net}} = B_1 + B_2$$

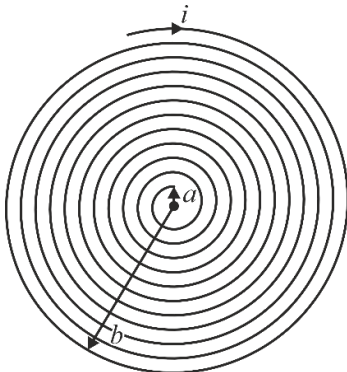
$$B = \frac{\mu_0 I}{4\pi} \left[\frac{1}{y} + \frac{x}{y\sqrt{x^2 + y^2}} + \frac{1}{x} + \frac{y}{x\sqrt{x^2 + y^2}} \right]$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi} \left[\frac{x+y}{xy} + \frac{x^2 + y^2}{xy\sqrt{x^2 + y^2}} \right]$$

$$= \frac{\mu_0 I}{4\pi} \left[\frac{x+y}{xy} + \frac{\sqrt{x^2 + y^2}}{xy} \right]$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi xy} [\sqrt{x^2 + y^2} + (x+y)]$$

45. (1)

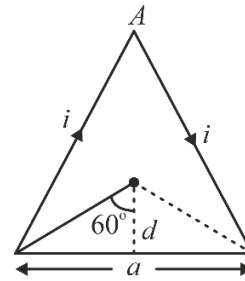


$$\text{Number of turns in } dr \text{ width} = \frac{N}{b-a} dr$$

$$B = \int dB = \int_a^b \left(\frac{N}{b-a} \right) dr \frac{\mu_0 I}{2x}$$

$$\therefore B = \frac{N\mu_0 I}{2(b-a)} \ln \left(\frac{b}{a} \right)$$

46. (4)



$$B = 3 \left[\frac{\mu_0 i}{4\pi d} (\sin 60^\circ + \sin 60^\circ) \right]$$

$$\tan 60^\circ = \frac{a/2}{d}$$

$$\text{Put } a = 9 \times 10^{-2} \text{ m} \Rightarrow d = \frac{9 \times 10^{-2}}{2\sqrt{3}} \text{ m}$$

$$B = 3 \times 10^{-5} \text{ T.}$$

$\vec{B}$ is into the plane of triangle.

47. (3)

$$\text{Number of turn in triangle, } N_1 = \frac{24a}{3a} = 8$$

$$\text{Number of turns in square, } N_2 = \frac{24a}{4a} = 8$$

$$\frac{M_1}{M_2} = \frac{N_1 I A_1}{N_2 I A_2} \quad [\text{As } i_1 = i_2 = I]$$

$$= \frac{8 \times \frac{\sqrt{3}}{4} \times a^2}{6 \times a^2}$$

$$\frac{M_1}{M_2} = \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{y}} \text{ so, } y = 3$$

48. (4)

Magnetic field at centre (B_c) is

$$B_c = \frac{\mu_0 i}{2a}$$

magnetic field at distance r from centre B_r on axis is

$$B_r = \frac{\mu_0 i a^2}{2(a^2 + r^2)^{3/2}}$$

fractional change in magnetic field

$$\frac{\frac{\mu_0 i}{2a} - \frac{\mu_0 i a^2}{2(a^2 + r^2)^{3/2}}}{\frac{\mu_0 i}{2a}}$$

$$= 1 - \frac{1}{\left[1 + \left(\frac{r^2}{a^2}\right)\right]^{3/2}} \approx 1 - \left[1 - \frac{3}{2} \frac{r^2}{a^2}\right] = \frac{3r^2}{2a^2}$$

Note: $\left(1 + \frac{r^2}{a^2}\right)^{-3/2} \approx \left(1 - \frac{3r^2}{2a^2}\right)$

49. (3)

$$(B_{\text{surface}})_A = \frac{\mu_0 I}{2\pi a}$$

$$(B_{\text{surface}})_B = \frac{\mu_0 I}{2\pi b}$$

As $a < b$ So, $(B_{\text{surface}})_A > (B_{\text{surface}})_B$

50. (1)

Magnetic field at P ,

$B_P = 2(\text{Magnetic field due to straight wire})$
 $+ (\text{Magnetic field due to semicircle})$

$$B_P = 2\left(\frac{\mu_0 I}{4\pi r}\right) + \frac{\mu_0 I}{4r}$$

$$\therefore B_P = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4\pi r}(2 + \pi)$$

51. (4)

Given:

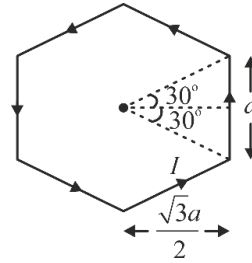
$$I_A = 2 \text{ A}, R_A = 2 \text{ cm}, \theta_A = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$$

$$I_B = 3 \text{ A}, R_B = 4 \text{ cm}, \theta_B = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

Using, magnetic field, $B = \frac{\mu_0 I \theta}{4\pi r}$

$$\frac{B_A}{B_B} = \frac{I_A}{I_B} \times \frac{\theta_A R_B}{\theta_B R_A} = \frac{2 \times \frac{3\pi}{2} \times 4}{3 \times \frac{5\pi}{3} \times 2} = \frac{6}{5}$$

52. (3)



Magnetic field due to one side of hexagon

$$B = \frac{\mu_0 I}{4\pi \frac{\sqrt{3}a}{2}} (\sin 30^\circ + \sin 30^\circ)$$

$$\Rightarrow B = \frac{\mu_0 I}{2\sqrt{3}a} \left(\frac{1}{2} + \frac{1}{2}\right) = \frac{\mu_0 I}{2\sqrt{3}\pi a}$$

Now, magnetic field due to one hexagon coil

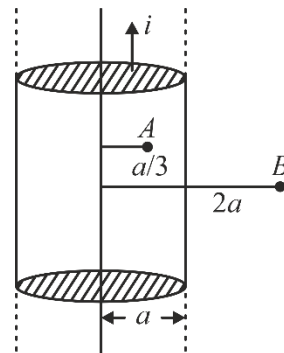
$$B = 6 \times \frac{\mu_0 I}{2\sqrt{3}\pi a}$$

Again magnetic field at the centre of hexagonal shape coil of 50 turns,

$$B = 50 \times 6 \times \frac{\mu_0 I}{2\sqrt{3}\pi a} \left[\because a = \frac{10}{100} = 0.1 \text{ m} \right]$$

$$\text{or, } B = \frac{150\mu_0 I}{\sqrt{3} \times 0.1 \times \pi} = 500\sqrt{3} \frac{\mu_0 I}{\pi}$$

53. (1)



Let a be the radius of the wire

Magnetic field at point A (inside)

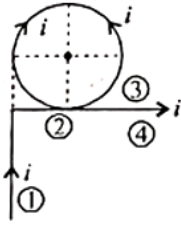
$$B_A = \frac{\mu_0 i r}{2\pi a^2} = \frac{\mu_0 i \frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i a}{2\pi a^2 \cdot 6} = \frac{\mu_0 i}{6\pi a}$$

Magnetic field at point B (outside)

$$B_B = \frac{\mu_0 i}{2\pi(2a)}$$

$$\therefore \frac{B_A}{B_B} = \frac{\frac{\mu_0 i}{6\pi a}}{\frac{\mu_0 i}{2\pi(2a)}} = \frac{4}{6} = \frac{2}{3}$$

54. (1)



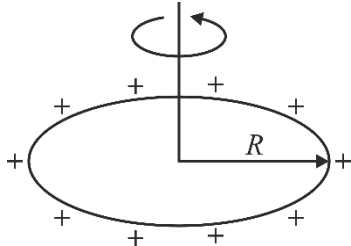
$$\begin{aligned} B_0 &= B_1 + B_2 + B_3 + B_4 \\ &= -\frac{\mu_0 I}{4\pi R} [\sin 90^\circ - \sin 45^\circ] \\ &\quad + \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{4\pi R} [\sin 45^\circ + \sin 90^\circ] \\ &= -\frac{\mu_0 I}{4\pi R} \left(1 - \frac{1}{\sqrt{2}}\right) + \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{4\pi R} \left(1 + \frac{1}{\sqrt{2}}\right) \\ B_0 &= \frac{\mu_0 I}{2\pi R} \left(\pi + \frac{1}{\sqrt{2}}\right) \end{aligned}$$

55. (1)

$$\vec{B} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{2\pi} \cdot \left(\frac{1}{d} \cdot \hat{k} + \frac{1}{d} (-\hat{k})\right) = 0$$

56. (2)

If q is the charge on the ring, then



$$i = \frac{q}{T} = \frac{q\omega}{2\pi}$$

Magnetic field,

$$B = \frac{\mu_0 i}{2R} = \frac{\mu_0 \left(\frac{q\omega}{2\pi}\right)}{2R}$$

$$\text{or } 3.8 \times 10^{-9} = \left(\frac{\mu_0}{4\pi}\right) \frac{q\omega}{R} = (10^{-7}) \frac{q \times 40\pi}{0.10}$$

$$\therefore q = 3 \times 10^{-5} \text{ C}$$

57. (2)

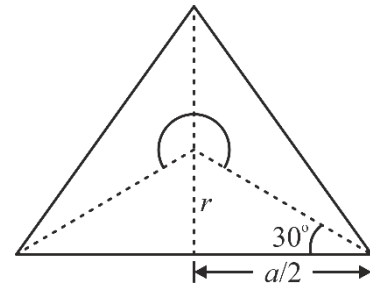
$$B = \frac{\mu_0 i}{4\pi r} (\sin \alpha + \sin \beta)$$

$$\text{Here, } r = \sqrt{5^2 - 3^2} = 4 \text{ cm}$$

$$\alpha = \beta = 37^\circ$$

$$\therefore B = 10^{-7} \times \frac{5}{4} \times 2 \sin 37^\circ = 1.5 \times 10^{-5} \text{ T}$$

58. (1)



$$\tan 30^\circ = \frac{r}{a/2}$$

$$\therefore r = \frac{a}{2\sqrt{3}}$$

$$B_0 = 3 \left[\frac{\mu_0 I}{4\pi r} (\sin 60^\circ + \sin 60^\circ) \right]$$

$$= \frac{3\mu_0 I}{4\pi \left(\frac{a}{2\sqrt{3}}\right)} \times (2) \left(\frac{\sqrt{3}}{2}\right) = \frac{9}{2} \left(\frac{\mu_0 I}{\pi a}\right)$$

$$= \frac{9 \times 2 \times 10^{-7} \times 10}{1} = 18 \mu\text{T}$$

59. (3)

Let i be the current in each wire.

Magnetic field at 'O' due to LP and QM will be zero.

$$\text{i.e., } B_0 = B_{PS} + B_{QN}$$

$\therefore$ Net magnetic field

$$B_0 = \frac{\mu_0 i}{4\pi d} + \frac{\mu_0 i}{4\pi d}$$

$$\Rightarrow 10^{-4} = \frac{\mu_0 i}{2\pi d} = \frac{2 \times 10^{-7} \times i}{4 \times 10^{-2}}$$

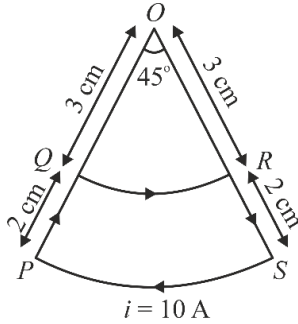
$\therefore i = 20 \text{ A}$ and the direction of magnetic field is perpendicular into the plane.

60. (4)

The will be no magnetic field at O due to wire PQ and RS

Magnetic field at ' O ' due to arc

$$QR = \frac{\mu_0}{4\pi} \frac{\left(\frac{\pi}{4}\right) \cdot I}{r_1}$$



Magnetic field at ' O ' due to arc

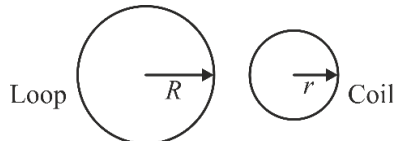
$$PS = \frac{\mu_0}{4\pi} \frac{\left(\frac{\pi}{4}\right) \cdot I}{r_2}$$

$\therefore$ Net magnetic field at ' O '

$$B = \frac{\mu_0 (\pi/4) \times 10}{4\pi} \left[\frac{1}{(3 \times 10^{-2})} - \frac{1}{(5 \times 10^{-2})} \right]$$

$$= \frac{\pi}{3} \times 10^{-5} \text{ T} \approx 1 \times 10^{-5} \text{ T}$$

61. (4)



$$L = 2\pi R \quad L = N \times 2\pi r$$

$$R = Nr \Rightarrow r = \frac{R}{N}$$

$$B_{\text{Loop}} = \frac{\mu_0 i}{2R}, B_{\text{coil}} = \frac{\mu_0 Ni}{2r} = \frac{\mu_0 Ni}{2\left(\frac{R}{N}\right)} = \frac{\mu_0 N^2 i}{2R}$$

$$\therefore \frac{B_L}{B_C} = \frac{1}{N^2}$$

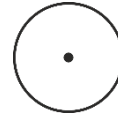
62. (2)

Point P is situated at the mid-point of the line joining the centres of the circular wires which have same radii (R). The magnetic fields ($\vec{B}$) at P due to the currents in the loops are in same direction. Magnitude of magnetic field at point, P .

$$B = 2 \left\{ \frac{\mu_0 N I R^2}{2(R^2 + \frac{R^2}{4})^{3/2}} \right\} = \frac{8\mu_0 N I}{5^{3/2} R}$$

63. (2)

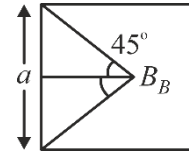
Case (1):



$$B_A = \frac{\mu_0 I}{4\pi R} \times 2\pi = \frac{\mu_0 I}{4\pi \ell / 2\pi} \times 2\pi (\because 2\pi R = \ell)$$

$$= \frac{\mu_0}{4\pi} \frac{I}{l} \times (2\pi)^2$$

Case (2):



$$B_B = 4 \times \frac{\mu_0}{4\pi} \frac{I}{a/2} [\sin 45^\circ + \sin 45^\circ]$$

$$= 4 \times \frac{\mu_0}{4\pi} \times \frac{I}{l/8} \times \frac{2}{\sqrt{2}} = \frac{\mu_0 I}{4\pi l} \times \frac{64}{\sqrt{2}}$$

$$= \frac{\mu_0 I}{4\pi l} 32\sqrt{2} \quad [4a = l]$$

$$\Rightarrow \frac{B_A}{B_B} = \frac{\pi^2}{8\sqrt{2}}$$

64. (2)

$$\text{For } B = \frac{\mu_0 n I}{2a}$$

where, a is the radius of loop.

$$\text{Then, } B_1 = \frac{\mu_0 I}{2a}$$

$$\text{Now, for coil } B = \frac{\mu_0 I}{4\pi} \cdot \frac{2nA}{x^3}$$

at the centre x = radius of loop

$$B_2 = \frac{\mu_0}{4\pi} \frac{2 \times 3 \times (I/3) \times \pi(a/3)^2}{(a/3)^3} = \frac{\mu_0 \cdot 3I}{2a}$$

$$\therefore \frac{B_1}{B_2} = \frac{\mu_o I / 2a}{\mu_o \cdot 3I / 2a} = 1/3$$

$$B_1 : B_2 = 1 : 3$$

65. (1)

Magnetic field at any point lies on axial position of current carrying conductor $B = 0$

66. (2)

Given: Radius = R

Distance $x = 2\sqrt{2}R$

$$\frac{B_{\text{centre}}}{B_{\text{axis}}} = \left(1 + \frac{x^2}{R^2}\right)^{3/2} = \left(1 + \frac{(2\sqrt{2}R)^2}{R^2}\right)^{3/2} = (9)^{3/2} = 27$$

67. (1)

The magnetic field due to a disc is given as

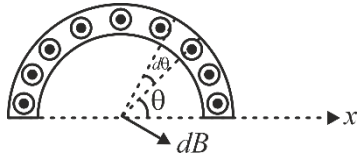
$$B = \frac{\mu_o \omega Q}{2\pi R} \quad \text{i.e., } B \propto \frac{1}{R}$$

68. (4)

Let R be the radius of semicircular ring. Let an elementary length dl is cut for finding magnetic field. So,

$$\text{Current in a small element, } dI = \frac{d\theta}{\pi} I$$

Magnetic field due to the element



$$dB = \frac{\mu_o}{4\pi} \frac{2dI}{R} = \frac{\mu_o I}{2\pi^2 R}$$

The component $dB \sin \theta$, of the field is cancelled by another opposite component.

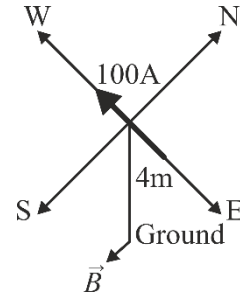
Therefore,

$$B_{\text{net}} = 2 \int dB \cos \theta = \frac{\mu_o I}{\pi^2 R} \int_0^{\pi/2} \cos \theta d\theta = \frac{\mu_o I}{\pi^2 R}$$

69. (1)

The magnetic field varies inversely with the distance for a long conductor. That is, $B \propto \frac{1}{d}$ so, graph in option (1) is the correct one.

70. (3)



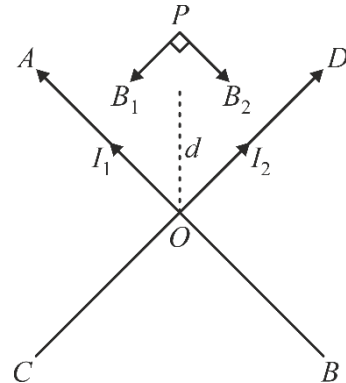
The magnetic field is

$$B = \frac{\mu_o}{4\pi} \frac{2I}{r} = 10^{-7} \times \frac{2 \times 100}{4} = 5 \times 10^{-6} \text{ T}$$

[$\because$ Since power line will behave as infinite conductor]
Current flows from east to west. Point is below the power line, using right hand thumb rule, the magnetic field is directed towards south.

71. (3)

The direction of magnetic field induction due to current through AB and CD at P are indicated as B_1 and B_2 . The magnetic fields at a point P , equidistant from AOB and COD will have directions perpendicular to each other, as they are placed normal to each other.



Magnetic field at P due to current through AB ,

$$B_1 = \frac{\mu_o I_1}{2\pi d}$$

Magnetic field at P due to current through CD ,

$$B_2 = \frac{\mu_o I_2}{2\pi d}$$

$$\therefore \text{Resultant field, } B = \sqrt{B_1^2 + B_2^2}$$

$$\therefore B = \sqrt{\left(\frac{\mu_o}{2\pi d}\right)^2 (I_1^2 + I_2^2)}$$

$$\text{or } B = \frac{\mu_o}{2\pi d} (I_1^2 + I_2^2)^{1/2}$$

72. (1)

Magnetic field due to long solenoid is given by $B = \mu_0 n I$. In first case $B_1 = \mu_0 n_1 I_1$.

In second case, $B_2 = \mu_0 n_2 I_2$

$$\therefore \frac{B_2}{B_1} = \frac{\mu_0 n_2 I_2}{\mu_0 n_1 I_1} \Rightarrow \frac{B_2}{6.28 \times 10^{-2}} = \frac{100 \times \frac{i}{3}}{200 \times i}$$

$$\Rightarrow B_2 = \frac{6.28 \times 10^{-2}}{6} = 1.05 \times 10^{-2} \text{ Wb/m}^2$$

73. (2)

The circumference of the first loop $= 2\pi R$. If it is bent into n circular coil of radius r .

$$n \times (2\pi r) = 2\pi R$$

$$\Rightarrow nr = R \quad \dots(i)$$

$$\text{New magnetic field, } B' = \frac{n \cdot \mu_0 i}{2r} \quad \dots(ii)$$

From (i) and (ii),

$$B' = \frac{n \mu_0 i \cdot n}{2R} = n^2 B$$

74. (3)

The magnetic field at a point on the axis of a circular loop at a distance x from centre is,

$$B = \frac{\mu_0 i a^2}{2(x^2 + a^2)^{3/2}}$$

Magnetic field at the centre of loop is

$$B' = \frac{\mu_0 i}{2a} \Rightarrow B' = \frac{B \cdot (x^2 + a^2)^{3/2}}{a^3}$$

$$\text{Put } x = 4 \text{ \& } a = 3 \Rightarrow a = 3 \Rightarrow B' = \frac{54(5^3)}{3 \times 3 \times 3} = 250 \mu\text{T}$$

75. (1)

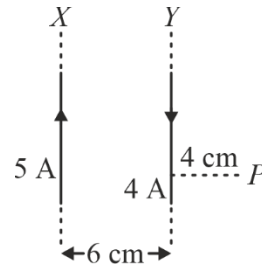
Magnetic field induction at the centre of current carrying circular coil of radius r is

$$B = \frac{\mu_0 I}{4\pi R} \times 2\pi$$

$$\text{Here } B_A = \frac{\mu_0 I}{4\pi R} \times 2\pi$$

$$\text{and } B_B = \frac{\mu_0 2 I}{4\pi 2R} \times 2\pi \Rightarrow \frac{B_A}{B_B} = \frac{I/R}{2I/2R} = 1.$$

76. (1)



The net magnetic field at point P is $B_p = B_y - B_x$

$$B_p = \frac{\mu_0 (4)}{2\pi \times 0.04} - \frac{\mu_0 (5)}{2\pi \times 0.1} = \frac{100\mu_0}{4\pi}$$

$$= 100 \times 10^{-7} = 1 \times 10^{-5} \text{ T}$$

77. (160)

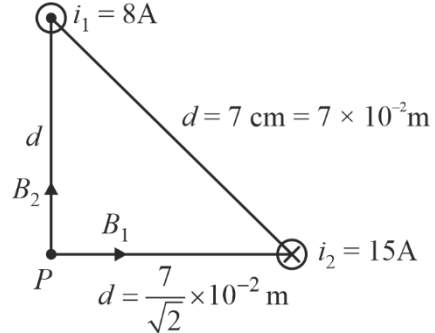
Magnetic field midway between the wires, B_p

$$= \left(\frac{\mu_0 i}{2\pi a} \right) \times 2 = \frac{4 \times 10^{-7} \times 10}{\pi \times \left(\frac{5}{2} \times 10^{-2} \right)} = 16 \times 10^{-5} = 160 \mu\text{T}$$

78. (68)

Magnetic field due to both wires will be perpendicular to each other

$$B_1 = \frac{\mu_0 i_1}{2\pi d} \quad B_2 = \frac{\mu_0 i_2}{2\pi d}$$



$$B_{net} = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0}{2\pi d} \sqrt{i_1^2 + i_2^2}$$

$$= \frac{4\pi \times 10^{-7}}{2\pi \times (7/\sqrt{2}) \times 10^{-2}} \times \sqrt{8^2 + 15^2}$$

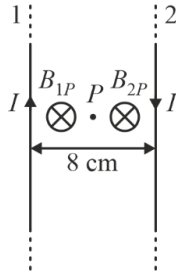
$$= 68 \times 10^{-6} \text{ T}$$

79. (2)

Here, current cannot be in same direction in both wire because $B_p = 0$. So current should be in opposite direction.

$$B_p = B_{1p} + B_{2p}$$

$$B_p = 2 \frac{\mu_0 I}{2\pi r}$$



$$300 \times 10^{-6} = 2 \times 2 \times 10^{-7} \times \frac{I}{4 \times 10^{-2}}$$

$$3 \times 10^{-4} = 10^{-5} \times I$$

$$I = 30 \text{ A}$$

80. (1)

The magnetic field at O due to current in DA is

$$B_1 = \frac{\mu_0 I}{4\pi a} \times \frac{\pi}{6} \text{ (directed vertically downwards)}$$

The magnetic field at O due to current in BC is

$$B_2 = \frac{\mu_0 I}{4\pi b} \times \frac{\pi}{6} \text{ (directed vertically downwards)}$$

The magnetic field due to current AB and CD at O is

zero Therefore the net magnetic field is

$B = B_1 - B_2$ (directed vertically upwards)

$$= \frac{\mu_0 I}{4\pi a} \times \frac{\pi}{6} - \frac{\mu_0 I}{4\pi b} \times \frac{\pi}{6} = \frac{\mu_0 I}{24} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{\mu_0 I}{24ab} (b - a)$$

81. (4)

$$\vec{F} = I(\vec{l} \times \vec{B})$$

The force on AD and BC due to current I_1 is zero.

This is because the directions of current element $I\vec{dl}$ and magnetic field $\vec{B}$ are parallel

82. (4)

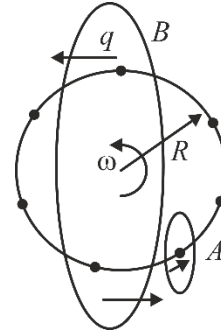
The field is maximum at the surface ($r = R$) and decreases when moving further away. Hence, statement C is incorrect.

The field distribution depends on the distance from the axis, not the length/midpoint of the conductor. Hence, statement (A) is incorrect. Since, the field varies linearly inside and inversely outside; it is not uniform. Hence, option E is incorrect.

Therefore, only statement D is correct.

Hence, the correct option is (4).

83. (3)



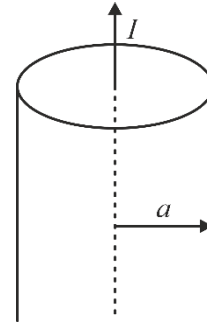
For given Amperian loops A and B,

$$I_A = \frac{Nq}{T} = \frac{Nq}{\frac{2\pi}{\omega}} = \frac{Nq\omega}{2\pi}$$

From loop B, Incoming current is equal to outgoing current, So, $I_B = 0$

$$\therefore I_A - I_B = \frac{Nq\omega}{2\pi}$$

84. (1)



For given straight circular wire,

$$B_{\text{in}} = \left(\frac{\mu_0 I}{2\pi R^2} \right) r$$

$$\therefore B_{\text{in}} \propto r \text{ for } r < R$$

$$B_{\text{out}} = \frac{\mu_0 I}{2\pi r}$$

$$\therefore B_{\text{net}} \propto \frac{1}{r} \text{ for } r > R$$

85. (3)

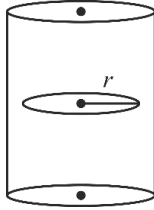
Using Ampere's circuital law, magnetic field for long straight wire of circular cross-section is given by

$$B = \frac{\mu_0 I r}{2\pi a^2} \text{ (For } r < a) \therefore B \propto r$$

For $r > a$

$$B = \frac{\mu_0 I}{2\pi r} \therefore B \propto \frac{1}{r}$$

86. (2)



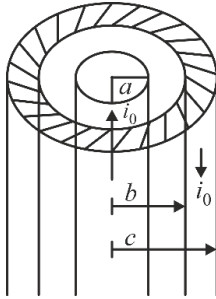
Let us draw an amperian loop radius 'r' as shown then.

By Ampere's law

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 I_{\text{in}}$$

$$\Rightarrow B \times 2\pi r = \mu_0 I \frac{\pi r^2}{\pi R^2} \times \pi r^2 \Rightarrow B = \frac{\mu_0 I r}{2\pi R^2} \Rightarrow B \propto r$$

87. (1)



For $x < a$

Using ampere's circuital law

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 (\text{current enclosed})$$

$$\Rightarrow B_1 (2\pi x) = \mu_0 \left(\frac{i_0}{\pi a^2} \right) \pi x^2$$

$$\Rightarrow B_1 (2\pi x) = \frac{\mu_0 i_0 x^2}{a^2}$$

$$\Rightarrow B_1 = \frac{\mu_0 i_0 x}{2\pi a^2} \quad \dots(i)$$

For $a < x < b$

$$B_2 (2\pi x) = \mu_0 i_0 \Rightarrow B_2 = \frac{\mu_0 i_0}{2\pi x} \quad \dots(ii)$$

$$\therefore \frac{B_1}{B_2} = \frac{\mu_0 i_0 \frac{x}{2\pi a^2}}{\frac{\mu_0 i_0}{2\pi x}} = \frac{x^2}{a^2}$$

88. (4)

There is no current inside the pipe. From Ampere's circuital law

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 I \quad [I = 0]$$

$$\therefore B = 0$$

89. (2)

From Ampere's circuital law

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 i \Rightarrow B \times 2\pi r = \mu_0 i$$

Here i is zero, for $r < R$, whereas R is the radius.

$$\therefore B = 0$$

90. (2)

Wires P and Q carry currents in opposite directions (3 A up and 1 A down). Parallel wires with opposite currents repel each other. Therefore, wire P pushes wire Q to the right (towards wire R) with a force

$$F_{PQ}$$

$$\text{Given } I_P = 3 \text{ A}, I_Q = 1 \text{ A}, r_{PQ} = 3 \text{ cm} = 0.03 \text{ m}$$

So, the magnitude of force on 15 cm of wire Q due to wire P is,

$$F_{PQ} = \frac{(4\pi \times 10^{-7}) \times 3 \times 1 \times 0.15}{2\pi \times 0.03} \text{ N} = 3 \times 10^{-6} \text{ N}$$

Wires Q and R carry currents in the same direction (1 A down and 2 A down). Parallel wires with same direction currents attract each other. Therefore, wire R pulls wire Q to the right (towards wire R) with force F_{RQ} .

$$\text{Given that } I_R = 2 \text{ A}, I_Q = 1 \text{ A}, r_{RQ} = 2 \text{ cm} = 0.02 \text{ m}$$

$$F_{RQ} = \frac{(4\pi \times 10^{-7}) \times 2 \times 1 \times 0.15}{2\pi \times 0.02}$$

$$\Rightarrow F_{RQ} = \frac{2 \times 10^{-7} \times 0.30}{0.02} = 2 \times 10^{-7} \times 15$$

$$= 3 \times 10^{-6} \text{ N}$$

Since both forces acting on wire Q are directed towards the right, the net force will be towards R.

$$F_{\text{net}} = F_{PQ} + F_{RQ}$$

$$\Rightarrow F_{\text{net}} = 3 \times 10^{-6} + 3 \times 10^{-6}$$

$$F_{\text{net}} = 6 \times 10^{-6} \text{ N}$$

Hence, the net force acting on wire Q is 6×10^{-6} N towards R.

91. (48)

The magnetic force on a current carrying wire placed in magnetic field is $F = I l B$

Here, $I = 8$ A, $B = 0.15$ T, $l = 4$ cm $= 4 \times 10^{-2}$ m

$$\therefore F = 8 \times \frac{4}{100} \times 0.15 = 48 \times 10^{-3} \text{ N} = 48 \text{ mN}$$

92. (5)

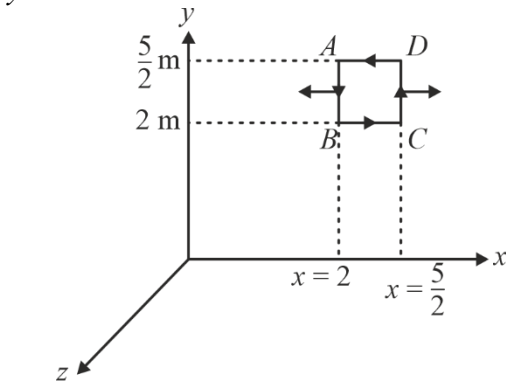
At equilibrium, $F_{\text{magnetic}} = mg$

$$\Rightarrow iLB = mg \text{ and } L = \frac{AR}{\rho}$$

$$\therefore B = \frac{mg\rho}{iAR} = \frac{0.5 \times 10 \times 2 \times 10^{-6}}{2 \times 10 \times 10^{-6} \times 1} = 0.5 \text{ T} = 5 \times 10^{-1} \text{ T}$$

93. (50)

Force on segment parallel to x-axis will cancel each other. Hence F_{net} will be due to portion parallel to y-axis



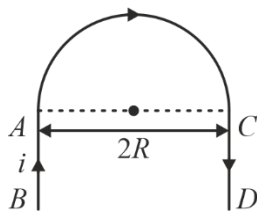
$$\vec{F}_{BC} + \vec{F}_{DA} = 0$$

$$\vec{F}_{AB} = i l B = 0.5 \times 0.5 (0.2 \times 5) = 0.25 \text{ N}$$

$$\vec{F}_{CD} = 0.5 \times 0.5 (0.2 \times 6) = 0.3 \text{ N}$$

$$\vec{F}_{\text{net}} = 0.05 \text{ N} = 50 \text{ mN}$$

94. (1)



Force due to AB and CD will cancel each other magnetic field is in $+\hat{k}$ direction

Effective length of semicircular loop, $\vec{l}_{\text{eff}} = 2R\hat{i}$

$$\vec{F} = i\vec{l} \times \vec{B}$$

$$\Rightarrow \vec{F} = i\vec{l}_{\text{eff}} B$$

$$\Rightarrow \vec{F} = i(2R\hat{i} \times B_0\hat{k}) = -2iBR\hat{j}$$

95. (35)

Magnetic field, $B_H = 3.5 \times 10^{-5}$ T, Current, $i = \sqrt{2}$ A

Force on the conductor, $F = i l B \sin \theta$

$$\frac{F}{l} = i B \sin 45^\circ = \sqrt{2} \times 3.5 \times 10^{-5} \times \frac{1}{\sqrt{2}}$$

$$= 35 \times 10^{-6} \text{ N/m}$$

96. (2)

For equilibrium, force $Mg = I l B$

$$\therefore I = \frac{mg}{lB} = \frac{40 \times 10^{-3} \times 10}{50 \times 10^{-2} \times 0.4} = 2 \text{ A}$$

97. (4)

Given that resistance of wire, $R = 10 \Omega$

Supporting mass $m = 10^{-3}$ kg

Magnetic field, $B = 10^3 \text{ G} = 10^3 \times 10^{-4} \text{ T}$

From the question magnetic force = weight

($\because V = IE$ using Ohm's Law)

$$F_m = mg \therefore I l B = mg$$

$$\therefore \left(\frac{V}{R} \right) l B = mg \therefore V = \frac{mgR}{lB}$$

$$= \frac{(1 \times 10^{-3} \text{ kg})(10 \text{ m/s}^2)(10 \Omega)}{(0.1 \text{ m})(10^3 \times 10^{-4} \text{ T})} = 10 \text{ V}$$

98. (2)

Magnetic force per unit length on wire PQ due to wire RS is given by

$$\frac{F}{l} = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{r} \therefore F_1 \propto I_1 I_2 \propto (I)^2$$

When ammeter current reading is changed from I to 2I.

$$\therefore F_1 : F_2 = 1 : 4$$

99. (1)

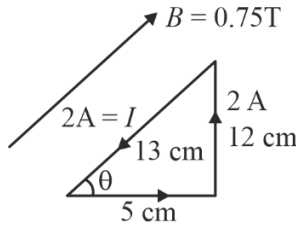
We have

$$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} \Rightarrow F = \frac{\mu_0}{2\pi} \frac{I_1 I_2 l}{d}$$

So, $F \propto \frac{I_1 I_2}{d}$

$$\frac{F_2}{F_1} = \frac{\left(\frac{2I_1 \cdot 2I_2}{d/2}\right)}{\left(\frac{I_1 I_2}{d}\right)} = 8 \Rightarrow F_2 = 8F_1$$

100. (9)



$$\vec{F} = I(\vec{L} \times \vec{B})$$

Force on 5 cm side is

$$|\vec{F}| = ILB \sin \theta$$

$$= (2)(5 \times 10^{-2}) \times \frac{3}{4} \times \frac{12}{13} = \frac{9}{130} \text{ N} = \frac{x}{130} \text{ N (given)}$$

So, $x = 9$

101. (1)

Given, length of wire X, $l_1 = 50 \text{ cm} = 0.5 \text{ m}$

Length of wire Y, $l_2 = 5 \text{ m}$

Distance between two wire, $r = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$

Force of interaction $= I_1 l_1 B_{12}$

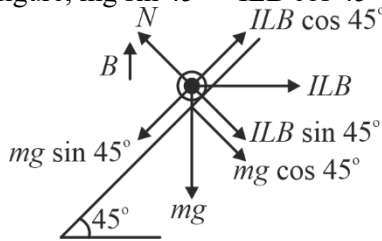
$$= \frac{\mu_0 I_1 I_2}{2\pi r} \quad \left(\because B_{12} = \frac{\mu_0 I_2}{2\pi r} \right)$$

$$= \frac{4\pi \times 10^{-7} \times 6 \times 0.5}{2\pi \times 5 \times 10^{-2}} = 1.2 \times 10^{-5} \text{ N towards X}$$

102. (1)

Force, $F = ILB \sin \theta$

From figure, $mg \sin 45^\circ = ILB \cos 45^\circ$



$$\therefore I = \left(\frac{m}{L}\right) \frac{g}{B}$$

$$= \frac{(0.45)(10)}{0.15} = 30 \text{ A}$$

103. (3)

$$F_M \text{ on CD} = BIl_{\text{eff}}$$

$$= 0.5 \times (10) \times (5 \sin 60 \times 10^{-2}) = 0.216 \text{ N}$$

104. (3)

$$\text{We know that, } \frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d}$$

$$\Rightarrow 2 \times 10^{-6} = 2 \times 10^{-7} \times \frac{x^2}{0.2}$$

$$\Rightarrow 2 = x^2 \Rightarrow x = \sqrt{2} = 1.4$$

105. (5)

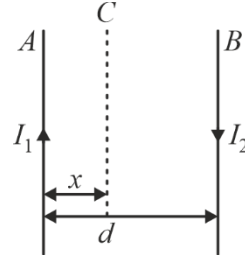
$$\text{We have } \frac{F}{l} = \frac{\mu_0}{2\pi} \frac{i_1 i_2}{r} \Rightarrow r = \frac{\mu_0}{2\pi} i_1 i_2 \times \frac{l}{F}$$

$$\Rightarrow r = 2 \times 10^{-7} \times 25 \times \frac{0.1}{10^{-5}}$$

$$= 50 \times 10^{-7} \times 10^4 = 50 \times 10^{-3} \text{ m} = 5 \text{ cm}$$

106. (4)

As net force on the third wire C is zero.



$$\Rightarrow F = \frac{\mu_0 I_1}{2\pi x} + \frac{\mu_0 I_2}{2\pi(d-x)} = 0$$

$$\Rightarrow \frac{\mu_0 I_1}{2\pi x} = \frac{\mu_0 I_2}{2\pi(x-d)}$$

$$\Rightarrow I_1 x - I_1 d = I_2 x$$

$$x = \frac{I_1 d}{I_1 - I_2} \text{ or } \frac{I_1 d}{I_2 - I_1}$$

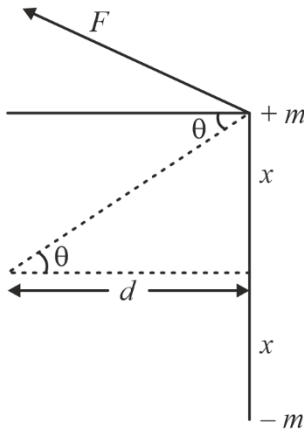
Two cases may be possible if $I_1 > I_2$ or $I_2 > I_1$

107. (3)

$$F = \frac{\mu_0}{2\pi} \left(\frac{i_1 i_2}{a} - \frac{i_1 i_2}{2a} \right) \times a = \frac{\mu_0 i_1 i_2}{4\pi}$$

108. (4)

Force on one pole,



$$F = m \times \frac{\mu_0 I}{2\pi\sqrt{d^2 + x^2}}$$

Total force, $F_{\text{total}} = 2F \sin \theta$

$$= 2 \times \frac{\mu_0 I m}{2\pi\sqrt{d^2 + a^2}} \times \frac{x}{\sqrt{d^2 + a^2}}$$

$$= \frac{\mu_0 I m x}{\pi(d^2 + a^2)}$$

Magnetic moment $M = I\pi a^2 = m \times 2x$

$$\therefore m = \frac{I\pi a^2}{2x}$$

Or, Total force,

$$F_{\text{total}} = \frac{\mu_0 I^2 a^2}{2(d^2 + a^2)} = \frac{\mu_0 I^2 a^2}{2d^2} [\because d \gg a]$$

$$\text{Clearly } F_{\text{total}} \propto \frac{a^2}{d^2}$$

109. (3)

$\vec{F}_1 = \vec{F}_2 = 0$ because of action and reaction pair

110. (1)

$I_1 I_1 = \text{Positive (attract)}$

$F = \text{Negative}$

$I_1 I_2 = \text{Negative (repell)}$

$F = \text{Positive}$

Hence, Option (1) is the correct answer

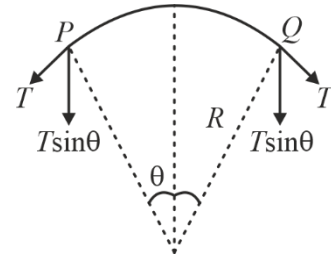
111. (3)

For small arc length

$2T \sin \theta = BIR \cdot 2\theta$ (As $F = BIL$ and $L = R2\theta$)

For small θ , $\sin \theta \approx \theta$

$\therefore T = BIR$

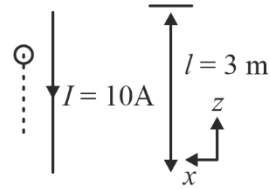


112. (2)

Work done in moving the conductor is,

$$W = \int_0^2 F dx = \int_0^2 3.0 \times 10^{-4} e^{-0.2x} \times 10 \times 3 dx$$

$$= 9 \times 10^{-3} \int_0^2 e^{-0.2x} dx$$



$$= \frac{9 \times 10^{-3}}{0.2} [-e^{-0.2 \times 2} + 1] = \frac{9 \times 10^{-3}}{0.2} \times [1 - e^{-0.4}]$$

$$= \frac{9 \times 10^{-3} \times (0.33)}{2} = \frac{2.97 \times 10^{-3}}{2}$$

Power required to move the conductor is,

$$P = \frac{W}{t} \Rightarrow P = \frac{2.97 \times 10^{-3}}{(0.2) \times 5 \times 10^{-3}} = 2.97 W$$

113. (1)

Force acting on conductor B due to conductor A is given by relation

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$$

l -length of conductor B

r -distance between two conductors

$$\therefore F = \frac{4\pi \times 10^{-7} \times 10 \times 2 \times 2}{2 \times \pi \times 0.1} = 8 \times 10^{-5} N$$

114. (1)

Force acting between two long conductor carrying current,

$$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d} \times l \quad \dots\dots(i)$$

Where d = distance between the conductors

l = length of conductor

In second case, $F' = \frac{\mu_0}{4\pi} \frac{2(2I_1)I_2}{3d} l$ (ii)

From equation (i) and (ii), we have

$$\therefore \frac{F'}{F} = \frac{-2}{3}$$

115. (2)

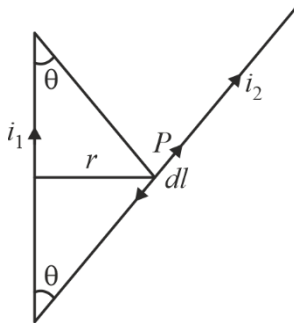
When current is passed through a spring then current flows parallel in the adjacent turns in the same direction. As a result the various turn attract each other and spring get compress.

116. (3)

Magnetic field due to current in wire 1 at point P distant r from the wire is

$$B = \frac{\mu_0}{4\pi} \frac{i_1}{r} [\cos \theta + \cos \theta]$$

$$B = \frac{\mu_0}{2\pi} \frac{i_1 \cos \theta}{r}$$



This is magnetic field is directed

Perpendicular to the plane of paper, inwards.

The force exerted due to this magnetic field on current element $i_2 dl$ is

$$dF = i_2 dl B \sin 90^\circ$$

$$\therefore dF = i_2 dl B$$

$$\Rightarrow dF = i_2 dl \left(\frac{\mu_0}{2\pi} \frac{i_1 \cos \theta}{r} \right)$$

$$= \frac{\mu_0}{2\pi r} i_1 i_2 dl \cos \theta$$

117. (4)

The magnitude of the magnetic force is the cross product of $Q(\vec{v} \times \vec{B})$,

$$F_m = QvB \sin \theta$$

Since the angle θ between the velocity and the magnetic field is 0° :

$$F_m = QvB \sin 0^\circ$$

The magnetic field exerts no force on the particle moving along the axis.

The solenoid is placed vertically. When the particle of mass m is released, the only non-zero force acting on it is the force of gravity ($\vec{F}_g$):

$$\vec{F}_g = mg$$

This force acts vertically downwards along the axis of the solenoid.

According to Newton's Second Law ($F_{\text{net}} = ma$):

$$mg = ma$$

$$\Rightarrow a = g$$

118. 2

The distance moved along the field lines in 1 revolution is called the Pitch.

$$\text{Pitch} = v_{\parallel} \times T$$

For 5 revolutions, the total distance (a) is:

$$a = 5 \times v_{\parallel} \times T = 5 \times v_{\parallel} \times \left(\frac{2\pi m}{qB} \right)$$

The mass of charged particle is $m = 5mg = 5 \times 10^{-6} \text{ kg}$.

The magnitude of charge on the particle is $q = 5\pi \times 10^{-6} \text{ C}$

The parallel component of velocity to magnetic field is $v_{\parallel} = 2 \times 10^{-2} \text{ m/s}$

The magnitude of magnetic field in the region is $B = 0.1 \text{ T}$

So, the distance travelled is;

$$a = 5 \times (2 \times 10^{-2}) \times \frac{2\pi \times (5 \times 10^{-6})}{(5\pi \times 10^{-6}) \times 0.1} = 2\text{m}$$

Therefore, the distance a moved along the $\hat{k}$ direction is 2 meters.

119. 12

If two vectors are perpendicular, their dot product must be zero:

$$\vec{a} \cdot \vec{B} = 0$$

The magnetic field is given as $\vec{B} = 3\hat{i} + 2\hat{j}$ and

$$\vec{a} = 4\hat{i} - \frac{x}{2}\hat{j} :$$

$$\left(4\hat{i} - \frac{x}{2}\hat{j}\right) \cdot (3\hat{i} + 2\hat{j}) = 0$$

$$\Rightarrow (4 \times 3) + \left(-\frac{x}{2} \times 2\right) = 0$$

$$\Rightarrow 12 - x = 0$$

$$\Rightarrow x = 12$$

Therefore, the value of x is 12.

120. 171

The magnetic force acting on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given by:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\vec{F} = q \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\vec{F} = 1 \times 10^{-6} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 3 & -5 \end{vmatrix}$$

$$\begin{aligned} \vec{F} &= 1 \times 10^{-6} [\hat{i}((-2)(-5) - (3)(3)) - \hat{j} \\ &[(1)(-5) - (3)(2)] + \hat{k}[(1)(3) - (-2)(2)]] \\ \vec{F} &= 1 \times 10^{-6} [\hat{i}(10 - 9) - \hat{j}(-5 - 6) + \hat{k}(3 + 4)] \end{aligned}$$

$$\vec{F} = 1 \times 10^{-6} [1\hat{i} + 11\hat{j} + 7\hat{k}]$$

So, the magnitude $|\vec{F}|$ is:

$$|\vec{F}| = 10^{-6} \sqrt{1^2 + 11^2 + 7^2}$$

$$|\vec{F}| = 10^{-6} \sqrt{1 + 121 + 49}$$

$$|\vec{F}| = \sqrt{171} \times 10^{-6} \text{ N}$$

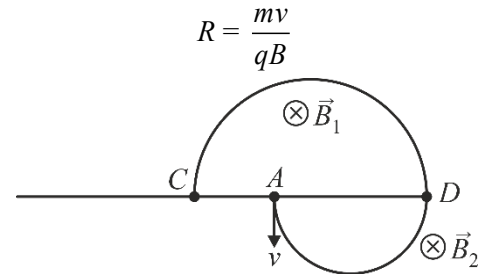
$$\Rightarrow \sqrt{\alpha} \times 10^{-6} \text{ N} = \sqrt{171} \times 10^{-6} \text{ N}$$

$$\Rightarrow \alpha = 171$$

Therefore, the value of α is 171. Thus, the correct answer is 171.

121. (4)

The radius of circular path for a charged particle moving perpendicularly in a magnetic field is



The displacement of the charge particle is

$$\begin{aligned} AC &= CD - AD = 2R_1 - 2R_2 \\ &= \frac{2mv}{qB_1} - \frac{2mv}{qB_2} = \frac{2mv}{qB_1} \left[1 - \frac{B_1}{B_2}\right] \end{aligned}$$

122. (4)

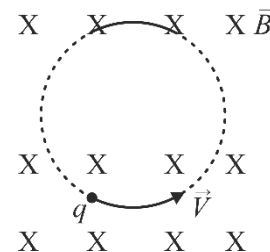
The radius of circular arc for a charge particle enters in magnetic field perpendicularly is

$$r = \frac{mv}{Bq} = \frac{\sqrt{2mE}}{Bq}$$

$$E = \left(\frac{q^2 B^2}{2m}\right) r^2 \quad \therefore E \propto r^2$$

Hence, r vs E graph is

123. (0 or 10)



When a charged particle moves perpendicular to a magnetic field, it undergoes circular motion, and the time it takes to complete one full revolution is Time period.

$$T = \frac{2\pi m}{qB} = \frac{2\pi \times 16 \times 10^{-9} \text{ kg}}{1.6 \times 10^{-6} \times 6.28}$$

$$T = 0.01 \text{ seconds}$$

NTA Answer is 10.

Correct Answer is 0 (nearest integer).

124. (1)

The radius of curvature of a charged particle in magnetic field is

$$r = \frac{mv}{qB} = \frac{P}{qB} \Rightarrow r \propto \frac{1}{q} \text{ (For same momentum } p)$$

In first case, $q_0 > qH \Rightarrow r_o = r_H$
 In second case, $q_e = q_p \Rightarrow r_e = r_p$
 $\therefore$ Assertion is true reason is false.

125. (2)

Magnetic force, $\vec{F} = q(\vec{v} \times \vec{B}) = 0$
 $\Rightarrow qvB \sin\theta = 0 \therefore \theta = 0^\circ$ or 180° .

126. (3)

In the uniform magnetic field, radius of the circular path is given by;

$$R = \frac{mv}{qB} = \frac{\sqrt{2m(\text{K.E})}}{qB}$$

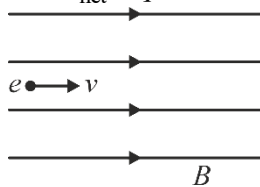
$$\Rightarrow R \propto \frac{\sqrt{m}}{q}; (\because \text{KE is same})$$

$$\therefore \frac{R_{\text{deuteron}}}{R_{\text{proton}}} = \frac{\sqrt{m_d}}{\sqrt{m_p}} \times \frac{q_p}{q_d} \because m_d = 2m_p \text{ and } q_d = q_p$$

$$\therefore \frac{R_d}{R_p} = \sqrt{\frac{2m_p}{m_p}} \times \frac{q_p}{q_p} = \sqrt{2} \therefore R_d : R_p = \sqrt{2} : 1$$

127. (2)

Electron is moving along the direction of magnetic field, i.e., $\vec{v} \parallel \vec{B}$ so force on electron due to magnetic field is zero i.e., $F_{\text{net}} = qvB \sin\theta = qvB \sin 0^\circ = 0$.



So it will move along the axis with uniform velocity.

128. (5)

Magnetic force acting on electron in a uniform magnetic field is given by

$$\begin{aligned} \vec{F} &= q(\vec{v} \times \vec{B}) \\ \Rightarrow 5e\hat{k} &= e(3\hat{i} + 5\hat{j}) \times (B_0\hat{i} + 2\hat{j}) \\ \Rightarrow 5e\hat{k} &= e(6B_0\hat{k} - 5B_0\hat{k}) \Rightarrow B_0 = 5T \end{aligned}$$

129. (4)

The radius of circular path is

$$R = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2m(\text{KE})}}{qB} = \frac{\sqrt{2mqV}}{qB}$$

$$\Rightarrow m \propto R^2 \therefore \frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)^2$$

130. (32)

$$q = 4 \mu\text{C}, \vec{v} = 4 \times 10^6 \hat{j} \text{ m/s}, \vec{B} = 2\hat{k} \text{ T}$$

The force acting on charge is

$$\vec{F} = q(\vec{v} \times \vec{B}) = 4 \times 10^{-6} (4 \times 10^6 \hat{j} \times 2\hat{k})$$

$$= 4 \times 10^{-6} \times 8 \times 10^6 \hat{i}$$

$$\vec{F} = 32\hat{i} \text{ N} = x\hat{i} \text{ N (given)}$$

$$\therefore x = 32$$

131. (1)

Magnetic force, $\vec{F} = -e(\vec{v} \times \vec{B})$ counters centripetal force. As magnetic force is $\perp$ to velocity, path of electron must be a circle. Force will be along -ve y-axis.

132. (2)

$$\vec{F} = q(\vec{v} \times \vec{B}) = qvB \sin\theta$$

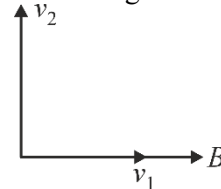
As $\vec{v}$ is parallel to $\vec{B}$, angle between $\vec{v}$ and $\vec{B}$ is 0° .

$$\therefore \vec{F} = 0.$$

Hence, electron will continue to move with constant velocity.

133. (3)

Let v_1 is component of velocity parallel to the magnetic field and v_2 is the component of velocity perpendicular to the magnetic field.



Due to component v_1 , magnetic force $F = qv_1B \sin\theta = 0$ ($\theta = 0$)

So v_1 remains unchanged, but due to component v_2 magnetic force act towards centre i.e. moving it along circular path with forward motion. So path is helical with the axis parallel to magnetic field B .

134. (40)

Given,

$$\text{Magnetic field, } B = \frac{\pi}{2} \times 10^{-3} \text{ T}$$



Kinetic energy of proton, $KE = 2 \text{ eV}$

Mass of proton, $m = 1.6 \times 10^{-27} \text{ kg}$

Charge on proton, $e = 1.6 \times 10^{-19} \text{ C}$

Pitch = $v \parallel \times$ time period of one rotation

$$= v \cos 60^\circ \times \frac{2\pi m}{eB} \quad (\because \theta = 60^\circ)$$

$$\therefore KE = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2KE}{m}}$$

$$\therefore \text{Pitch} = \sqrt{\frac{2KE}{m}} \cos 60^\circ \times \frac{2\pi m}{eB}$$

$$= \sqrt{\frac{2 \times 2 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-27}}} \times \cos 60^\circ \times \frac{2\pi \times 1.6 \times 10^{-27}}{1.6 \times 10^{-19} \times \frac{\pi}{2} \times 10^{-3}}$$

$$= 2 \times 10^4 \times \frac{1}{2} \times 4 \times 10^{-5} = 4 \times 10^{-1} \text{ m} = 40 \text{ cm}$$

135. (144)

Given,

Charge, $q = 2\mu\text{C} = 2 \times 10^{-6} \text{ C}$

Magnetic field, $B = 4 \times 10^{-3} \text{ T}$

Radius $r = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$.

$$r = \frac{mv}{qB} = \frac{\sqrt{2Km}}{qB}, m = \frac{r^2 q^2 B^2}{2K}$$

Kinetic energy $K = qV = 2 \times 10^{-6} \times 100$

$$m = \frac{(3 \times 10^{-2})^2 \times (2 \times 10^{-6})^2 \times (4 \times 10^{-3})^2}{2 \times 10^{-6} \times 100}$$

$$= 144 \times 10^{-18} \text{ kg}$$

136. (2)

Given that uniform magnetic field, $\vec{B} = (2\hat{i} + 3\hat{j}) \text{ T}$

Acceleration $\vec{a} = (\alpha\hat{i} - 4\hat{j}) \text{ ms}^{-2}$

We know that $\vec{F} = q(\vec{v} \times \vec{B}) \Rightarrow m\vec{a} = q(\vec{v} \times \vec{B})$

Here, $\vec{a} \perp \vec{B}$, so, $\vec{a} \cdot \vec{B} = 0$

$$\alpha\hat{i} - 4\hat{j} \cdot (2\hat{i} + 3\hat{j}) = 0 \Rightarrow 2\alpha - 12 = 0 \Rightarrow \alpha = 6$$

137. (2)

$$\text{As } R = \frac{mv}{qB} = \frac{\sqrt{2m(\text{K.E})}}{qB} \Rightarrow R \propto \frac{\sqrt{m}}{q}$$

$$\Rightarrow \frac{q_1}{q_2} = \frac{\sqrt{m_1}}{\sqrt{m_2}} \times \frac{R_2}{R_1} = \sqrt{\frac{9}{4}} \times \frac{5}{6} = \frac{3}{2} \times \frac{5}{6} = \frac{5}{4}$$

138. (3)

$$f = \frac{1}{T} = \frac{eB}{2\pi m} \quad \left[\because T = \frac{2\pi m}{qB} \right]$$

$$f = \frac{1.6 \times 10^{-19} \times 10^{-4}}{2\pi \times 9 \times 10^{-31}} = 2.8 \times 10^6 \text{ Hz}$$

139. (10)

We know that,

$$r = \frac{mV}{qB} = \frac{\sqrt{2m(\text{K.E})}}{qB}$$

$$= \frac{\sqrt{2 \times 24 \times 1.67 \times 10^{-27} \times 5000 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times 0.5}$$

$$= 0.1 \text{ m} = 10 \text{ cm}$$

140. (2)

$$\text{We have, } r = \frac{mv}{qB} = \frac{\sqrt{2m(\text{K.E})}}{qB}$$

$$r \propto \frac{\sqrt{m}}{q} \Rightarrow \frac{r_d}{r_p} = \frac{\sqrt{m_d}}{\sqrt{m_p}} \times \frac{q_p}{q_d} = \sqrt{\frac{2}{1}} \times \frac{1}{1} = \frac{\sqrt{2}}{1}$$

141. (3)

$$\text{As } R = \frac{mv}{qB}$$

$$\Rightarrow R \propto \frac{m}{q} \Rightarrow \frac{R_\alpha}{R_p} = \left(\frac{m_\alpha}{m_p} \right) \times \frac{q_p}{q_\alpha} \Rightarrow \frac{R_\alpha}{R_p} = \left(\frac{4}{1} \right) \times \frac{1}{2} = 2$$

142. (1)

In uniform magnetic field

$$\vec{F} = q(\vec{v} \times \vec{B}) \text{ So, } \vec{F} \perp \vec{v}.$$

So, speed and kinetic energy remains same.

143. (4)

Radius of circular path in uniform magnetic field is given by,

$$r = \frac{mv}{qB} = \frac{\sqrt{2m(\text{K.E})}}{qB}$$

As K.E constant

$$\text{So, } r \propto \frac{\sqrt{m}}{q}$$

$$\text{So, } r_p : r_d : r_\alpha = \frac{\sqrt{m}}{e} : \frac{\sqrt{2m}}{e} : \frac{\sqrt{4m}}{2e} = 1 : \sqrt{2} : 1$$

144. (2)

Given,

$$\frac{q_1}{q_2} = \frac{1}{2} \text{ \& } \frac{v_1}{v_2} = \frac{2}{3}$$

Radius of circular path,

$$R = \frac{mv}{qB} \text{ where,}$$

m = mass of charged particle

B = magnetic field

$$\frac{R_1}{R_2} = \frac{\frac{mv_1}{q_1 B}}{\frac{mv_2}{q_2 B}} = \frac{v_1}{v_2} \times \frac{q_2}{q_1} = \frac{2}{3} \times \frac{2}{1} = \frac{4}{3}$$

145. (2)

Radius of circular path in uniform magnetic field is

$$r = \frac{\sqrt{2mE}}{qB} \Rightarrow r \propto \frac{\sqrt{m}}{q}$$

$$m_\alpha = 2m_d \text{ and } q_\alpha = 2q_d$$

$$\therefore \frac{r_d}{r_\alpha} = \frac{\sqrt{m_d}}{\sqrt{m_\alpha}} \times \frac{2q_d}{2q_\alpha} = \sqrt{2}$$

146. (4)

Radius of circular path in uniform magnetic field is

$$r = \frac{\sqrt{2mK}}{qB} \quad (\because p = mV = \sqrt{2mK})$$

$$\Rightarrow \frac{r^2 q^2 B^2}{2m} = K$$

$$K_p = \frac{r_p^2 q_p^2 B^2}{2m_p} \text{ and } K_\alpha = \frac{r_\alpha^2 q_\alpha^2 B^2}{2m_\alpha}$$

$$\therefore \frac{K_p}{K_\alpha} = \frac{r_p^2 q_p^2 m_\alpha}{r_\alpha^2 q_\alpha^2 m_p} = \left(\frac{2}{1}\right)^2 \left(\frac{1}{2}\right)^2 \left(\frac{4}{1}\right)$$

$$\text{or, } \frac{K_p}{K_\alpha} = 4:1$$

147. (2)

Force on a charge particle of charge q moving with velocity $\vec{v}$ in the magnetic field of strength $\vec{B}$ is given by

$$\vec{F}_m = q\vec{v} \times \vec{B}$$

Since, force on a point charge by magnetic field is always perpendicular to $\vec{v}$ So, work done by magnetic force on the point charge is zero.

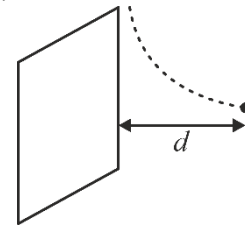
148. (3)

If $r \leq d$, particle will not hit screen

$$r = \frac{mv}{qB_0} \quad \left[\because \frac{mv^2}{r} = qvB_0 \right]$$

Hence, minimum value of v for which the particle will not hit the screen.

$$v = \frac{qB_0 d}{m}$$



149. (2)

Given: $q = 1\mu C = 1 \times 10^{-6} C$;

$$\vec{V} = (2\hat{i} + 3\hat{j} + 4\hat{k}) \text{ m/s and}$$

$$\vec{B} = (5\hat{i} + 3\hat{j} - 6\hat{k}) \times 10^{-3} \text{ T}$$

$$\vec{F} = q(\vec{V} \times \vec{B}) = 10^{-6} \times 10^{-3} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 5 & 3 & -6 \end{vmatrix}$$

$$= (-30\hat{i} + 32\hat{j} - 9\hat{k}) \times 10^{-9} \text{ N}$$

$$\therefore \vec{F} = (-30\hat{i} + 32\hat{j} - 9\hat{k})$$

150. (4)

$$\text{Pitch} = (V \cos \theta)T \text{ and } T = \frac{2\pi m}{qB}$$

$$\therefore \text{Pitch} = (V \cos \theta) \frac{2\pi m}{qB}$$

$$= (4 \times 10^5 \cos 60^\circ) \frac{2\pi \left(\frac{1.67 \times 10^{-27}}{1.6 \times 10^{-19}} \right)}{0.3} = 4 \text{ cm}$$

151. (3)

Time period of one revolution of proton, $T = \frac{2\pi m}{qB}$

Here, m = mass of proton,
 q = charge of proton,
 B = magnetic field.

Linear distance travelled in one revolution,
 $p = T(v \cos \theta)$ (Here, v = velocity of proton)

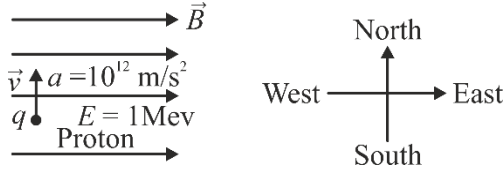
$\therefore$ Length of region, $l = 10 \times (v \cos \theta) T$

$$\Rightarrow l = 10 \times v \cos 60^\circ \times \frac{2\pi m}{qB}$$

$$\Rightarrow l = \frac{10\pi mv}{qB} = \frac{10 \times 3.14 \times 1.67 \times 10^{-27} \times 4 \times 10^5}{1.6 \times 10^{-19} \times 0.3}$$

$$\Rightarrow l = 0.44 \text{ m}$$

152. (1)



As we know, magnetic force $F = qvB = ma$

$$\therefore a = \left(\frac{qvB}{m} \right) \text{ perpendicular to velocity.}$$

$$\therefore \text{Also } v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times e \times 10^6}{m}}$$

$$\therefore a = \frac{qvB}{m} = \frac{eB}{m} \sqrt{\frac{2 \times e \times 10^6}{m}}$$

$$\therefore 10^{12} = \left(\frac{1.6 \times 10^{-19}}{1.67 \times 10^{-27}} \right)^{\frac{3}{2}} \sqrt{2} \times 10^3 B$$

$$\therefore B = \frac{1}{\sqrt{2}} \times 10^{-3} \text{ T} = 0.71 \text{ mT (approx)}$$

153. (2)

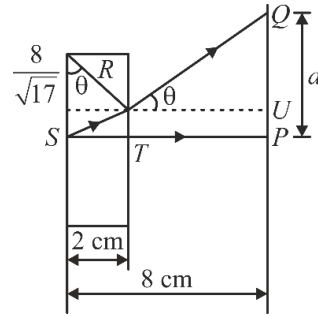
The radius of the circular path is,

$$R = \frac{mv}{Bq} = \frac{\sqrt{2mE_k}}{Bq}$$

$$\Rightarrow R = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 100 \times 1.6 \times 10^{-19}}}{1.5 \times 10^{-3} \times 1.6 \times 10^{-19}}$$

$$\Rightarrow R = 2.25 \text{ cm}$$

$$\text{Now, } \sin \theta = \frac{2}{2.25} = \frac{8}{9}$$



$$\Rightarrow \cos \theta = \frac{\sqrt{17}}{9} \text{ and } \tan \theta = \frac{8}{\sqrt{17}}$$

$$PU = R(1 - \cos \theta) = 2.25 \times \left(1 - \frac{\sqrt{17}}{9} \right)$$

$$\Rightarrow PU = 1.22 \text{ cm}$$

$$\tan \theta = \frac{QU}{6}$$

$$\Rightarrow QU = 6 \times \frac{8}{\sqrt{17}} = 11.64 \text{ cm}$$

$$\therefore PQ = PU + QU = 1.22 + 11.64 = 12.86 \text{ cm}$$

154. (2)

$$\text{As } mvr = qvB \Rightarrow r = \frac{mv}{qB} = \frac{\sqrt{2mK.E.}}{qB}$$

$$\text{As } \frac{1}{2}mv^2 = K.E.$$

$$\Rightarrow m^2v^2 = 2mK.E. \Rightarrow mv = \sqrt{2mK.E.}$$

For proton, electron and α -particle, $m_{He} = 4m_p$ and

$$m_p \gg m_e.$$

$$\text{Also } q_{He} = 2q_p \text{ and } q_p = q_e$$

$$\therefore \text{As KE of all the particles is same then, } r \propto \frac{\sqrt{m}}{q}$$

$$\therefore r_{He} > r_p > r_e \Rightarrow r_e < r_p = r_{He}$$

155. (1)

Radius of the circular path will be $r = \frac{mv}{qB}$

$$\Rightarrow r = \frac{\sqrt{2mKE}}{qB} \quad (\because p = mv = \sqrt{2mKE})$$

$$\therefore KE = qAV$$

$$\therefore r = \frac{\sqrt{2mqAV}}{qB} \Rightarrow r \propto \sqrt{\frac{m}{q}} \quad \therefore \frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$$

156. (4)

Radius of the path is given by $r = \frac{mv}{qB}$

$$r = \frac{\sqrt{2mk}}{eB} \quad (\because k = mv = \sqrt{2mk})$$

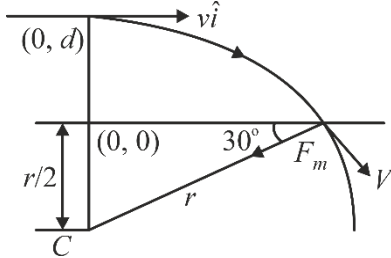
$$= \frac{\sqrt{2meV}}{eB} \quad (\because k = eV)$$

$$r = \frac{\sqrt{\frac{2m}{e}V}}{B} = \frac{\sqrt{\frac{2 \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19}}(500)}}{100 \times 10^{-3}}$$

$$\Rightarrow r = \frac{\sqrt{\frac{9.1}{0.16} \times 10^{-10}}}{10^{-1}} = \frac{3}{4} \times 10^{-4} = 7.5 \times 10^{-4} \text{ m}$$

157. (None)

Assuming particle enters from (0, d)



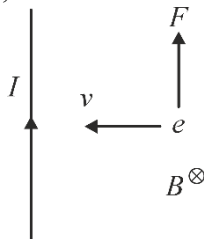
$$r = \frac{mv}{qB}, d = \frac{r}{2}$$

$$a = \frac{qVB}{m} \left[\frac{-\sqrt{3}i - j}{2} \right]$$

this option is not given in the all above four choices.

158. (2)

The force is parallel to the direction of current in magnetic field,



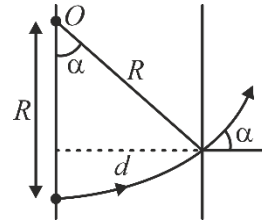
Hence, $F = q(v \times \vec{B})$

According to Fleming's left hand rule, we have, the direction of motion of charge is towards the wire.

159. (4)

From figure, $\sin \alpha = d/R$

And we know, $\frac{mv^2}{R} = qvB$



$$\Rightarrow R = \frac{mv}{qB}$$

$$\therefore \sin \alpha = \frac{dqB}{mv}$$

$$\sin \alpha = Bd \sqrt{\frac{q}{2mV}} \left[\because qV = \frac{1}{2}mv^2 \right]$$

160. (4)

When a charged particle enters the magnetic field in perpendicular direction then it experiences a force in a perpendicular direction.

$$\text{i.e. } F = Bqv \sin \theta$$

Due to which it moves in a circular path with constant speed as well as kinetic energy.

161. (2)

When a charged particle enters a magnetic field at a direction perpendicular to the direction of motion, the path of the motion is circular. In circular motion the direction of velocity changes at every point (the magnitude remains constant).

Therefore, the momentum will change at every point, but kinetic energy will remain constant as it is given by $\frac{1}{2}mv^2$ and v^2 is the square of the magnitude of velocity which does not change.

162. (3)

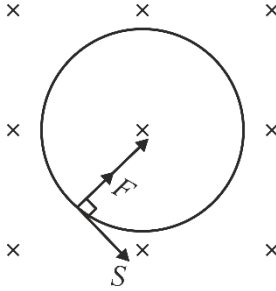
Equating magnetic force to centripetal force,

$$\frac{mv^2}{r} = qvB \sin 90^\circ \Rightarrow \frac{mv}{r} = Bq \Rightarrow v = \frac{Bqr}{m}$$

Time to complete one revolution,

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

163. (2)



The workdone, $dW = Fds \cos\theta$

The angle between force and displacement is 90° .
Therefore work done is zero.

164. (1)

When a moving charged particle is subjected to a perpendicular magnetic field, then it describes a circular path of radius.

$$r = \frac{p}{qB}$$

where q = Charge of the particle
 p = Momentum of the particle
 B = Magnetic field

Here, p , q and B are constant for electron and proton, therefore the radius will be same.

165. (1)

The time period of a charged particle of charge q and mass m moving in a magnetic field (2) is $T = \frac{2\pi m}{qB}$

Clearly time period is independent of speed of the particle.

166. (2)

Magnetic field inside the solenoid is given by

$$B = \mu_0 n I \quad \dots (i)$$

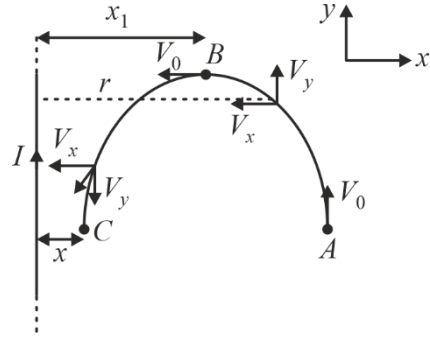
Here, n = number of turns per unit length

The path of charge particle is circular. The maximum possible radius of electrons = $\frac{R}{2}$

$$\therefore \frac{mV_{\max}}{qB} = \frac{R}{2}$$

$$\Rightarrow V_{\max} = \frac{qBR}{2m} = \frac{eR\mu_0 n I}{2m} \text{ (using (i))}$$

167. (4)



$A \rightarrow B$

$$\vec{V} = -v_x \hat{i} + v_y \hat{j}$$

The magnetic field at a distance r from an infinitely long straight wire is $\vec{B} = \frac{\mu_0 I}{2\pi r}$

$$\vec{B} = \frac{\mu_0 I}{2\pi r} (-\hat{k})$$

A charge q moving with velocity in magnetic field experiences a Lorentz force

$$\vec{F} = q(\vec{v} \times \vec{B}) = \frac{\mu_0 q I}{2\pi r} [-v_x \hat{j} - v_y \hat{i}]$$

$$a_x = -\frac{\mu_0 I q}{2\pi m} \cdot \frac{v_y}{r} \quad a_y = -\frac{\mu_0 I q}{2\pi m} \cdot \frac{v_x}{r}$$

$$\frac{v_x dv_x}{dr} = -\frac{\mu_0 I q}{2\pi m} \frac{v_y}{r} \quad \frac{v_x dv_x}{v_y} = -\frac{\mu_0 I q}{2\pi m} \frac{dr}{r}$$

$$\int_{v_0}^{v_0} \frac{v_x dv_x}{v_0^2 - v_x^2} = -\frac{\mu_0 I q}{2\pi m} \int_a^{x_1} \frac{dr}{r}$$

$$\text{Let, } z^2 = v_0^2 - v_x^2$$

$$2zdz = -2v_x dv_x$$

$$zdz = -v_x dv_x$$

$$\frac{v_x dv_x}{\sqrt{v_0^2 - v_x^2}} = \frac{-zdz}{z} = -dz$$

Then integral becomes

$$-\int_{v_0}^0 dz = -\frac{\mu_0 I q}{2\pi m} \ln \frac{x_1}{a}$$

$$v_0 = -\frac{\mu_0 I q}{2\pi m} \ln \frac{x_1}{a}$$



$$v_0 = -\frac{\mu_0 I q}{2\pi m} \ln \frac{x_1}{a}$$

$$x_1 = ae^{-\frac{2\pi m v_0}{\mu_0 I q}} \quad \dots\dots(i)$$

For $B \rightarrow C$

$$\vec{v} = -v_x \hat{i} - v_y \hat{j} ; \quad \vec{B} = \frac{\mu_0 I}{2\pi m} (-\hat{k})$$

$$\vec{F} = q(\vec{v} \times \vec{B}) = \frac{\mu_0 I q}{2\pi r} (-v_x \hat{j} - v_y \hat{i})$$

$$a_x = +\frac{\mu_0 I q}{2\pi m} \frac{v_y}{r} ; \quad a_y = -\frac{\mu_0 I q}{2\pi m} \frac{v_x}{r}$$

$$\frac{v_x dv_x}{dr} = \frac{\mu_0 I q}{2\pi m} \frac{v_y}{r}$$

$$\int_{v_0}^0 \frac{v_x dv_x}{v_0^2 - v_x^2} = \frac{\mu_0 I q}{2\pi m} \int_{x_1}^x \frac{dr}{r}$$

$$\frac{\mu_0 I q}{2\pi m} \ln \frac{x}{x_1} = -\int_{v_0}^0 dz = -v_0$$

$$x = x_1 e^{-\frac{2\pi m v_0}{\mu_0 I q}} \quad \dots\dots(ii)$$

From equation (i) and (ii)

$$X = ae^{-\frac{4\pi m v_0}{\mu_0 I q}}$$

168. (3)

We have,

$$B_p = \frac{\mu_0}{2\pi} \left[\frac{I_1}{r_1} - \frac{I_2}{r_2} \right] (-\hat{k}) = \frac{2 \times 10^{-7}}{10^{-2}} \left[\frac{4}{4} - \frac{2}{6} \right] (-\hat{k})$$

$$= 2 \times 10^{-5} \left[1 - \frac{1}{3} \right] (-\hat{k}) = \frac{4}{3} \times 10^{-5} (-\hat{k})$$

$$\text{So, } \vec{F} = q(\vec{V} \times \vec{B})$$

$$= 3\pi \left[(2\hat{i} + 3\hat{j}) \times \left(-\frac{4}{3} \times 10^{-5} \hat{k} \right) \right]$$

$$= 3\pi \left[\frac{8}{3} \hat{j} - 4\hat{i} \right] \times 10^{-5}$$

$$= [8\pi \hat{j} - 12\pi \hat{i}] \times 10^{-5} = 4\pi [2\hat{j} - 3\hat{i}] \times 10^{-5}$$

$$= 4\pi \times 10^{-5} [-3\hat{i} + 2\hat{j}] = 4\pi \times 10^{-5} [-x\hat{i} + 2\hat{j}]$$

So, $x = 3$

169. (1)

The electric force $\vec{F}_e$ is given by $\vec{F}_e = q\vec{E}$.

The electric field in the region is given as

$\vec{E} = E_0 e^{-\lambda x} \hat{x}$, the force is:

$$\vec{F}_e = qE_0 e^{-\lambda x}$$

The work done by the electric field as the particle moves from $x = 0$ to $x = L$ is:

$$W_e = \int_0^L \vec{F}_e \cdot d\vec{x}$$

$$\Rightarrow W_e = \int_0^L qE_0 e^{-\lambda x} \cdot dx$$

$$\Rightarrow W_e = qE_0 \int_0^L e^{-\lambda x} \cdot dx$$

$$\Rightarrow W_e = qE_0 \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^L$$

$$\Rightarrow W_e = -\frac{qE_0}{\lambda} [e^{-\lambda L} - e^0]$$

$$\Rightarrow W_e = -\frac{qE_0}{\lambda} [e^{-\lambda L} - 1]$$

$$\Rightarrow W_e = \frac{qE_0}{\lambda} [1 - e^{-\lambda L}]$$

Using work-energy theory, the total work done on particle is equal to total change in kinetic energy.

$$\Delta K = W_{\text{total}} = W_m + W_e$$

$$\Delta K = 0 + \frac{qE_0}{\lambda} [1 - e^{-\lambda L}]$$

$$\Delta K = \frac{qE_0}{\lambda} [1 - e^{-\lambda L}]$$

Therefore, the change in kinetic energy is $\frac{qE_0}{\lambda} [1 - e^{-\lambda L}]$.

Thus, the correct option is (1).

170. (2)

Given, Speed of proton, $v = 2 \times 10^5 \text{ ms}^{-1}$

Radius of circular path, $R = 2 \text{ cm}$

For undeflected motion of proton,

$$v = \frac{E}{B} \Rightarrow B = \frac{E}{v}$$

When electric field is switched off



$$R = \frac{mv}{eB} = \frac{mv^2}{eE}$$

$$\Rightarrow E = \frac{mv^2}{eR} = \frac{1.6 \times 10^{-27} \times 4 \times 10^{10}}{1.6 \times 10^{-19} \times 2 \times 10^{-2}} = 2 \times 10^4 \text{ NC}$$

171. (4)

Given, Mass of electron, $m = 9 \times 10^{-31} \text{ kg}$

Kinetic energy of electron, $= 5 \text{ eV}$

Magnetic field, $B = 3 \times 10^{-6} \text{ T}$

As the electron is moving undeflected, net force should be zero.

$$\therefore qE = qvB \Rightarrow E = vB$$

$$\therefore KE = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2KE}{m}}$$

$$\therefore E = \sqrt{\frac{2KE}{m}} \times B = \sqrt{\frac{2 \times 5 \times 1.6 \times 10^{-19}}{9 \times 10^{-31}}} \times 3 \times 10^{-6}$$

$$= \sqrt{\frac{16}{9}} \times 10^{12} \times 3 \times 10^{-6} = 4 \text{ N/C}$$

172. (3)

Electrostatic force, $\vec{F}_1 = q\vec{E}$

Magnetic force, $\vec{F}_2 = q(\vec{v} \times \vec{B})$

173. (3)

Net force $\vec{F}_{\text{net}}$ on particle must be zero

i.e., $q\vec{E} + q(\vec{v} \times \vec{B}) = 0$

Possible cases are

(i) $\vec{E} = 0$ & $\vec{B} = 0$

(ii) $\vec{v} \times \vec{B} = 0$, $\vec{B} \neq 0$, $\vec{E} = 0$

(iii) $q\vec{E} = -q(\vec{v} \times \vec{B})$ (i.e., $\vec{E} \neq 0$ and $\vec{B} \neq 0$)

174. (1)

Given, $\vec{E} = E\hat{k}$

$\vec{B} = B\hat{j}$, $B = 12 \text{ mT} = 12 \times 10^{-3} \text{ T}$. Energy $= 728 \text{ eV}$

$$\text{Energy} = \frac{1}{2}mv^2$$

$$728 \text{ eV} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$$

$$728 \times 1.6 \times 10^{-19} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$$

$$= v = \sqrt{\frac{2 \times 728 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}}$$

we have, $v = 16 \times 10^6 \text{ m/s}$

$$E = vB \Rightarrow E = 16 \times 10^6 \times 12 \times 10^{-3}$$

$$\Rightarrow E = 192 \times 10^3 \text{ V/m} = 192 \text{ kVm}^{-1}$$

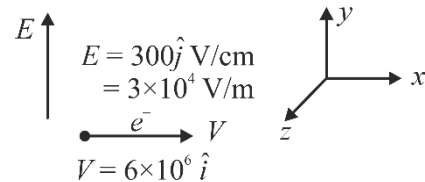
175. (3)

Electric field can change speed and kinetic energy as $\vec{F}$ is parallel or antiparallel to $\vec{v}$.

But, magnetic field cannot change speed and kinetic energy as magnetic force is perpendicular to velocity.

176. (3)

$$\vec{E} = 300\hat{j} \text{ V/cm} = 3 \times 10^4 \hat{j} \text{ V/cm.}$$



$$V = 6 \times 10^6 \hat{i}$$

$\vec{B}$ must be in $+z$ axis.

$$q\vec{E} + q\vec{v} \times \vec{B} = 0$$

$$E = VB$$

$$\therefore B = \frac{E}{V} = \frac{3 \times 10^4}{6 \times 10^6} = 5 \times 10^{-3} \text{ T}$$

Hence, magnetic field $B = 5 \times 10^{-3} \text{ T}$ along $+z$ direction.

177. (4)

As particle is moving along a circular path

$$\therefore R = \frac{mv}{qB} \quad \dots (i)$$

Path is straight line, then

$$qE = qvB$$

$$E = vB \Rightarrow v = \frac{E}{B} \quad \dots (ii)$$

From equation (i) and (ii)

$$m = \frac{qB^2 R}{E} = \frac{1.6 \times 10^{-19} \times (0.5)^2 \times 0.5 \times 10^{-2}}{100}$$

$$\therefore m = 2.0 \times 10^{-24} \text{ kg}$$



178. (3)

As velocity $v = \frac{E}{B} = \frac{7.7 \times 10^3}{0.14} = 55 \text{ km/s}$

179. (1)

The charge experiences both electric and magnetic force.

Electric force, $F_e = qE$

Magnetic force, $F_m = q(\vec{v} \times \vec{B})$

$\therefore$ Net force,

$$\begin{aligned}\vec{F} &= q[\vec{E} + \vec{v} \times \vec{B}] = q \left[3\hat{i} + \hat{j} + 2\hat{k} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & -3 \end{vmatrix} \right] \\ &= q[3\hat{i} + \hat{j} + 2\hat{k} + \hat{i}(-12-1) - \hat{j}(-9-1) + \hat{k}(3-4)] \\ &= q[3\hat{i} + \hat{j} + 2\hat{k} - 13\hat{i} + 10\hat{j} - \hat{k}] = q[-10\hat{i} + 11\hat{j} + \hat{k}] \\ F_y &= 11q\end{aligned}$$

180. (2)

As velocity is not changing, charge particle must go undeflected, then

$$qE = qvB \Rightarrow v = \frac{E}{B}$$

Also, $\vec{E} \times \vec{B} \parallel \vec{v}$

$$\text{and } \left| \frac{\vec{E} \times \vec{B}}{B^2} \right| = \frac{EB \sin \theta}{B^2} = \frac{EB \sin 90^\circ}{B^2} = \frac{E}{B} = v$$

181. (2)

The charged particle will move along the lines of electric field (and magnetic field). Magnetic field will exert no force. The force by electric field will be along the lines of uniform electric field. Hence the particle will move in a straight line.

182. (2)

Due to electric field, it experiences force and accelerates i.e. its velocity decreases.

183. (4)

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$$\Rightarrow \vec{\tau} = (IA \hat{n}) \times \vec{B}$$

The radius of circular loop is $r = 2 \text{ cm} = 0.02 \text{ m}$.

So, the area is $A = \pi r^2 = \pi(0.02)^2 = 4\pi \times 10^{-4} \text{ m}^2$

The unit normal is $\hat{n} = \frac{\hat{k} + \hat{i}}{\sqrt{2}}$

And the current in loop is $I = 100\sqrt{2} \text{ A}$

So, the magnetic moment vector is;

$$\vec{m} = (100\sqrt{2}) \times (4\pi \times 10^{-4}) \times \left(\frac{\hat{i} + \hat{k}}{\sqrt{2}} \right)$$

$$\Rightarrow \vec{m} = 400\pi \times 10^{-4} (\hat{i} + \hat{k}) = 0.04\pi (\hat{i} + \hat{k}) \text{ A} \cdot \text{m}^2$$

The magnetic field in the region is $\vec{B} = B_0 (3\hat{i} + 2\hat{k})$

with $B_0 = 4 \times 10^{-3} \text{ T}$

$$\vec{B} = 4 \times 10^{-3} (3\hat{i} + 2\hat{k}) = (12\hat{i} + 8\hat{k}) \times 10^{-3} \text{ T}$$

Using the formula for magnetic torque, $\vec{\tau} = \vec{m} \times \vec{B}$

$$\Rightarrow \vec{\tau} = [0.04\pi (\hat{i} + \hat{k})] \times [(12\hat{i} + 8\hat{k}) \times 10^{-3}]$$

$$\Rightarrow \vec{\tau} = (0.04\pi \times 10^{-3}) [(\hat{i} + \hat{k}) \times (12\hat{i} + 8\hat{k})]$$

$$\Rightarrow \vec{\tau} = (0.04\pi \times 10^{-3}) (4\hat{j})$$

$$\Rightarrow \vec{\tau} = 0.16\pi \times 10^{-3} \hat{j}$$

$$\Rightarrow \vec{\tau} = 0.16 \times 3.14 \times 10^{-3} \hat{j} = 0.5024 \times 10^{-3} \hat{j}$$

$$\Rightarrow \vec{\tau} = 5024 \times 10^{-7} \hat{j}$$

Therefore, the torque experienced by the loop is $5024 \times 10^{-7} \hat{j}$.

Hence, the correct option is (4).

184. (2)

For a single circular turn of radius r carrying current I , the magnetic moment is:

$$m = I \cdot A = I(\pi r^2)$$

In a flat coil where the turns are wound from r_1 to r_2 , the number of turns per unit radial length is

$$n = \frac{N}{r_2 - r_1}. \text{ The total magnetic moment (M) is the}$$

integral of the moments of all infinitesimal turns of thickness dr :

$$M = \int_{r_1}^{r_2} (I\pi r^2) \cdot n dr$$

$$\Rightarrow M = I\pi n \int_{r_1}^{r_2} r^2 dr = I\pi \left(\frac{N}{r_2 - r_1} \right) \left[\frac{r^3}{3} \right]_{r_1}^{r_2}$$



$$\Rightarrow M = \frac{NI\pi}{3(r_2 - r_1)}(r_2^3 - r_1^3)$$

Using the algebraic identity $r_2^3 - r_1^3 = (r_2 - r_1)(r_2^2 + r_1^2 + r_1r_2)$:

$$M = \frac{NI\pi}{3}(r_1^2 + r_2^2 + r_1r_2)$$

Substituting the values, where number of turns is $N = 200$, the current is $I = 20 \text{ mA} = 20 \times 10^{-3} \text{ A}$, the inner radius of coil is $r_1 = 3 \text{ cm} = 0.03 \text{ m}$ and the outer radius is $r_2 = 6 \text{ cm} = 0.06 \text{ m}$

$$M = \frac{200 \times 20 \times 10^{-3} \times 3.14}{3}$$

$$(0.03^2 + 0.06^2 + 0.03 \times 0.06)$$

$$\Rightarrow M = \frac{4 \times 3.14}{3}(0.0009 + 0.0036 + 0.0018)$$

$$\Rightarrow M = \frac{12.56}{3}(0.0063) = 12.56 \times 0.0021$$

$$\Rightarrow M = 0.026376 \approx 2.64 \times 10^{-2} \text{ A.m}^2$$

$$= \alpha \times 10^{-2} \text{ A.m}^2 \Rightarrow \alpha = 2.64$$

Therefore, the value of α is 2.64.

Thus, the correct option is (2).

185. (1)

The total force on the charged particle is found by adding the electric force and the magnetic force:

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$\vec{v} \times \vec{B} = (v_x \hat{i} + v_y \hat{j}) \times (4 \times 10^{-3} \hat{k})$$

Using the rules $\hat{i} \times \hat{k} = -\hat{j}$ and $\hat{j} \times \hat{k} = \hat{i}$:

$$\vec{v} \times \vec{B} = 4 \times 10^{-3} v_y \hat{i} - 4 \times 10^{-3} v_x \hat{j}$$

Substitute in the main force equation after dividing both sides by q :

$$\frac{(4\hat{i} + 2\hat{j}) \times 10^{-10}}{10^{-9}}$$

$$= 0.4\hat{j} + [(4 \times 10^{-3} v_y) \hat{i} - (4 \times 10^{-3} v_x) \hat{j}]$$

This simplifies to: $0.4\hat{i} + 0.2\hat{j}$

$$= (4 \times 10^{-3} v_y) \hat{i} - (4 \times 10^{-3} v_x + 0.4) \hat{j}$$

Comparing components along

$$\hat{i} : 0.4 = 4 \times 10^{-3} v_y$$

$$v_y = \frac{0.4}{4 \times 10^{-3}} = 100 \text{ m/s}$$

Comparing components along $\hat{j}$:

$$0.2 = 0.4 - 4 \times 10^{-3} v_x$$

$$4 \times 10^{-3} v_x = 0.2$$

$$v_x = 50 \text{ m/s}$$

Hence, the velocity of the particle is:

$$\vec{v} = 50\hat{i} + 100\hat{j} \text{ m/s}$$

Therefore, the correct answer is (1).

186. 314

The magnetic dipole moment of a single loop of wire carrying current I and enclosing area A is given by $m = IA$

The torque experienced by a magnetic dipole in a uniform magnetic field $\vec{B}$ is given by the vector cross product: $\vec{\tau} = \vec{m} \times \vec{B}$

The magnitude of this torque is: $\tau = mB \sin \theta$

Substituting the values into the torque formula:

$$\tau = (NI\pi r^2) B \sin \theta$$

$$\Rightarrow \tau = (125 \times 1 \times 3.14 \times (0.02)^2) \times 0.4 \times \sin 30^\circ$$

$$\Rightarrow \tau = (125 \times 3.14 \times 0.0004) \times 0.4 \times 0.5$$

$$\Rightarrow \tau = 314 \times 10^{-4} \text{ N.m} = \alpha \times 10^{-4} \text{ N} \cdot \text{m}$$

$$\Rightarrow \alpha = 314$$

Therefore, the value of α is 314.

187. (3)

The magnetic moment associated with circular coil,

$$M = M I A$$

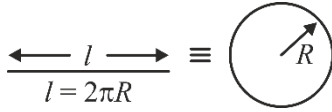
$$M_A = M_B$$

$$\therefore N_A I_A A_A = N_B I_B A_B$$

$$\therefore N_A I_A \pi (0.1)^2 = N_B I_B \pi (0.2)^2 \quad (\because A = \pi r^2)$$

$$\therefore N_A I_A = 4 N_B I_B$$

188. (11)



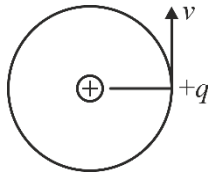
As, length of wire = $2\pi R$

$$\Rightarrow \frac{314}{100} = 2\pi R \Rightarrow R = 0.5 \text{ m}$$

Magnetic Moment = $I \times A = 14 \times \pi R^2$

$$= 14 \times (3.14) \times \frac{1}{4} = 1.99 \approx 11.00 \text{ A-m}^2$$

189. (4)



Length of the circular path, $l = 2\pi r$

$$\text{Current, } i = \frac{q}{T} = \frac{qv}{2\pi r}$$

Magnetic moment

$$M = \text{Current} \times \text{Area} = i \times \pi r^2 = \frac{qv}{2\pi r} \times \pi r^2$$

$$M = \frac{1}{2} qv.r$$

Radius of circular path in magnetic field, $r = \frac{mv}{qB}$

$$\therefore M = \frac{1}{2} qv \times \frac{mv}{qB} \Rightarrow M = \frac{mv^2}{2B}$$

Direction of $\vec{M}$ is opposite of $\vec{B}$ therefore

$$\vec{M} = \frac{-mv^2 \vec{B}}{-2B^2}$$

190. (4)

Let be the area of the square and r be the radius of circular loop.

For square

$$m = (I)a^2$$

For circular loop

$$M = (I)\pi^2 \Rightarrow M = (I)(\pi) \left(\frac{4a^2}{\pi^2} \right)$$

$$M = \frac{4Ia^2}{\pi} \Rightarrow M = \frac{4m}{\pi} \quad (\because m = Ia^2)$$

191. (3)

Magnetic field at the centre of loop, $B_1 = \frac{\mu_0 I}{2R}$

Dipole moment of circular loop is $m = IA$

$$m_1 = I.A = I.\pi R^2 \quad \{R = \text{Radius of the loop}\}$$

If magnetic moment is doubled (keeping current constant)

R becomes $\sqrt{2}R$

$$B_2 = \frac{\mu_0 I}{2(\sqrt{2}R)} \quad \therefore \frac{B_1}{B_2} = \frac{\frac{\mu_0 I}{2R}}{\frac{\mu_0 I}{2(\sqrt{2}R)}} = \sqrt{2}$$

192. (4)

Given,

$$\tau = 80\sqrt{3} \text{ Nm}, \theta = 60^\circ$$

The torque experienced by a magnetic dipole is

$$\tau = MB \sin\theta = MB \sin 60^\circ = \frac{\sqrt{3}}{2} MB = 80\sqrt{3}$$

$$\therefore MB = 160$$

The potential energy of the dipole is

$$U - MB \cos\theta = -MB \cos 60^\circ = -160 \times 1/2 = -80 \text{ J}$$

193. (2)

Given

Magnetic field, $B = 2 \times 10^{-3} \text{ T}$

Current in rectangular loop, $I = 5 \text{ A}$

Area of rectangular loop, $A = l \times b = 0.2 \times 0.1 \text{ m}^2$

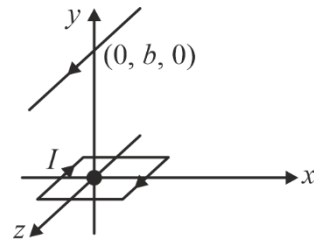
Magnetic moment of rectangular coil, $\vec{M} = i\vec{A}$

$$= 5 \times (0.2) \times (0.1) (-\hat{i}) = 0.1(-\hat{i})$$

$$\text{Torque, } \vec{\tau} = \vec{M} \times \vec{B} = 0.1(-\hat{i}) \times (2 \times 10^{-3})(\hat{j})$$

$$= 2 \times 10^{-4} (-\hat{k}) \text{ N-m}$$

194. (1)



$$\vec{\tau} = \vec{M} \times \vec{B} = MB\hat{n}$$

$$|\vec{\tau}| = I(2a)^2 \times \frac{\mu_0 I}{2\pi b}$$

$$= \frac{2\mu_0 I^2 a^2}{\pi b}$$

195. (2)

Magnetic moment of loop ABCD,

$M_1 = \text{area of loop} \times \text{current}$

$\vec{M}_1 = (abI)\hat{k}$ (Here, $ab = \text{area of rectangle}$)

Magnetic moment of loop DEFA,

$\vec{M}_2 = (abI)\hat{j}$

Net magnetic moment,

$\vec{M} = \vec{M}_1 + \vec{M}_2 \Rightarrow \vec{M} = abI(\hat{k} + \hat{j})$

$$\Rightarrow |\vec{M}| = \sqrt{2}abI \left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}} \right)$$

195. (3)

Torque on circular loop, $\tau = -MB \sin \theta$

Where, $M = \text{magnetic moment}$

$B = \text{magnetic field}$

Now using $\tau = I\alpha$

$\therefore \tau = -MB \sin \theta = I\alpha$

$$\Rightarrow -\pi a^2 IB \theta = \frac{ma^2 \alpha}{2}$$

($\because M = IA$ and moment of inertia of circular loop,

$$I = \frac{ma^2}{2})$$

$$\Rightarrow \alpha = -\left(\frac{2\pi IB}{m}\right)\theta$$

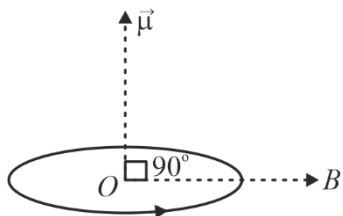
$$\Rightarrow \omega = \sqrt{\frac{2\pi IB}{m}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{2\pi IB}{m}} \Rightarrow T = \sqrt{\frac{2\pi m}{IB}}$$

197. (4)

$\tau = MB \sin 45^\circ = N(iA) B \sin 45^\circ$

$$= 100 \times 3(5 \times 2.5) \times 10^{-4} \times 1 \times \frac{1}{\sqrt{2}} = 0.27 \text{ N-m}$$

198. (2)



$$|\vec{\tau}| = |\vec{\mu} \times \vec{B}| \quad [\mu = NIA]$$

$$= NIA \times B \sin 90^\circ \quad [A = \pi r^2]$$

$$\Rightarrow \tau = N\pi r^2 B$$

199. (1)

Magnetic moment, $\mu = IA = \frac{qv}{2\pi r}(\pi r^2)$

$$\text{Or, } \mu = \frac{qr\omega}{2\pi r}(\pi r^2) = \frac{1}{2}qr^2\omega$$

200. (3)

Magnetic moment of current carrying rectangular loop of area A is given by $M = NIA$.

Magnetic moment of current carrying coil is a vector and its direction is given by right hand thumb rule,

for stable equilibrium of rectangular loop, $\vec{B}$ and $\vec{M}$ at centre due to current in loop are always parallel.

Hence, (3) corresponds to stable equilibrium.

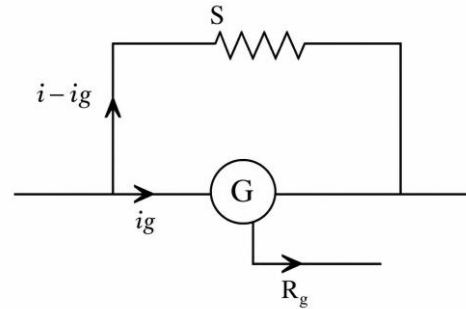
201. (1)

For stable equilibrium $\vec{M} \parallel \vec{B}$

For unstable equilibrium $\vec{M} \parallel (-\vec{B})$

202. 50

The value of resistance of galvanometer coil is 50Ω .



$$i_g R_g = 2 \left(\frac{1}{2} - i_g \right)$$

$$i_g = \frac{1}{R_g + 2}$$

$$10 = (R_g + 470)i_g$$

$$10 = \frac{R_g + 470}{R_g + 2}$$

$$10R_g + 20 = R_g + 470$$

$$R_g = 50\Omega$$

203. (1)

Surface energy released voltage sensitivity

$$= \frac{\theta}{V} = \frac{NAB}{KR}$$

$$\frac{V_s(M_1)}{V_s(M_2)} = \left(\frac{N_1 A_1 B_1}{N_2 A_2 B_2} \right) \frac{R_2}{R_1} = \frac{15 \times 3.6 \times 0.25}{21 \times 1.8 \times 0.5} \times \frac{7}{5} = \frac{1}{1}$$

204. (3)

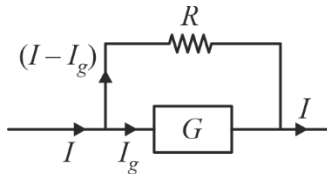
Here,

I_g = full scale deflection current of galvanometer

G = resistance of galvanometer

I = total current to be measured

R = required shunt resistance



$$I_g G = (I - I_g)R$$

$$20 \times 10^{-3} \times 30 = (3 - 0.02) \times R$$

$$R = \left(\frac{0.6}{2.98} \right) = \frac{30}{x}$$

$$x = \left(\frac{2.98 \times 30}{0.6} \right) = 149$$

205. (1)

A moving coil galvanometer works on the principle that when a current-carrying coil is placed in a magnetic field, it experiences a torque which is given by $\tau = BINA$ this torque causes the coil to rotate against restoring torque $\therefore \tau = C\theta$

$$(A) \tau = C\theta \Rightarrow [ML^2T^{-2}]$$

(B) The expression of current sensitivity is

$$C.S. = \frac{\theta}{I} = \frac{BNA}{C}$$

$\therefore$ If N increases, then R also increases. So,
V.S = constant

206. (2)

Given,

Resistance of galvanometer,

$$G = 200 \Omega$$

Current through galvanometer,

$$i_g = 20 \mu A$$

$$i = i_g \left(\frac{G}{S} + 1 \right)$$

$$\Rightarrow 20 \times 10^{-3} = 20 \times 10^{-6} \left(\frac{200}{S} + 1 \right)$$

$$10^3 = \frac{200}{S} + 1 \Rightarrow \frac{200}{S} = 999$$

$$\Rightarrow S \approx 0.2 \Omega$$

207. (3)

$$i_g = \frac{V}{(R + G)} = \frac{10}{400 + 100} = 20 \times 10^{-3} A$$

For ammeter,

Let shunt resistance = S

$$\therefore V_G = V_S \Rightarrow i_g R = (i - i_g)S$$

$$\Rightarrow 20 \times 10^{-3} \times 100 = 10S$$

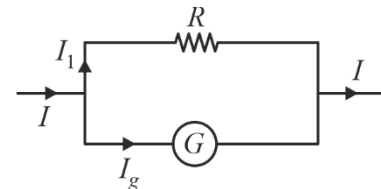
$$\therefore S = 20 \times 10^{-2} \Omega$$

208. (3)

$$V = I_g(R + R_g)$$

$$\Rightarrow R = \frac{V}{I_g} - R_g = \frac{100}{5 \times 10^{-3}} - 50 = 19950 \Omega$$

209. (4)



$$I_1 R = I_g G \Rightarrow \frac{95}{100} IR = \frac{51}{100} G \Rightarrow R = \frac{G}{19}$$

$$\text{Resistance of ammeter, } R_a = \frac{RG}{R + G} = \frac{\frac{G^2}{19}}{\frac{20G}{19}} = \frac{G}{20}$$

210. (4)

$$\text{Torque, } \tau = NIAB \sin \theta$$

$$\text{Torque is also given as, } \tau = K\theta$$

$$\text{On comparing, } NIAB \sin \theta = K\theta$$

$$K = \frac{NIAB \sin \theta}{\theta}$$

$$K = \frac{100 \times 0.01 \times 0.01 \times 2 \times 10^{-4}}{0.05} = 4 \times 10^{-5} \text{ N-m/rad}$$

$$= x \times 10^{-5} \text{ N-m/rad (given)} \therefore x = 4$$

211. (3)

$$I_g = 3\text{mA}, I = 8\text{A}, G = 10\Omega$$

Since S and G are in parallel,

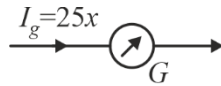
$$\therefore V_G = V_S \Rightarrow GI_g = S(I - I_g) \Rightarrow 10 \times 3 = 5(8000 - 3)$$

$$\therefore S = 3.75 \times 10^{-3} \Omega$$

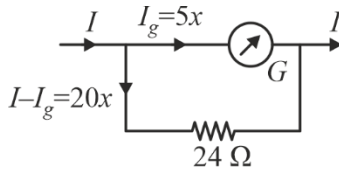
212. (4)

For moving coil galvanometer

Let x = current/division



After applying shunt



$$\therefore V_G = V_S$$

$$\text{Now, } 5x \times G = 20x \times 24$$

$$G = 4 \times 24 = 96 \Omega$$

213. (3)

Electric current,

$i \propto \theta$ (angle of deflection)

$$\therefore \frac{i_2}{i_1} = \frac{\theta_2}{\theta_1} \Rightarrow \frac{i_2}{200\mu\text{A}} = \frac{\pi/10}{\pi/3} = \frac{3}{10} \left[\because 60^\circ = \frac{\pi}{3} \right]$$

$$\therefore i_2 = 60\mu\text{A}$$

214. (3)

$$\text{Here, } I = 10 \times 10^{-3}\text{A}; I_g = 1 \times 10^{-3}\text{A}$$

For voltmeter,

$$R = \frac{V}{I_g} - G = \frac{50}{10^{-3}} - 54 \approx 50k\Omega$$

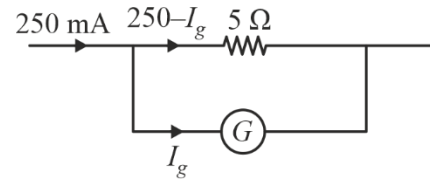
(A) Is correct.

For ammeter,

$$r = \frac{I_g G}{I - I_g} = \frac{10^{-3} \times 54}{(10 - 1) \times 10^{-3}} = 6\Omega(C)$$

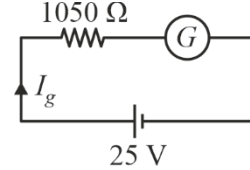
215. (50)

When a resistance of 5Ω is shunted with a MCG



$$I_g = \frac{250\text{mA} \times 5}{5 + R_G} \quad \dots\dots(i)$$

When a resistance of 1050Ω is shunted with a MCG



$$I_g = \frac{25}{1050 + R_G} \quad \dots\dots(ii)$$

From (i) and (ii)

$$\frac{25}{1050 + R_G} = \frac{5}{4(5 + R_G)}$$

$$100(5 + R_G) = 1050 \times 5 + R_G \times 5$$

$$\Rightarrow 95 R_G = 4750$$

$$\therefore R_G = 50\Omega$$

216. (1)

Current sensitivity,

$$I_s = \frac{NBA}{C} \text{ and}$$

$$\text{Voltage sensitivity } V_s = \frac{NBA}{CG}$$

$$\Rightarrow V_s = \frac{I_s}{G},$$

As G is constant,

$$\therefore V_s \propto I_s$$

$\therefore$ Percentage change in V_s is also 25%

217. (4)

$$\text{Current sensitive} = \frac{BNA}{k}$$

So, if N is doubled then current sensitivity is doubled.

Voltage sensitivity,

$$V = \frac{BNA}{Rk}, \text{ as } N \text{ is doubled } R \text{ is also doubled. So,}$$

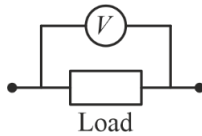
No change in voltage sensitivity.

218. (2)

Due to motion or oscillates of the coil, the eddy current develops in the metallic material which oppose the motion of the coil thus bringing the coil to rest.

219. (1)

For measuring potential difference, we always use voltmeter having high resistance. Because higher is the resistance, lesser is the current drawn by voltmeter which results in correct voltage drop across the load.



220. (4)

We have

$$C.S = \frac{NBA}{K} \text{ and } V.S = \frac{NBA}{KR}$$

As current sensitivity increases by 50%. So, no. of turns increases by 50%. As no. of turn increases by 50%, so resistance also increases by 50%. Therefore, V.S remains constant

221. (2)

In moving coil galvanometer

$$BINA = K\theta$$

$$A = \frac{K\theta}{NIB} = \frac{4 \times 10^{-5} \times 0.05}{200 \times 10 \times 10^{-3} \times 10^{-2}} = \frac{20 \times 10^{-7}}{2 \times 10^{-2}} \\ = \frac{10 \times 10^{-7}}{10^{-2}} = 10^{-4} \text{ m}^2 = 1 \text{ cm}^2$$

222. (4)

$$\text{Current } I = \left(\frac{K}{NAB} \right) \theta$$

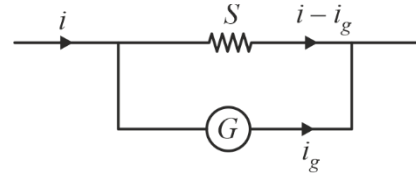
Where, K = torsional constant and B = magnetic field

$$\therefore \frac{\theta}{I} = \frac{NAB}{K} \Rightarrow \frac{\theta}{I} \propto B \text{ and } \frac{\theta}{I} \propto \frac{1}{K}$$

223. (2)

We have, $S(i - i_g) = i_g G$

$$\Rightarrow S \left(1 - \frac{i_g}{i} \right) = \frac{i_g}{i} G$$



$$\Rightarrow \frac{G}{S} = \frac{i}{i_g} - 1$$

$$\Rightarrow \frac{i}{i_g} = \frac{G}{S} + 1 = \frac{72}{8} + 1 = 10$$

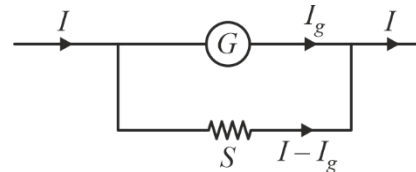
$$\Rightarrow \frac{i_g}{i} = \frac{1}{10}. \text{ So, } \% \frac{i_g}{i} = 10\%$$

224. (4)

By current division rule

$$I_g = I \times \frac{S}{G+S} \Rightarrow I = \frac{G+S}{S} I_g \dots\dots(i)$$

$$\text{Now, figure of merit}(k) = \frac{I_g}{\theta}$$



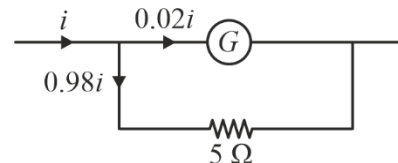
$$\Rightarrow I_g = K\theta$$

As, n = actual number of deflection

So, $I_g = kn$, [$\because k$ represent current per deflection]..(ii)

$$\text{So, } I = \frac{K_n}{S} (G+S) \quad [\text{From (i) \& (ii)}]$$

225. (3)



$$0.02i R_g = 0.98i \times 5$$

$$R_g = 245\Omega.$$

226. (4)

Given, current sensitivity = 2 div/mA

Full scale current,

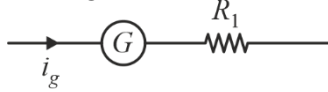
$$I_{\max} = \frac{50}{2} = 25 \text{ mA}$$



$$\therefore \text{Resistance, } R = \frac{V}{I} = \frac{50\text{mV}}{25\text{mA}} = 2\Omega$$

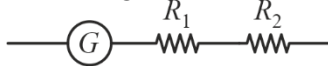
227. (4)

Galvanometer of resistance (G) converted into a voltmeter of range 0-1 V.



$$V = 1 = i_g(G + R_1) \quad \dots\dots(i)$$

To increase the range of voltmeter 0-2V



$$2 = i_g(R_1 + R_2 + G) \quad \dots\dots(ii)$$

Dividing eq. (i) by (ii),

$$\Rightarrow \frac{1}{2} = \frac{G + R_1}{G + R_1 + R_2} \Rightarrow G + R_1 + R_2 = 2G + 2R_1$$

$$\therefore R_2 = G + R_1$$

228. (4)

Given,

Current passing through galvanometer, $I = 6 \text{ mA}$

Deflection, $\theta = 2^\circ$

Figure of merit of galvanometer

$$= \frac{I}{\theta} = \frac{6 \times 10^{-3}}{2} = 3 \times 10^{-3} \text{ A/div}$$

229. (20)

Given,

Area of galvanometer coil, $A = 3 \times 10^{-4} \text{ m}^2$

Number of turns in the coil, $N = 500$

Current in the coil, $I = 0.5 \text{ A}$

$$\text{Torque } \tau = |\vec{M} \times \vec{B}| = NIAB \sin(90^\circ) = NIAB$$

$$\Rightarrow B = \frac{\tau}{NiA} = \frac{1.5}{500 \times 0.5 \times 3 \times 10^{-4}} = 20 \text{ T}$$

230. (3)

$$i_g = 20 \times 50 = 1000 \mu\text{A} = 1 \text{ mA}$$

Using, $V = i_g(G + R)$, we have

$$2 = 10^{-3}(100 + R_1)$$

$$R_1 = 1900 \Omega$$

When, $V = 10 \text{ volt}$

$$10 = 10^{-3}(100 + R_2 + R_1)$$

$$\Rightarrow 10000 = (100 + R_2 + 1900)$$

$$\Rightarrow R_2 = 8000 \Omega$$

231. (2)

$$\text{In an ammeter, } i_g = i_0 \frac{R_A}{R_A + G}$$

$$\text{And for voltmeter, } V = i_g(G + R_V) = G i_0$$

On solving above equations, we get, $R_A R_V = G^2$

$$\text{and } \frac{R_A}{R_V} = \left(\frac{i_g}{i_0 - i_g} \right)^2$$

232. (None)

$$v = i_g(R + G) \Rightarrow 5 = 10^{-4}(2 \times 10^6 + x)$$

$x = -195 \times 10^4 \Omega$. As the resistance of given galvanometer cannot be negative. Hence the given condition is impossible

233. (3)

$$V = i_g(G + R) = 4 \times 10^{-3}(50 + 5000) = 20 \text{ V}$$

234. (4)

$$C\theta = NBI A \sin 90^\circ$$

$$\text{Or } 10^{-6} \left(\frac{\pi}{180} \right) = 175 B (10^{-3}) \times 10^{-4}$$

$$\therefore B = 10^{-3} \text{ T}$$

235. (4)

$$\text{Using, } i_g = i \frac{S}{S + G}$$

$$0.002 = 0.5 \frac{S}{S + 50}$$

$$\text{On Solving, we get, } S = \frac{100}{498} \approx 0.2 \Omega$$

236. (2)

When key K_1 is closed and key K_2 is open

$$i_g = \frac{E}{220 + R_g} = C\theta_0 \quad \dots\dots(i)$$

When both the keys are closed

$$I_g = \left(\frac{E}{220 + \frac{5R_g}{5 + R_g}} \right) \times \frac{5}{(R_g + 5)} = \frac{C\theta_0}{5}$$

$$\Rightarrow \frac{5E}{225R_g + 1100} = \frac{C\theta_0}{5} \quad \dots\dots(ii)$$

Dividing (i) by (ii), we get

$$\Rightarrow \frac{225R_g + 1100}{1100 + 5R_g} = 5$$

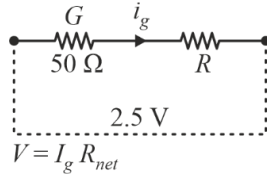
$$\Rightarrow 5500 + 25R_g = 225R_g + 1100$$

$$\Rightarrow 200R_g = 4400 \Rightarrow R_g = 22 \Omega$$

237. (2)

Galvanometer has 25 divisions

$$I_g = 4 \times 10^{-4} \times 25 = 10^{-2} \text{ A}$$



$$V = I_g R_{\text{net}}$$

$$V = I_g(G + R) \Rightarrow 2.5 = (50 + R) 10^{-2} \therefore R = 200 \Omega$$

238. (3)

Deflection current $= (I_g)_{\text{max}} = nk = 0.005 \times 30$

Where, n = Number of divisions = 30 and $k = 0.005$ amp/division $= 15 \times 10^{-2} = 0.15$

$$V = I_g[20 + R]$$

$$\Rightarrow 15 = 0.15 [20 + R] \Rightarrow 100 = 20 + R \Rightarrow R = 80 \Omega$$

239. (4)

Given

Resistance of galvanometer, $G = 100 \Omega$

Current, $i_g = 1 \text{ mA}$

A galvanometer can be converted into voltmeter by connecting a large resistance R in series with it.

Total resistance of the combination $= G + R$

According to Ohm's law, $V = i_g(G + R)$

$$\therefore 10 = 1 \times 10^{-3}(100 + R_0)$$

$$\Rightarrow R_0 = 10000 - 100 = 9900 \Omega = 9.9 \text{ k}\Omega$$

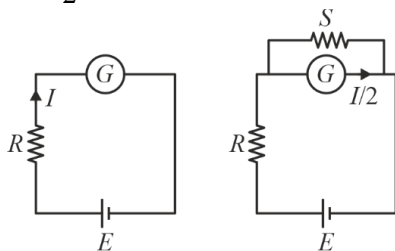
240. (2)

Figure of merit of a galvanometer is the current required to produce a deflection of one division in the galvanometer i.e., figure of merit $= I/\theta$

$$I = \frac{\epsilon}{R + G} \quad G = \frac{1}{9} \text{ K}\Omega$$

$$\frac{1}{2} = \frac{\epsilon}{R + \frac{GS}{G + S}} \times \frac{S}{S + G} \Rightarrow \frac{1}{2} = \frac{\epsilon S}{R(S + G) + GS}$$

$$S = \frac{RG \times \frac{1}{2}}{\epsilon - \frac{(R + G)I}{2}}$$

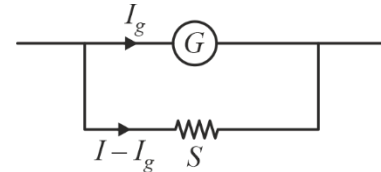


$$S = \frac{11 \times 10^3 \times \frac{1}{2} \times 10^2 \times 270 \times 10^{-6}}{6 - \left(\frac{6}{2}\right)} = 110 \Omega$$

241. (4)

According to question, current through galvanometer,

$$I_g = 1 \text{ mA}$$



Current through shunt, $(I - I_g) = 2 \text{ A}$

Galvanometer resistance $R_g = 25 \Omega$

Resistance of shunt, $S = ?$

$$I_g R_g = (I - I_g) S$$

$$S = \frac{10^{-3} \times 25}{2} \Rightarrow S \approx 1.25 \times 10^{-2} \Omega$$

242. (3)

Given: Current through the galvanometer,

$$i_g = 5 \times 10^{-3} \text{ A}$$

Galvanometer resistance $G = 15 \Omega$

Let resistance R to be put in series with the galvanometer to convert it into a voltmeter.

$$V = i_g(R + G)$$

$$10 = 5 \times 10^{-3}(R + 15)$$

$$\therefore R = 2000 - 15 = 1985 = 1.985 \times 10^3 \Omega$$

243. (3)

$$I_g G = (I - I_g) S$$

$$\therefore 10^{-3} \times 100 = (10 - 10^{-3}) \times S$$

$$\therefore S \approx 0.01 \Omega$$

244. (4)

$$\text{As we know, } I = \frac{V}{R} = \frac{5}{50} = 0.1 \text{ A}$$

$$I' = 0.099 \text{ A}$$

When Galvanometer is connected

$$R_{eq} = 50 + \frac{100S}{100 + S} = \frac{V}{I'}$$

$$\Rightarrow \frac{100S}{100 + S} = \frac{5}{0.099} - 50$$

$$\Rightarrow \frac{100S}{100 + S} = 50.50 - 50 \Rightarrow \frac{100S}{100 + S} = 0.5$$

$$\Rightarrow 100S = 50 + 0.5S \Rightarrow 99.5S = 50$$

$$\Rightarrow S = r_s = \frac{50}{99.05} = 0.5\Omega$$

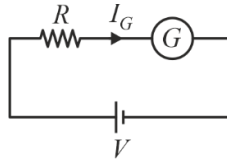
So, shunt of resistance = 0.5Ω is connected in parallel with the galvanometer.

245. (1)

According to Ohm's Law, $I = V/R$

$$I_g = \frac{V}{R+G}$$

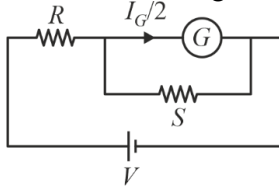
Where, I_g -Galvanometer current, G -Galvanometer resistance



When shunt of resistance S is connected parallel to the Galvanometer then $G = \frac{GS}{G+S}$

$$\therefore I = \frac{V}{R + \frac{GS}{G+S}}$$

Equal potential difference is given by



$$I'_g G = (I - I'_g)S$$

$$I'_g (G+S) = IS \Rightarrow \frac{I_g}{2} = \frac{IS}{G+S}$$

$$\Rightarrow \frac{V}{2(R+G)} = \frac{V}{R + \frac{GS}{G+S}} \times \frac{S}{G+S}$$

$$\Rightarrow \frac{1}{2(R+G)} = \frac{S}{R(G+S) + GS}$$

$$\Rightarrow R(G+S) + GS = 2S(R+G)$$

$$\Rightarrow RG + RS + GS = 2S(R+G)$$

$$\Rightarrow RG + 2S(R+G) - S(R+G)$$

$$\Rightarrow RG + S(R+G)$$

246. (2)

To measure AC voltage across a resistance a moving coil galvanometer is used.

247. (4)

Statements I is false and Statement II is true

$$\text{For ammeter shunt resistance, } S = \frac{I_g G}{I - I_g}$$

Therefore for I to increase, S should decrease, So additional S can be connected across it.

248. (4)

The current that will given full scale deflection in the absence of the shunt is nearly equal to the current through the galvanometer when shunt is connected i.e. I_g

$$\text{As } I_g = \frac{IS}{G+S} = \frac{5.5 \times 1}{120+1} = 0.045 \text{ ampere.}$$

249. (3)

Resistance of Galvanometer

$$G = \frac{\text{Current sensitivity}}{\text{Voltage sensitivity}} \Rightarrow G = \frac{10}{2} = 5\Omega$$

Here i_g = Full scale deflection current

$$= \frac{150}{10} = 15\text{mA}$$

V = voltage to be measured = 150 volts
($\therefore$ each division reads 1 volt)

$$\Rightarrow R = \frac{150}{15 \times 10^{-3}} - 5 = 9995\Omega$$

